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Fiber-coupled Terahertz transceiver system

Abstract

Transport networks, network elements, and methods of use are described herein, including a transmitter comprising a client-side input, transmitter circuitry, and antennas. The client-side input is configured to receive baseband signals having client data encoded therein. The transmitter circuitry is configured to receive the baseband signals from the client-side input and generate antenna feed signals based on the baseband signals. The antennas are configured to receive the antenna feed signals from the transmitter circuitry, generate radiated signals based on the antenna feed signals, and couple the radiated signals into a hollow waveguide. Each of the radiated signals is a radiated electromagnetic wave configured for coherent detection and has a frequency in a range between 300 Gigahertz (GHz) and 10 Terahertz (THz).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of the patent application filed on Oct. 25, 2024, and identified by U.S. Ser. No. 18/927,535, which claims priority under 35 U.S.C. 119(e) to the provisional application identified by U.S. Ser. No. 63/655,823, filed on Jun. 4, 2024; to the provisional application identified by U.S. Ser. No. 63/658,162, filed on Jun. 10, 2024; to the provisional application identified by U.S. Ser. No. 63/658,176, filed on Jun. 10, 2024; to the provisional application identified by U.S. Ser. No. 63/661,437, filed on Jun. 18, 2024; to the provisional application identified by U.S. Ser. No. 63/666,886, filed on Jul. 2, 2024; to the provisional application identified by U.S. Ser. No. 63/575,162, filed on Apr. 5, 2024; to the provisional application identified by U.S. Ser. No. 63/683,007, filed on Aug. 14, 2024; and to the provisional application identified by U.S. Ser. No. 63/593,874, filed on Oct. 27, 2023; the entire contents of all of which are hereby incorporated by reference herein.

BACKGROUND ART

(1) Optical networking is a means of communication that uses signals encoded in light to transmit information in various types of telecommunications networks, including limited range local-area networks (LANs) or wide-area networks (WANs). It is a form of optical communication that relies on optical amplifiers, lasers, or LEDs and wavelength-division multiplexing (WDM) to transmit large quantities of data, generally across fiber-optic cables. Because it is capable of achieving extremely high bandwidth, it is an enabling technology for the Internet and telecommunication

networks that transmit the vast majority of all human and machine-to-machine information. However, further development and optimization of optical networking systems face certain limiting factors, namely, power dissipation, thermal requirements, and mechanical tolerances.
(2) Optical components generate photons by exciting electrons in a gain medium, and the electrons emit photons as they return to lower energy levels. Despite efforts to improve efficiency, optical components generate some amount of heat during the electron excitation process, and such heat is referred to as power dissipation. Excessive power dissipation may lead to thermal management problems and may affect the performance and longevity of the optical components.
(3) Optical components are sensitive to temperature fluctuations and often require lower operating temperatures than purely electronic components to maintain optimal performance. Elevated temperatures may result in increased signal noise, diminished signal quality, and reduced service life for optical components. Accordingly, optical components often require cooling systems (e.g., heat sinks, fans, or thermoelectric devices) to dissipate excess heat and maintain the optical components within a safe temperature range.
(4) Optical networking systems typically operate in micrometer wavelengths, demanding extreme precision in component fabrication, assembly, and alignment. Even slight deviations from the required mechanical tolerances may lead to signal degradation, loss, or the introduction of optical crosstalk, negatively impacting network performance. Achieving and maintaining the necessary mechanical tolerances necessitates advanced manufacturing techniques and stringent quality control measures.
SUMMARY OF THE INVENTION
(5) Transport networks, network elements, and methods of use are disclosed herein. The problems of power dissipation, thermal requirements, and mechanical tolerances are addressed through a Terahertz (THz) radio frequency (RF) transmission system in which RF signals are coupled into hollow waveguides for transmission.
(6) In terms of power dissipation, RF transceivers lack optical components, thereby eliminating power requirements associated with activating optical components and generating photons. Further, transmission of RF signals in the THz frequency band involves longer wavelengths than transmission of optical signals in higher frequency bands, meaning that less energy is required to create and modulate the signals. Finally, no optical-electrical conversion is required, as RF transceivers operate entirely in the electrical domain. Thus, power dissipation is reduced in the fiber-coupled THz RF transceiver system. RF transceivers also entail relaxed thermal requirements, as the RF transceivers lack optical components that are sensitive to temperature fluctuations. As a result, no temperature control is required, and no direct current (DC) bias controls are required. Further, because of the relaxed thermal requirements, RF transceivers may be more easily integrated into existing processes or technologies. In terms of mechanical tolerances, antennas do not require the precise alignment that optics do (i.e., coupling RF signals into hollow waveguides requires less precision than coupling optical signals into hollow waveguides). Further, operating in the THz frequency band means that wavelengths of signals being transmitted are much longer, which also contributes to relaxed mechanical tolerances. Finally, in terms of spectral efficiency, RF systems are generally more spectrally efficient than optical systems, thus allowing for an increased throughput.
(7) In one aspect, the present disclosure includes a transmitter, comprising: a client-side input configured to receive one or more baseband signals having client data encoded therein; transmitter circuitry configured to receive the one or more baseband signals from the client-side input and generate one or more antenna feed signals based on the one or more baseband signals; and one or more antennas configured to receive the one or more antenna feed signals from the transmitter circuitry, generate one or more radiated signals based on the one or more antenna feed signals, and couple the one or more radiated signals into a hollow waveguide, each of the one or more radiated signals being radiated electromagnetic waves configured for coherent detection and having a

frequency in a range between 300 Gigahertz (GHz) and 10 Terahertz (THz).

(8) In another aspect, the present disclosure includes a receiver, comprising: one or more antennas configured to detect one or more radiated signals received from a hollow waveguide and generate one or more antenna output signals based on the one or more radiated signals, each of the one or more radiated signals being radiated electromagnetic waves configured for coherent detection, having a frequency in a range between 300 Gigahertz (GHz) and 10 Terahertz (THz), and having client data encoded therein; receiver circuitry configured to receive the one or more antenna output signals from the one or more antennas and generate one or more baseband signals based on the one or more antenna output signals; and a client-side output configured to receive the one or more baseband signals from the receiver circuitry and transmit the one or more baseband signals.

(9) In another aspect, the present disclosure includes a transport network, comprising: one or more hollow waveguides; a transmitter, comprising: a client-side input configured to receive one or more first baseband signals having client data encoded therein; transmitter circuitry configured to receive the one or more first baseband signals from the client-side input and generate one or more antenna feed signals based on the one or more first baseband signals; and one or more first antennas configured to receive the one or more antenna feed signals from the transmitter circuitry, generate one or more radiated signals based on the one or more antenna feed signals, and couple the one or more radiated signals into at least one of the one or more hollow waveguides, each of the one or more radiated signals being radiated electromagnetic waves configured for coherent detection and having a frequency in a range between 300 Gigahertz (GHz) and 10 Terahertz (THz); and a receiver, comprising: one or more second antennas configured to detect the one or more radiated signals received from the at least one of the one or more hollow waveguides and generate one or more antenna output signals based on the one or more radiated signals; receiver circuitry configured to receive the one or more antenna output signals from the one or more second antennas and generate one or more second baseband signals based on the one or more antenna output signals, the one or more second baseband signals having the client data; and a client-side output configured to receive the one or more second baseband signals from the receiver circuitry and transmit the one or more second baseband signals.

(10) In another aspect, the present disclosure includes a transceiver, comprising: a transmitter, comprising: a client-side input configured to receive one or more first baseband signals having first client data; transmitter circuitry configured to receive the one or more first baseband signals from the client-side input and generate one or more antenna feed signals based on the one or more first baseband signals; and one or more first antennas configured to receive the one or more antenna feed signals from the transmitter circuitry, generate one or more first radiated signals based on the one or more antenna feed signals, and couple the one or more first radiated signals into a first hollow waveguide, each of the one or more first radiated signals being radiated electromagnetic waves configured for coherent detection and having a first frequency in a range between 300 Gigahertz (GHz) and 10 Terahertz (THz); and a receiver, comprising: one or more second antennas configured to detect one or more second radiated signals received from one of the first hollow waveguide and a second hollow waveguide and generate one or more antenna output signals based on the one or more second radiated signals, each of the one or more second radiated signals being radiated electromagnetic waves configured for coherent detection, having a second frequency in a range between 300 GHz and 10 THz, and having second client data; receiver circuitry configured to receive the one or more antenna output signals from the one or more second antennas and generate one or more second baseband signals based on the one or more antenna output signals; and a client-side output configured to receive the one or more second baseband signals from the receiver circuitry and transmit the one or more second baseband signals.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate one or more implementation described herein and, together with the description, explain these implementations. The drawings are not intended to be drawn to scale, and certain features and certain views of the figures may be shown exaggerated, to scale or in schematic in the interest of clarity and conciseness. Not every component may be labeled in every drawing. Like reference numerals in the figures may represent and refer to the same or similar element or function. In the drawings:

(2) FIG. **1** is a diagrammatic view of an electromagnetic (EM) spectrum;

(3) FIG. **2** is a block diagram of an exemplary implementation of a transport network constructed in accordance with the present disclosure;

(4) FIG. **3**A is a cross-sectional view of an exemplary implementation of a first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows;

(5) FIG. **3**B is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide lacks an optional dielectric layer;

(6) FIG. **3**C is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide lacks an optional support layer;

(7) FIG. **3**D is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide lacks the optional dielectric layer and the optional support layer;

(8) FIG. **3**E is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a photonic-bandgap fiber;

(9) FIG. **3**F is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide has a hollow waveguide core with an elliptical cross-section;

(10) FIG. **3**G is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the hollow waveguide core of the first hollow waveguide has a rectangular cross-section;

(11) FIG. **3**H is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the hollow waveguide core of the first hollow waveguide has a square cross-section;

(12) FIG. **3**I is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the hollow waveguide core of the first hollow waveguide has a cross-shaped cross-section;

(13) FIG. **3**J is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a solid rod fiber;

(14) FIG. **3**K is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a microstructured optical fiber;

(15) FIG. **3**L is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a porous fiber;

(16) FIG. **3**M is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a suspended porous-core fiber;

(17) FIG. **3**N is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a suspended slotted core fiber;
(18) FIG. **3**O is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a hollow-core bandgap fiber;
(19) FIG. **3**P is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a hollow-core tube fiber;
(20) FIG. **3**Q is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a hollow-core fiber with negative curvature;
(21) FIG. **3**R is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a hollow-core fiber based on anti-resonances and inhibited coupling;
(22) FIG. **3**S is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a hollow-core nested anti-resonant nodeless fiber;
(23) FIG. **3**T is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a 3D-printed hollow-core fiber based on anti-resonances and inhibited coupling;
(24) FIG. **3**U is a cross-sectional view of another exemplary implementation of the first hollow waveguide shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows, wherein the first hollow waveguide is a Bragg fiber;
(25) FIG. **4**A is a block diagram of an exemplary implementation of a first transmitter shown in FIG. **2**;
(26) FIG. **4**B is a block diagram of another exemplary implementation of the first transmitter shown in FIG. **2**, wherein the first transmitter comprises a serializer;
(27) FIG. **4**C is a block diagram of another exemplary implementation of the first transmitter shown in FIG. **2**, wherein the first transmitter comprises a deserializer;
(28) FIG. **4**D is a block diagram of an exemplary implementation of transmitter circuitry shown in FIG. **4**A;
(29) FIG. **4**E is a block diagram of another exemplary implementation of the transmitter circuitry shown in FIG. **4**A, wherein the transmitter circuitry comprises a combiner;
(30) FIG. **4**F is a block diagram of another exemplary implementation of the first transmitter shown in FIG. **2**;
(31) FIG. **4**G is a block diagram of another exemplary implementation of the first transmitter shown in FIG. **2**;
(32) FIG. **5**A is a block diagram of an exemplary implementation of a first receiver shown in FIG. **2**;
(33) FIG. **5**B is a block diagram of another exemplary implementation of the first receiver shown in FIG. **2**, wherein the first receiver comprises a deserializer;
(34) FIG. **5**C is a block diagram of another exemplary implementation of the first receiver shown in FIG. **2**, wherein the first receiver comprises a serializer;
(35) FIG. **5**D is a block diagram of an exemplary implementation of receiver circuitry shown in FIG. **5**A;
(36) FIG. **5**E is a block diagram of another exemplary implementation of the receiver circuitry shown in FIG. **5**A, wherein the receiver circuitry comprises a splitter;
(37) FIG. **5**F is a block diagram of another exemplary implementation of the first receiver shown in

FIG. **2**;
(38) FIG. **5**G is a block diagram of another exemplary implementation of the first receiver shown in FIG. **2**;
(39) FIG. **6**A is a block diagram of an exemplary implementation of a transceiver shown in FIG. **2**;
(40) FIG. **6**B is a block diagram of another exemplary implementation of the transceiver shown in FIG. **2**;
(41) FIG. **7** is a schematic diagram of a folded modulator constructed in accordance with the present disclosure;
(42) FIG. **8** is a schematic diagram of a rectifying detector constructed in accordance with the present disclosure;
(43) FIG. **9**A is a side view of an exemplary implementation of an antenna constructed in accordance with the present disclosure for generating circularly polarized signals;
(44) FIG. **9**B is a side view of another exemplary implementation of the antenna shown in FIG. **9**A;
(45) FIG. **10** is a perspective view of another exemplary implementation of the antenna shown in FIG. **9**A, wherein the antenna is a bifilar helix antenna;
(46) FIG. **11** is a perspective view of another exemplary implementation of the bifilar helix antenna shown in FIG. **10**, wherein the bifilar helix antenna is enclosed within a conductive cone;
(47) FIG. **12** is a partial cross-sectional view of the bifilar helix antenna shown in FIG. **11**, taken from the line 12-12′ and in the direction of the arrows;
(48) FIG. **13** is a diagrammatic view of an electric field produced by the bifilar helix antenna enclosed within the conductive cone shown in FIG. **12**;
(49) FIG. **14** is a diagrammatic view of a radiation pattern of the bifilar helix antenna enclosed within the conductive cone shown in FIG. **12**;
(50) FIG. **15** is a side view of an exemplary implementation of a non-uniform bifilar helix antenna constructed in accordance with the present disclosure;
(51) FIG. **16** is a side view of another exemplary implementation of the non-uniform bifilar helix antenna;
(52) FIG. **17** is a graphical view of a polarization discrimination of the non-uniform bifilar helix antenna shown in FIG. **15**;
(53) FIG. **18** is a graphical view of a polarization discrimination of the non-uniform bifilar helix antenna shown in FIG. **16**;
(54) FIG. **19** is a side view of another exemplary implementation of the non-uniform bifilar helix antenna;
(55) FIG. **20** is a side view of another exemplary implementation of the non-uniform bifilar helix antenna;
(56) FIG. **21** is a diagrammatic view of an exemplary implementation of a differential waveguide probe antenna constructed in accordance with the present disclosure;
(57) FIG. **22**A is a partial cross-sectional view of the differential waveguide probe antenna shown in FIG. **21**, taken from the line 22-22′ and in the direction of the arrows;
(58) FIG. **22**B is another partial cross-sectional view of the differential waveguide probe antenna shown in FIG. **22**A, taken from the line 23-23′ and in the direction of the arrows;
(59) FIG. **22**C is another partial cross-sectional view of the differential waveguide probe antenna shown in FIG. **22**B, taken from the line 24-24′ and in the direction of the arrows;
(60) FIG. **22**D is a graphical view of a polarization discrimination of the differential waveguide probe antenna shown in FIG. **21**;
(61) FIG. **23** a diagrammatic view of an exemplary implementation of a differential tapered antenna constructed in accordance with the present disclosure;
(62) FIG. **24**A is a partial cross-sectional view of the differential tapered antenna shown in FIG. **23**, taken from the line 27-27′ and in the direction of the arrows;
(63) FIG. **24**B is another partial cross-sectional view of the differential tapered antenna shown in

FIG. **24**A, taken from the line 28-28′ and in the direction of the arrows;
(64) FIG. **24**C is a graphical view of a polarization discrimination of the differential tapered antenna shown in FIG. **23**;
(65) FIG. **25** is a diagrammatic view of an exemplary implementation of a differential microstrip patch antenna constructed in accordance with the present disclosure;
(66) FIG. **26** is a diagrammatic view of an exemplary implementation of a single-ended waveguide probe antenna constructed in accordance with the present disclosure;
(67) FIG. **27**A is a cross-sectional view of the single-ended waveguide probe antenna shown in FIG. **26**, taken along the line 55-55′ and in the direction of the arrows;
(68) FIG. **27**B is another cross-sectional view of the single-ended waveguide probe antenna shown in FIG. **26**, taken along the line 56-56′ and in the direction of the arrows;
(69) FIG. **27**C is a partial cross-sectional view of the single-ended waveguide probe antenna shown in FIG. **27**B, taken along the line 57-57′ and in the direction of the arrows;
(70) FIG. **28** is a diagrammatic view of an exemplary implementation of a slot antenna constructed in accordance with the present disclosure;
(71) FIG. **29**A is a cross-sectional view of the slot antenna shown in FIG. **28**, taken along the line 59-59′ and in the direction of the arrows;
(72) FIG. **29**B is a partial cross-sectional view of the slot antenna shown in FIG. **29**A, taken along the line 60-60′ and in the direction of the arrows;
(73) FIG. **29**C is another partial cross-sectional view of the slot antenna shown in FIG. **29**A, taken along the line 61-61′ and in the direction of the arrows;
(74) FIG. **30**A is a cross-sectional view of another implementation of the slot antenna shown in FIG. **28**, taken along the line 59-59′ and in the direction of the arrows, wherein the slot antenna is a double slot antenna;
(75) FIG. **30**B is a partial cross-sectional view of the slot antenna shown in FIG. **30**A, taken along the line 63-63′ and in the direction of the arrows;
(76) FIG. **30**C is another partial cross-sectional view of the slot antenna shown in FIG. **30**A, taken along the line 64-64′ and in the direction of the arrows;
(77) FIG. **31** is a diagrammatic view of another exemplary implementation of a transport network constructed in accordance with the present disclosure;
(78) FIG. **32**A is a diagrammatic view of an exemplary implementation of a transmitter shown in FIG. **31**;
(79) FIG. **32**B is a diagrammatic view of an exemplary implementation of a receiver shown in FIG. **31**;
(80) FIG. **33** is a diagrammatic view of an exemplary implementation of an antenna array shown in FIG. **31**;
(81) FIG. **34** is a diagrammatic view of another exemplary implementation of the transport network constructed in accordance with the present disclosure;
(82) FIG. **35**A is a diagrammatic view of an exemplary implementation of the antenna array shown in FIG. **34**, wherein a first antenna, a second antenna, a third antenna, and a fourth antenna are arranged in an n×m grid pattern;
(83) FIG. **35**B is a diagrammatic view of another exemplary implementation of the antenna array shown in FIG. **34**, wherein the first antenna, the second antenna, the third antenna, and the fourth antenna are arranged in a 1×m grid pattern;
(84) FIG. **36** is a diagrammatic view of a method of using the transport network shown in FIG. **31**;
(85) FIG. **37**A is a diagrammatic view of an exemplary implementation of a dual-polarization (dual-pol) network element constructed in accordance with the present disclosure;
(86) FIG. **37**B is a diagrammatic view of another exemplary implementation of a dual-pol network element constructed in accordance with the present disclosure;
(87) FIG. **38** is a diagrammatic view of a dual-pol signal in accordance with the present disclosure;

(88) FIG. **39** is a diagrammatic view of an exemplary implementation of a dual-pol transport network constructed in accordance with the present disclosure, wherein the dual-pol transport network comprises a dual-pol RF antenna;
(89) FIG. **40** is a diagrammatic view of another exemplary implementation of a dual-pol transport network constructed in accordance with the present disclosure, wherein the dual-pol transport network comprises a first pair of RF antennas and a second pair of RF antennas;
(90) FIG. **41** is a diagrammatic view of another exemplary implementation of a dual-pol transport network constructed in accordance with the present disclosure, wherein the dual-pol transport network comprises a plurality of first RF antennas and a plurality of second RF antennas;
(91) FIG. **42** is a diagrammatic view of an exemplary implementation of a first modulator shown in FIG. **37**A;
(92) FIG. **43** is a diagrammatic view of an exemplary implementation of a first demodulator shown in FIG. **37**B;
(93) FIG. **44** is a diagrammatic view of another exemplary implementation of a dual-pol network element constructed in accordance with the present disclosure, wherein the dual-pol network element comprises an equalizer;
(94) FIG. **45** is a diagrammatic view of a method of use in accordance with the present disclosure;
(95) FIG. **46**A is a diagrammatic view of another exemplary implementation of a network element constructed in accordance with the present disclosure, wherein the network element is configured to perform direct conversion from a first modulation format in a first electrical signal to a second modulation format in a second electrical signal in the THz frequency band;
(96) FIG. **46**B is a diagrammatic view of another exemplary implementation of a network element constructed in accordance with the present disclosure, wherein the network element is configured to perform direct conversion from the first modulation format to the second modulation format in the THz frequency band and includes an RF antenna;
(97) FIG. **47**A is a diagrammatic view of an exemplary implementation of a demodulator shown in FIG. **46**A;
(98) FIG. **47**B is a diagrammatic view of another exemplary implementation of a demodulator constructed in accordance with the present disclosure, wherein the demodulator includes a clock-and-data-recovery circuit (CDR);
(99) FIG. **48** is a diagrammatic view of an exemplary implementation of a first phase demodulator shown in FIG. **47**A;
(100) FIG. **49** is a diagrammatic view of an exemplary implementation of a first amplitude demodulator shown in FIG. **47**A;
(101) FIG. **50**A is a diagrammatic view of an exemplary implementation of a modulator shown in FIGS. **46**A and **46**B;
(102) FIG. **50**B is a diagrammatic view of another exemplary implementation of the modulator shown in FIGS. **46**A and **46**B, wherein the modulator includes a local oscillator (LO) generator;
(103) FIG. **51** is a diagrammatic view of an exemplary implementation of a method for performing direct modulation from the first modulation format to the second modulation format in in an electrical signal in the THz frequency band;
(104) FIG. **52** is a diagrammatic view of an exemplary implementation of a first phase modulator shown in FIGS. **50**A and **50**B, wherein the first phase modulator comprises a crossbar switch;
(105) FIG. **53**A is a diagrammatic view of an exemplary implementations of a first amplitude modulator shown in FIGS. **50**A and **50**B, wherein the first amplitude modulator comprises a PI-type switched attenuator;
(106) FIG. **53**B is a diagrammatic view of another exemplary implementations of the first amplitude modulator shown in FIGS. **50**A and **50**B, wherein the first amplitude modulator comprises a T-type switched attenuator;
(107) FIG. **53**C is a diagrammatic view of another exemplary implementations of the first

amplitude modulator shown in FIGS. **50**A and **50**B, wherein the first amplitude modulator comprises a bridged T-type switched attenuator;
(108) FIG. **54** is a diagrammatic view of another exemplary implementation of a transceiver constructed in accordance with the present disclosure
(109) FIG. **55** is a diagrammatic view of another exemplary implementation of a transceiver constructed in accordance with the present disclosure;
(110) FIG. **56** is a diagrammatic view of another exemplary implementation of a transceiver constructed in accordance with the present disclosure;
(111) FIG. **57** is a diagrammatic view of another exemplary implementation of a transmitter constructed in accordance with the present disclosure.
(112) FIG. **58** is a diagrammatic view of another exemplary implementation of a receiver constructed in accordance with the present disclosure.
(113) FIG. **59** is a diagrammatic view of another exemplary implementation of a transmitter constructed in accordance with the present disclosure.
(114) FIG. **60** is a diagrammatic view of another exemplary implementation of a transmitter constructed in accordance with the present disclosure.
(115) FIG. **61** is a diagrammatic view of an exemplary implementations of a differential circuit constructed in accordance with the present disclosure;
(116) FIG. **62** is a diagrammatic view of another exemplary implementation of the differential circuit shown in FIG. **61**;
(117) FIG. **63** is a diagrammatic view of another exemplary implementation of the differential circuit shown in FIG. **61**;
(118) FIG. **64** is a diagrammatic view of another exemplary implementation of an antenna array constructed in accordance with the present disclosure;
(119) FIG. **65** is a perspective view of an exemplary implementation of an electromagnetic absorber used and constructed in accordance with the present disclosure;
(120) FIG. **66** is a cross-section view of another exemplary implementation of an electromagnetic absorber constructed in accordance with the present disclosure;
(121) FIG. **67** is a cross-section view of another exemplary implementation of an electromagnetic absorber constructed in accordance with the present disclosure;
(122) FIG. **68** is a cross-section view of another exemplary implementation of an electromagnetic absorber constructed in accordance with the present disclosure;
(123) FIG. **69** is a diagrammatic view of another exemplary implementation of an electromagnetic absorber constructed in accordance with the present disclosure;
(124) FIG. **70** is a flow diagram of an exemplary implementation of a process in accordance with the present disclosure;
(125) FIG. **71** is a flow diagram of another exemplary implementation of a process in accordance with the present disclosure; and
(126) FIG. **72** is a process flow diagram of an exemplary implementation of a construction process constructed in accordance with the present disclosure.

DETAILED DESCRIPTION

(127) The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.
(128) As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by anyone of the following: A is true (or present) and B is false (or not present), A is false (or not

present) and B is true (or present), and both A and B are true (or present).
(129) In addition, use of the “a” or “an” are employed to describe elements and components of the implementations herein. This is done merely for convenience and to give a general sense of the inventive concept. This description should be read to include one or more and the singular also includes the plural unless it is obvious that it is meant otherwise. The term “implementation” as used herein is synonymous with the term “embodiment”.
(130) Further, use of the term “plurality” is meant to convey “more than one” unless expressly stated to the contrary.
(131) As used herein, qualifiers like “substantially,” “about,” “approximately,” and combinations and variations thereof, are intended to include not only the exact amount or value that they qualify, but also some slight deviations therefrom, which may be due to manufacturing tolerances, measurement error, wear and tear, stresses exerted on various parts, and combinations thereof, for example.
(132) The use of the term “at least one” or “one or more” will be understood to include one as well as any quantity more than one. In addition, the use of the phrase “at least one of X, V, and Z” will be understood to include X alone, V alone, and Z alone, as well as any combination of X, V, and Z.
(133) The use of ordinal number terminology (i.e., “first”, “second”, “third”, “fourth”, etc.) is solely for the purpose of differentiating between two or more items and, unless explicitly stated otherwise, is not meant to imply any sequence or order or importance to one item over another or any order of addition.
(134) Finally, as used herein any reference to “one implementation” or “an implementation” means that a particular element, feature, structure, or characteristic described in connection with the implementation is included in at least one implementation. The appearances of the phrase “in one implementation” in various places in the specification are not necessarily all referring to the same implementation.
(135) As used herein, all numerical values or ranges include fractions of the values and integers within such ranges and fractions of the integers within such ranges unless the context clearly indicates otherwise. Thus, to illustrate, reference to a numerical range, such as 1-10 includes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, as well as 1.1, 1.2, 1.3, 1.4, 1.5, etc., and so forth. Reference to a range of 1-50 therefore includes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, etc., up to and including 50, as well as 1.1, 1.2, 1.3, 1.4, 1.5, etc., 2.1, 2.2, 2.3, 2.4, 2.5, etc., and so forth. Reference to a series of ranges includes ranges which combine the values of the boundaries of different ranges within the series. Thus, to illustrate reference to a series of ranges, for example, of 1-10, 10-20, 20-30, 30-40, 40-50, 50-60, 60-75, 75-100, 100-150, 150-200, 200-250, 250-300, 300-400, 400-500, 500-750, 750-1,000, includes ranges of 1-20, 10-50, 50-100, 100-500, and 500-1,000, for example.
(136) As used herein, “circuitry” may refer to analog and/or digital components, or one or more suitably programmed processor (e.g., a microprocessor) and associated hardware and software, or hardwired logic. Also, “components” may perform one or more function. The term “component” may include hardware, such as a processor (e.g., microprocessor), a combination of hardware and software, and/or the like. Software may include one or more processor-executable instruction that when executed by one or more processor cause the one or more processor to perform a specified function. It should be understood that the algorithms described herein may be stored on one or more non-transitory memory. Exemplary non-transitory memory may include random access memory, read only memory, flash memory, and/or the like. Such non-transitory memory may be electrically based, optically based, and/or the like.
(137) As used herein, a “mode” refers to a unique distribution of electric and magnetic fields which repeat along the length of a hollow waveguide by which electromagnetic energy may be transported through the hollow waveguide. “Single-mode” refers to a hollow waveguide designed to carry only one mode of electromagnetic wave. This is achieved by having a narrow core

diameter, which allows only one mode of light to propagate at a time. On the other hand, “multi-mode” refers to a hollow waveguide designed to carry multiple modes of electromagnetic waves simultaneously. This is possible due to its larger core diameter, which enables multiple modes to be propagated.

(138) As used herein, “Amplitude Modulation” (AM) refers to a form of signal modulation in which data is encoded in an amplitude of a carrier signal.

(139) As used herein, “Amplitude-Shift Keying” (ASK) refers to a form of AM in which digital data is encoded in an amplitude of a carrier signal, and each symbol (i.e., representing one or more data bit) is sent by transmitting a fixed-amplitude carrier wave at a fixed frequency for a specific time period.

(140) As used herein, “Phase-Shift Keying” (PSK) is a form of signal modulation in which signal data is encoded in a phase of a carrier signal having a constant frequency. “Quadrature PSK” (PSK) Is a form of PSK in which two data bits (i.e., 00, 01, 10, or 11) are modulated at once, selecting one of four possible carrier phase shifts (i.e., 0°, 90°, 180°, or 270°).

(141) As used herein, “Pulse-Amplitude Modulation” (PAM) refers to a form of AM in which a data signal is encoded in an amplitude of a series of carrier signal pulses. “PAM4” refers to a form of PAM in which a data signal is encoded in an amplitude of a series of carrier signal pulses, in which the amplitude of the carrier signal pulses may be one of four discrete values (i.e., 0, 1, 2, or 3) and each carrier signal pulse represents two data bits (i.e., 00, 01, 10, or 11).

(142) As used herein, “Non-Return-to-Zero” (NRZ) refers to a form of signal modulation in which a binary data signal is encoded in a carrier signal such that ones are represented by a first significant condition (e.g., a positive voltage) and zeroes are represented by a second significant condition (e.g., a negative voltage). “Non-return-to-Zero, Inverted” (NRZI) refers to a form of signal modulation in which the data bits are represented by the presence or absence of a transition at a clock boundary.

(143) As used herein, “Quadrature Amplitude Modulation” (QAM) refers to a form of AM in which two analog message signals or two digital bit streams are encoded in amplitudes of two carrier waves, using either ASK or AM, and the two carrier signals are out of phase with each other by 90°. “QAM16” refers to a form of QAM in which the carrier signals may exist in one of sixteen discrete states (i.e., symbols) having one of sixteen different amplitude and phase levels representing four data bits (i.e., from 0000 to 1111).

(144) As used herein, “Trellis Coded Modulation” (TCM) refers to a form of signal modulation in which a binary data signal is encoded in a phase of a constant amplitude carrier signal. The transmitted signal is created by convolutionally encoding the binary data signal and mapping the result to a signal constellation.

(145) As used herein, “Rayleigh range” refers to the distance along the propagation direction of a beam from the waist to the place where the area of the cross section is doubled.

(146) As used herein, “hollow waveguide” refers to a structure that guides waves by restricting transmission of energy in a particular direction. In the context of the present disclosure, “hollow waveguide” may refer to an optical fiber having a waveguide core operable to propagate RF signals in the THz frequency band or a routed waveguide operable to propagate RF signals in the THz frequency band.

(147) As used herein, “diameter” refers to a straight line passing from side to side through the center of a body or figure. In some implementations, the body or figure has a circular or elliptical shape.

(148) As used herein, “data” refers to quantities, characters, or symbols on which operations are performed by a computer. Data can be recorded on a non-transitory computer readable medium, such as random-access memory and/or read only memory. The random-access memory and/or read only memory may be implemented on semiconductor, magnetic, optical, or mechanical recording media. An example of data is client data, e.g., data provided by a client in connection with a

telecommunication service and/or a storage service.
(149) Referring now to the drawings, and in particular to FIG. **1**, shown therein is a diagrammatic view of an electromagnetic (EM) spectrum **100** in accordance with the present disclosure. The present disclosure is generally related to network elements that communicate using radiated signals comprising radiated electromagnetic waves coupled into hollow waveguides. The radiated signals described herein generally have a transmission frequency in what is referred to as a Terahertz (THz) frequency band **104** (i.e., frequencies between 0.1 THz and 10 THz corresponding to wavelengths between 3 millimeters (mm) and 30 micrometers (μm)). However, in some implementations described herein, the transmission frequency of the radiated signals is in a range between 300 Gigahertz (GHz) and 10 THz. The radiated signals described herein are generally configured for coherent detection and generally have a bandwidth in a range between 10% and 40% of the transmission frequency.
(150) Referring now to FIG. **2**, shown therein is a block diagram of an exemplary implementation of a transport network **200** (hereinafter, the "transport network **200**") constructed in accordance with the present disclosure. The transport network **200** is depicted as comprising a plurality of network elements **204***a-n* (hereinafter, the "network elements **204**") (e.g., a first network element **204***a*, a second network element **204***b*, a third network element **204***c*, and a fourth network element **204***d* shown in FIG. **2**). While only four of the network elements **204** are shown in FIG. **2** for exemplary purposes, it should be understood that the transport network **200** may comprise a number of the network elements **204** that may be greater or fewer than four.
(151) The transport network **200** may further comprise one or more hollow waveguides **208***a-n* (hereinafter, the "hollow waveguides **208**") (e.g., a first hollow waveguide **208***a*, a second hollow waveguide **208***b*, a third hollow waveguide **208***c*, and a fourth hollow waveguide **208***d* shown in FIG. **2**). While only four of the hollow waveguides **208** are shown in FIG. **2** for exemplary purposes, it should be understood that the transport network **200** may comprise a number of the hollow waveguides **208** that may be greater or fewer than four.
(152) Radiated signals transmitted within the transport network **200** from the first network element **204***a* to the fourth network element **204***d* or vice versa may travel along (1) a first path formed by the first hollow waveguide **208***a*, the second network element **204***b*, and the second hollow waveguide **208***b* or (2) a second path formed by the third hollow waveguide **208***c*, the third network element **204***c*, and the fourth hollow waveguide **208***d*.
(153) In some implementations, each of the hollow waveguides **208** is configured to support propagation of radiated signals in only a single direction. However, in other implementations, one or more of the hollow waveguides **208** may be configured to support propagation of radiated signals in a plurality of directions (i.e., two opposing directions). In implementations where one or more of the hollow waveguides **208** are configured to support propagation of radiated signals in a plurality of directions, a first radiated signal being propagated through the hollow waveguide **208** in a first direction may be differentiated from a second radiated signal being propagated through the hollow waveguide **208** in a second direction opposite the first direction by being provided with a different polarization, frequency, etc. In some such implementations, one or more circulator may be included to achieve such differentiation.
(154) Each of the network elements **204** may comprise one or more of a transmitter **212** (e.g., a first transmitter **212***a* and a second transmitter **212***b* shown in FIG. **2**) operable to transmit radiated signals comprising radiated electromagnetic waves having client data encoded therein via the hollow waveguides **208**, a receiver **216** (e.g., a first receiver **216***a* and a second receiver **216***b* shown in FIG. **2**) operable to receive radiated signals comprising radiated electromagnetic waves having client data encoded therein via the hollow waveguides **208**, and/or a transceiver **220** (e.g., a first transceiver **220***a* shown in FIG. **2** and a second transceiver **220***b* shown in FIG. **6**B) operable to transmit first radiated signals comprising first radiated electromagnetic waves having first client data encoded therein via particular ones of the hollow waveguides **208** and/or receive second

radiated signals comprising second radiated electromagnetic waves having second client data encoded therein via other ones of the hollow waveguides **208**.
(155) Each of the network elements **204** may further comprise a control module **224** (e.g., a first control module **224***a*, a second control module **224***b*, a third control module **224***c*, and a fourth control module **224***d* shown in FIG. **2**) (collectively, the "control modules **224**") operable to regulate one or more operating parameter of the network element **204** to which the control module **224** is coupled.
(156) In some implementations, one or more of the network elements **204** may communicate with each other via a communication network **228**. The communication network **228** may permit bidirectional communication of information and/or data between one or more of the network elements **204** of the transport network **200**. The communication network **228** may interface with one or more of the network elements **204** in a variety of ways. For example, in some implementations, the communication network **228** may interface by optical and/or electronic interfaces, and/or may use a plurality of network topographies and/or protocols including, but not limited to, Ethernet, TCP/IP, circuit switched path, combinations thereof, and/or the like. The communication network **228** may utilize a variety of network protocols to permit bidirectional interface and/or communication of data and/or information between one or more of the network elements **204**.
(157) The communication network **228** may be almost any type of network. For example, in some implementations, the communication network **228** may be a version of an Internet network (e.g., exist in a TCP/IP-based network). In one implementation, the communication network **228** is the Internet. It should be noted, however, that the communication network **228** may be almost any type of network and may be implemented as the World Wide Web (i.e., the Internet), a local area network (LAN), a wide area network (WAN), a metropolitan network, a wireless network, a cellular network, a Bluetooth network, a Global System for Mobile Communications (GSM) network, a code division multiple access (CDMA) network, a 3G network, a 4G network, an LTE network, a 5G network, a satellite network, a radio network, an optical network, a cable network, a public switched telephone network, an Ethernet network, combinations thereof, and/or the like.
(158) If the communication network **228** is the Internet, a primary user interface of the transport network **200** may be delivered through a series of web pages or private internal web pages of a company or corporation, which may be written in hypertext markup language, JavaScript, or the like, and accessible by the user. It should be noted that the primary user interface of the transport network **200** may be another type of interface including, but not limited to, a Windows-based application, a tablet-based application, a mobile web interface, a VR-based application, an application running on a mobile device, and/or the like. In one implementation, the communication network **228** may be connected to one or more of the network elements **204**.
(159) The number of devices and/or networks illustrated in FIG. **2** is provided for exemplary purposes. In practice, there may be additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than are shown in FIG. **2**. Furthermore, two or more of the devices illustrated in FIG. **2** may be implemented within a single device, or a single device illustrated in FIG. **2** may be implemented as multiple, distributed devices. Additionally, or alternatively, one or more of the devices of the transport network **200** may perform one or more functions described as being performed by another one or more of the devices of the transport network **200**.
(160) The network elements **204** may take many different forms. For example, the network elements **204** may be integrated circuits (ICs). In this example, the network elements **204** (e.g., ICs) may communicate via signals comprising radiated electromagnetic waves having client data encoded therein via the hollow waveguides **208** without requiring electrical data busses. In other implementations, the network elements **204** may be incorporated into components in a data center, such as servers, routers, switches, firewalls, storage systems, application delivery controllers,

and/or the like to establish communication between such components in the data center via signals comprising radiated electromagnetic waves having client data encoded therein propagated through the hollow waveguides **208**. The hollow waveguides **208** may thus extend from one integrated circuit to another integrated circuit, or from one component to another component, and such may be implemented in a variety of ways, such as IC-to-IC communications, printed circuit board (PCB)-to-PCB communications, component-to-component communications, and/or combinations thereof. In the example of PCB-to-PCB communications, the network elements **204** may each include a PCB.

(161) Referring now to FIGS. **3**A-**3**H and **4**A-**4**L, shown therein are cross-sectional views of various exemplary implementations of the first hollow waveguide **208***a* shown in FIG. **2**, taken along the line 3-3′ and in the direction of the arrows. However, it should be understood that the description referring to FIGS. **3**A-**3**H and **4**A-**4**L may be applicable to any of the hollow waveguides **208** described herein. In the implementations shown in FIGS. **3**A-**3**H and **4**A-**4**L, the first hollow waveguide **208***a* is a hollow fiber. However, it should be understood that in other implementations, the first hollow waveguide **208***a* may be another form of hollow waveguide, such as a substrate-integrated waveguide, for example.

(162) The first hollow waveguide **208***a* (and, therefore, each of the hollow waveguides **208**) generally comprises a hollow waveguide core **304** and a tubular sidewall **306** having an inner surface **312** in some implementations defining the hollow waveguide core **304** or in other implementations simply surrounding the hollow waveguide core **304**.

(163) Generally, the hollow waveguide core **304** may be composed of any material capable of propagating radiated electromagnetic waves within the THz frequency band **104** or, in some implementations, in the range between 300 GHz and 10 THz. More particularly, the hollow waveguide core **304** may be composed of any materials having a low absorption loss (i.e., an absorption loss in a range between 1 dB/km and 10,000 dB/km) within the THz frequency band **104**, or in some implementations, in the range between 300 GHz and 10 THz.

(164) In some implementations, the hollow waveguide core **304** may be composed of a polymer (e.g., cyclo olefin polymer (COP), cyclic olefin co-polymer (COC), polytetrafluoroethylene (PTFE), high-density polyethylene (HDPE), polymethylpentene (PMP), polypropylene (PP), polystyrene, polycarbonate, poly(methyl methacrylate) (PMMA), Picarin, or ultraviolet (UV) resin) or glass (e.g., silica glass, crown glass, or borosilicate glass).

(165) In other implementations, the hollow waveguide core **304** may be composed of a gas, a vacuum, or a porous material (i.e., a material having a porosity in a range between 25% and 99%). In such implementations, the hollow waveguide core **304** may have a refractive index in a range between 1.0 and 1.4, for example. As discussed in more detail below, the hollow waveguide core **304** may have a refractive index n.sub.1.

(166) In some implementations, the hollow waveguide core **304** may have a cross-section configured to support propagation of radiated signals having only a single polarization at a given time. However, in other implementations, the hollow waveguide core **304** may have a cross-section configured to support propagation of radiated signals having a plurality of polarizations at a given time. In either case, the hollow waveguide core **304** may have a cross-section configured to support propagation of radiated signals having one or more linear polarizations or one or more circular polarizations.

(167) In some implementations, the hollow waveguide core **304** may have a cross-section configured to support propagation of radiated signals having only a single mode at a given time. However, in other implementations, the hollow waveguide core **304** may have a cross-section configured to support propagation of radiated signals having a plurality of modes at a given time.

(168) The tubular sidewall **306** of the first hollow waveguide **208***a* (and, therefore, each of the hollow waveguides **208**) may comprise a conductive layer **316** (shown in FIGS. **3**A-**3**I) surrounding the hollow waveguide core **304**, a dielectric layer **308** (shown in FIGS. **3**A, **3**C, and

3F-**3**I) optionally disposed between the hollow waveguide core **304** and the conductive layer **316**, and a support layer **320** (shown in FIGS. **3**A, **3**B, and **3**E-**3**I) optionally surrounding the conductive layer **316**.
(169) In some implementations, the tubular sidewall **306** of the first hollow waveguide **208***a* (and, therefore, each of the hollow waveguides **208**) may comprise a plurality of the conductive layer **316** interleaved with a plurality of the dielectric layer **308**.
(170) In some implementations, the tubular sidewall **306** of the first hollow waveguide **208***a* (and, therefore, each of the hollow waveguides **208**) may further comprise one or more strength members (not shown) (hereinafter, the "strength members") surrounding the conductive layer **316** configured to enhance resilience of the first hollow waveguide **208***a*. In such implementations, the support layer **320** may surround the strength members.
(171) Generally, the conductive layer **316** may be composed of any material having a refractive index n.sub.3 greater than the refractive index of the hollow waveguide core **304** (i.e., n.sub.1). More particularly, the conductive layer **316** may be composed of a non-oxidizing metallic material (e.g., silver, gold, or indium tin oxide (ITO)). Providing the conductive layer **316** with a refractive index greater than the refractive index of the hollow waveguide core **304** may cause an effective index Δn of the first hollow waveguide **208***a* to increase, thereby causing more radiated signals to be confined and propagated within the hollow waveguide core **304**.
(172) Generally, in implementations in which the dielectric layer **308** is disposed between the conductive layer **316** and the hollow waveguide core **304**, the dielectric layer **308** may be composed of any material having a refractive index n.sub.2 greater than the refractive index of the hollow waveguide core **304** (i.e., n.sub.1). More particularly, the dielectric layer **308** may be composed of a polymer (e.g., cyclo olefin polymer (COP), cyclic olefin co-polymer (COC), polytetrafluoroethylene (PTFE), high-density polyethylene (HDPE), polymethylpentene (PMP), polypropylene (PP), polystyrene, polycarbonate, poly(methyl methacrylate) (PMMA), Picarin, or ultraviolet (UV) resin) or glass (e.g., silica glass, crown glass, or borosilicate glass), but particularly a material having a refractive index n.sub.2 greater than the refractive index of the hollow waveguide core **304** (i.e., n.sub.1) in that implementation. Providing the dielectric layer **308** with a refractive index greater than the refractive index of the hollow waveguide core **304** may cause an effective index Δn of the first hollow waveguide **208***a* to increase, thereby causing more radiated signals to be confined and propagated within the hollow waveguide core **304**.
(173) The support layer **320** may be configured to shield the inner layers of the first hollow waveguide **208***a* (and, therefore, any of the hollow waveguides **208**) from external environmental factors, provide flexibility to the first hollow waveguide **208***a*, and/or enhance a tensile strength of the first hollow waveguide **208***a*. In some implementations, the support layer **320** may be composed of polymer materials, such as acrylate polymer or polyimide, for example.
(174) In some implementations, the cross-section of the hollow waveguide core **304** may have a circular shape (i.e., having a diameter d.sub.1 that is equal along both the x-axis and the y-axis) (shown in FIGS. **3**A-**3**D). In some such implementations, the diameter d.sub.1 of the hollow waveguide core **304** may be between 30 μm and 6 mm. In some such implementations, the diameter d.sub.1 of the hollow waveguide core **304** may be between 30 μm and 3 mm. In at least one such implementation, the diameter d.sub.1 of the hollow waveguide core **304** may be 1 mm.
(175) In some implementations, as shown in FIG. **3**E, the first hollow waveguide **208***a* may be a photonic-bandgap fiber comprising a plurality of air channels **324** (hereinafter the "air channels **324**") periodically spaced throughout the conductive layer **316**.
(176) In other implementations, the cross-section of the hollow waveguide core **304** may have an elliptical shape (i.e., having a first diameter x.sub.1 along the x-axis and a second diameter y.sub.1 along the y-axis, wherein the first diameter is not equal to the second diameter) (shown in FIG. **3**F), a rectangular shape (shown in FIG. **3**G) (i.e., having a first length x.sub.1 along the x-axis and a second length y.sub.1 along the y-axis, wherein the first length is not equal to the second length), a

square shape (i.e., having a length **11** that is equal along both the x-axis and the y-axis) (shown in FIG. **3**H), or a cross shape (i.e., having a length **11** that is equal along both the x-axis and the y-axis) (shown in FIG. **3**I), for example.
(177) In other implementations, the first hollow waveguide **208***a* (and, therefore, any of the hollow waveguides **208**) may be implemented as a solid rod fiber (shown in FIG. **3**J), a microstructured optical fiber (shown in FIG. **3**K), a porous fiber (shown in FIG. **3**L), a suspended porous-core fiber (shown in FIG. **3**M), a suspended slotted core fiber (shown in FIG. **3**N), a hollow-core bandgap fiber (shown in FIG. **3**O), a hollow-core tube fiber (shown in FIG. **3**P), a hollow-core fiber with negative curvature (shown in FIG. **3**Q), a hollow-core fiber based on anti-resonances and inhibited coupling (shown in FIG. **3**R), a hollow-core nested anti-resonant nodeless fiber (shown in FIG. **3**S), a 3D-printed hollow-core fiber based on anti-resonances and inhibited coupling (shown in FIG. **3**T), or a Bragg fiber (shown in FIG. **3**U), for example.
(178) Referring now to FIG. **4**A, shown therein is a block diagram of an exemplary implementation of the first transmitter **212***a* shown in FIG. **2**. However, it should be understood that the description of any particular one of the transmitter **212** may be applicable to any of the transmitters **212** described herein. The first transmitter **212***a* (and, therefore, each of the transmitters **212**) generally comprises a client-side input **400** configured to receive one or more baseband signals **404** (hereinafter, the "baseband signals **404**") having client data encoded therein from one or more external component (e.g., a control module **224**), transmitter circuitry **408** configured to receive the baseband signals **404** from the client-side input **400** and generate one or more antenna feed signals **412** (hereinafter, the "antenna feed signals **412**") based on the baseband signals **404**, and one or more first antennas **416** configured to receive the antenna feed signals **412** from the transmitter circuitry **408**, generate one or more radiated signals **420** (hereinafter, the "radiated signals **420**") based on the antenna feed signals **412**, and couple the radiated signals **420** into the first hollow waveguide **208***a*.
(179) In some implementations, the client-side input **400** is a pair of inputs configured to receive a differential signal. In some such implementations, the client-side input **400** may be a low voltage differential signaling (LVDS) link configured to receive LVDS signals, and the baseband signals **404** may be LVDS signals indicative of client data.
(180) In some implementations, the antenna feed signals **412** are provided to the first antennas **416** on one or more transmission lines (not shown) (hereinafter, the "transmission lines"), wherein each of the transmission lines has two or more conductors (not shown) (hereinafter, the "conductors"). In some implementations, the transmission lines have a first transmission loss and the first hollow waveguide **208***a* has a second transmission loss that is less than the first transmission loss. In some implementations, the second transmission loss is in a range between 0.001 and 20.00 decibels (dB) per meter (m) per Terabit (Tb) per second (s).
(181) In some implementations, as shown in FIG. **4**A, each of the client-side input **400**, the transmitter circuitry **408**, and the first antennas **416** may be disposed on a substrate **424**. However, in other implementations, one or more of the client-side input **400**, the transmitter circuitry **408**, and the first antennas **416** may be disposed on a first substrate (not shown), and one or more of the client-side input **400**, the transmitter circuitry **408**, and the first antennas **416** may not be disposed on the first substrate. For example, the one or more of the client-side input **400**, the transmitter circuitry **408**, and the first antennas **416** may be disposed on a second substrate (not shown). In such implementations, the first substrate and the second substrate may be in a stacked arrangement.
(182) In some implementations, the substrate **424** may have a plurality of layers (not shown). In such implementations, one or more of the client-side input **400**, the transmitter circuitry **408**, and the first antennas **416** may be disposed on a first layer (not shown), and one or more of the client-side input **400**, the transmitter circuitry **408**, and the first antennas **416** may be disposed on a second layer (not shown).
(183) In some implementations, one or more of the client-side input **400**, the transmitter circuitry

408, and the first antennas **416** may be integrated into a monolithic semiconductor die (not shown). In some implementations, one or more of the client-side input **400**, the transmitter circuitry **408**, and the first antennas **416** may implemented using one or more of complementary metal-oxide semiconductor (CMOS) technology, silicon-germanium (SiGe) semiconductor technology, and Ill-V compound semiconductor technology.

(184) In some implementations, the baseband signals **404** are digital bitstreams. In some implementations, the client data may be encoded in the baseband signals **404** using an encoding protocol conforming to requirements of one or more of return-to-zero (RZ) code, non-return-to-zero (NRZ) code, pulse-amplitude modulation (PAM), and quadrature-amplitude modulation (QAM). In some implementations, the client data may be encoded in the radiated signals **420** using an encoding protocol conforming to requirements of one or more of RZ, NRZ, quadrature phase-shift keying (QPSK), QAM, trellis coded modulation (TCM), and Bose-Chaudhuri-Hocquenghem (BCH) code.

(185) In some implementations, the radiated signals **420** include a first complementary radiated signal (not shown) having a first polarization and a second complementary radiated signal (not shown) having a second polarization different from the first polarization. In such implementations, the first antennas **416** may be configured to generate the radiated signals **420** including the first complementary radiated signal and the second complementary radiated signal based on the antenna feed signals **412**. The first polarization and the second polarization may be orthogonal to each other.

(186) In some implementations, each of the first polarization and the second polarization may be a linear polarization. In such implementations, the first antennas **416** may include one or more of a differential waveguide probe antenna, a differential tapered antenna, and a differential patch antenna. In other implementations, each of the first polarization and the second polarization may be a circular polarization. In such implementations, the first antennas **416** may include one or more of a helix antenna and a spiral antenna. It should be understood that any of the signals described herein may be single-ended signals or differential signals.

(187) In some implementations, the radiated signals **420** include a first complementary radiated signal (not shown) having a first polarization and a second complementary radiated signal (not shown) having a second polarization different from the first polarization, and the first antennas **416** are further configured to couple the first complementary radiated signal and the second complementary radiated signal into the first hollow waveguide **208***a* such that the first complementary radiated signal and the second complementary radiated signal interact in the first hollow waveguide **208***a* to form the combined radiated signal (not shown) having a third polarization different from the first polarization and the second polarization. In such implementations, the first antennas **416** may include an antenna array.

(188) Referring now to FIG. **4**B, in some implementations, the first transmitter **212***a* (and, therefore, any of the transmitters **212**) further comprises a first serializer **426** configured to receive a plurality of parallel baseband signals **428***a-n* (hereinafter, the "parallel baseband signals **428**") and combine the parallel baseband signals **428** into a serial baseband signal (i.e., the baseband signals **404**). In such implementations, the client-side input **400** may be configured to receive the baseband signals **404** from the first serializer **426**. In some such implementations, combining the parallel baseband signals **428** into the baseband signals **404** utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM), and wavelength division multiplexing (WDM).

(189) Referring now to FIG. **4**C, in some implementations, the first transmitter **212***a* (and, therefore, any of the transmitters **212**) further comprises a first deserializer **432** configured to receive a serial baseband signal (i.e., the baseband signals **404**) and split the baseband signals **404** into parallel baseband signals **428**. In such implementations, the client-side input **400** may be configured to receive the parallel baseband signals **428** from the first deserializer **432**. In some such

implementations, splitting the baseband signals **404** into the parallel baseband signals **428** utilizes at least one of PDM, TDM, and WDM.

(190) Referring now to FIG. **4**D, shown therein is an exemplary implementation of the transmitter circuitry **408** shown in FIGS. **4**A-**4**C. In some implementations, the transmitter circuitry **408** comprises one or more local oscillators **436***a-n* (hereinafter, the "LO **436**") configured to generate one or more carrier signals **440** (hereinafter, the "carrier signals **440**") having a baseband frequency less than the transmission frequency, one or more modulation circuits **444** (hereinafter, the "modulator **444**") configured to receive the baseband signals **404** from the client-side input **400** and the carrier signals **440** from the LO **436** and modulate the baseband signals **404** onto the carrier signals **440** to generate one or more modulated signals **448** (hereinafter, the "modulated signals **448**"), and one or more up-conversion circuits **452** (hereinafter, the "up-convertor **452**") configured to receive the modulated signals **448** from the modulator **444** and up-convert the modulated signals **448** (i.e., raise a frequency of the modulated signals **448** from the baseband frequency to the transmission frequency) to generate the antenna feed signals **412**.

(191) Referring now to FIG. **4**E, in implementations in which the client-side input **400** is configured to receive the parallel baseband signals **428**, the transmitter circuitry **408** may be configured to receive the parallel baseband signals **428** from the client-side input **400**. In such implementations, the modulator **444** may be configured to receive the parallel baseband signals **428** from the client-side input **400** and the carrier signals **440** from first LO **436** and modulate the parallel baseband signals **428** onto the carrier signals **440** to generate the modulated signals **448**. In such implementations, the up-converter **452** may be configured to receive the modulated signals **448** from the modulator **444** and up-convert the modulated signals **448** to generate one or more up-converted signals **460** (hereinafter, the "up-converted signals **460**").

(192) In some implementations, the transmitter circuitry **408** may further comprise a combiner **456** configured to receive the up-converted signals **460** from the up-converter **452** and combine the up-converted signals **460** into the antenna feed signals **412**. However, in other implementations, the first antennas **416** may be configured to receive the antenna feed signals **412** from the up-converter **452**, generate the radiated signals **420** based on the antenna feed signals **412**, and couple the radiated signals **420** into the first hollow waveguide **208***a* such that the radiated signals **420** interact in the first hollow waveguide **208***a* to form a combined radiated signal (not shown).

(193) In some implementations, coupling the radiated signals **420** into the first hollow waveguide **208***a* such that the radiated signals **420** interact in the first hollow waveguide **208***a* to form the combined radiated signal utilizes at least one of PDM, TDM, and WDM.

(194) Referring now to FIG. **4**F, shown therein is a block diagram of another exemplary implementation of the first transmitter **212***a* shown in FIG. **2**. However, it should be understood that the description of any particular one of the transmitters **212** may be applicable to any of the transmitters **212** described herein.

(195) In the implementation shown in FIG. **4**F, the first transmitter **212***a* comprises the client-side input **400** configured to receive the baseband signals **404** from one or more external component (e.g., a control module **224**) and send the baseband signals **404** to the transmitter circuitry **408**, the transmitter circuitry **408** configured to receive the baseband signals **404** from the client-side input **400**, generate the antenna feed signals **412** based on the baseband signals **404**, and send the antenna feed signals **412** to an RF interface **464** configured to receive the antenna feed signals **412** from the transmitter circuitry **408** and transmit the antenna feed signals **412**, and a digital enhancement and control unit **468** configured to provide digital control and/or processing capabilities for one or more of the components of the first transmitter **212***a*.

(196) In the implementation shown in FIG. **4**F, the transmitter circuitry **408** comprises one or more modulation block **444***a* (hereinafter, the "modulation block **444***a*"), a frequency synthesizer **472** comprising a phase-locked loop (PLL) **476** and a first LO **436***a*, a second LO **436***b*, a first frequency mixer **480***a*, a second frequency mixer **480***b*, a first amplifier **484***a*, and a second

amplifier **484***b*.

(197) The modulation block **444***a* may be configured to receive the baseband signals **404** from the client-side input **400** and encode the baseband signals **404** in a format suitable for modulation onto a carrier signal. In some implementations, the modulation block **444***a* may include one or more digital-to-analog converter (DAC), one or more Serializer/Deserializer (SerDes), one or more folded modulator **700** (shown in FIG. **7**), and/or circuitry operable to encode the baseband signals **404** in a modulation format, such as AM, ASK, PSK, QAM, QAM16, or variations thereof, for example. In some implementations, the modulation block **444***a* may include circuitry operable to perform forward error correction (FEC). The modulation block **444***a* may be further configured to send the encoded input signals having the data encoded therein to the second frequency mixer **480***b*.

(198) In some implementations, the modulation block **444***a* is configured to simply receive the baseband signals **404** (i.e., the baseband signals **404** having been previously encoded in a modulation format) from the client-side input **400** and send the baseband signals **404** to the second frequency mixer **480***b*.

(199) The second LO **436***b* may be configured to generate second carrier signals having a continuous waveform (e.g., a sinusoidal waveform) having a predetermined frequency (i.e., a baseband (BB) frequency). In some implementations, the predetermined frequency of the second carrier signals (i.e., the BB frequency) is in an RF band (i.e., in a range between 30 Hertz (Hz) and 300 GHz). In some implementations, the predetermined frequency of the second carrier signals (i.e., the BB frequency) is in a range between 1 Megahertz (MHz) and 300 GHz. In some implementations, the predetermined frequency of the second carrier signals (i.e., the BB frequency) is in a range between 5 GHz and 30 GHz. The second LO **436***b* may be further configured to send the second carrier signals to the second frequency mixer **480***b*.

(200) The second frequency mixer **480***b* may be configured to receive the encoded baseband signals from the modulation block **444***a*, receive the second carrier signals from the second LO **436***b*, up-convert the encoded baseband signals with the second carrier signals to produce first modulated signals having client data encoded therein and having the predetermined frequency of the second carrier signals (i.e., the BB frequency), and send the first modulated signals to the third amplifier **484***c*.

(201) The third amplifier **484***c* may be configured to receive the first modulated signals from the second frequency mixer **480***b*, adjust an amplitude of the first modulated signals such that the amplified first modulated signals can drive the first frequency mixer **480***a*, and send the amplified first modulated signals to the first frequency mixer **480***a*.

(202) The frequency synthesizer **472** (i.e., the first LO **436***a* and the PLL **476**) may be configured to generate first carrier signals having a continuous waveform (e.g., a sinusoidal waveform) having a predetermined frequency (e.g., within the THz frequency band **104** or, in some implementations, in a range between 300 GHz and 10 THz). In some implementations, the predetermined frequency of the first carrier signals is in a range between 30 GHz and 300 GHz. In some such implementations, the predetermined frequency of the first carrier signals is 240 GHz. In other implementations, the predetermined frequency of the first carrier signals is in a range between 300 GHz and 3 THz. The frequency synthesizer **472** may be further configured to send the first carrier signals to the second amplifier **484***b*.

(203) The second amplifier **484***b* may be configured to receive the first carrier signals from the first LO **436***a*, adjust an amplitude of the first carrier signals to generate amplified carrier signals that can drive the first frequency mixer **480***a*, and send the amplified carrier signals to the first frequency mixer **480***a*.

(204) The first frequency mixer **480***a* may be configured to receive the amplified carrier signals from the second amplifier **484***b*, receive the amplified first modulated signals from the third amplifier **484***c*, up-convert the amplified first modulated signals with the amplified carrier signals

to produce second modulated signals having the client data encoded therein and having the predetermined frequency of the amplified carrier signals (i.e., within the THz frequency band **104** or, in some implementations, in a range between 300 GHz and 10 THz), and send the second modulated signals to the first amplifier **484***a*.

(205) The first amplifier **484***a* may be configured to receive the second modulated signals from the first frequency mixer **480***a*, adjust an amplitude of the second modulated signals such that the amplified second modulated signals can be transmitted by the RF interface **464**, and send the amplified second modulated signals to the RF interface **464**. The first amplifier **484***a* may be configured to generate the amplified second modulated signals to have a power in a range between 0.05 watts (W) and 0.4 W, for example.

(206) The RF interface **464** may be configured to receive the amplified second modulated signals with the client data encoded therein from the first amplifier **484***a* and send the amplified second modulated signals as the antenna feed signals **412** (i.e., having the client data encoded therein) within a predetermined frequency range (e.g., the THz frequency band **104** or, in some implementations, in a range between 300 GHz and 10 THz). In some implementations, the RF interface **464** may be electrically connected to one of the first antennas **416** and configured to send the antenna feed signals **412** to the first antenna **416**. In other implementations, however, the first antennas **416** may be included in place of the RF interface **464**.

(207) Referring now to FIG. **4**G, shown therein is a block diagram of another exemplary implementation of the first transmitter **212***a* shown in FIG. **2**. In the implementation shown in FIG. **5**B, the first transmitter **212***a* comprises a plurality of inputs including an in-phase (I)-BB client-side input **400***a* and a quadrature (Q)-BB client-side input **400***b* configured to receive I-BB baseband signals **404***a* and Q-BB baseband signals **404***b*, respectively, from one or more external component (e.g., a control module **224**) and an LO input **400***c* configured to receive one or more carrier signals **488** (hereinafter, the “carrier signals **488**”) from an external LO, the transmitter circuitry **408** configured to generate the antenna feed signals **412** based on the I-BB baseband signals **404***a*, the Q-BB baseband signals **404***b*, and the carrier signals **488**, and the RF interface **464** configured to transmit the antenna feed signals **412**.

(208) In the implementation shown in FIG. **4**G, the transmitter circuitry **408** comprises a balancing unit (Balun) **492**, a third frequency mixer **480***c*, a fourth frequency mixer **480***d*, a fifth frequency mixer **480***e*, and a sixth frequency mixer **480***f*, a fourth amplifier **484***d*, a fifth amplifier **484***e*, a sixth amplifier **484***f*, a seventh amplifier **484***g*, and eighth amplifier **484***h*, a quadrature coupler (e.g., branchline coupler) **494**, and a power combiner (e.g., Wilkinson power combiner) **498**.

(209) The I-BB baseband signals **404***a* and the Q-BB baseband signals **404***b* may be I and Q components of baseband signals **404** having client data encoded therein. The I-BB client-side input **400***a* may be configured to send the I-BB baseband signals **404***a* to the sixth amplifier **484***f*. The Q-BB client-side input **400***b* may be configured to send the Q-BB baseband signals **404***b* to the seventh amplifier **484***g*.

(210) The LO input **400***c* may be configured to receive the carrier signals **488** from an external LO, the carrier signals **488** having a continuous waveform (e.g., a sinusoidal waveform) having a predetermined frequency. The LO input **400***c* may be further configured to send the carrier signals **488** to the Balun **492**.

(211) The Balun **492** may be configured to isolate and/or maintain impedance differences between balanced transmission lines and unbalanced transmission lines. The Balun **492** may be further configured to send the carrier signals **488** to the third frequency mixer **480***c*.

(212) The third frequency mixer **480***c* may be configured to receive the carrier signals **488** from the Balun **492**, multiply the carrier signals **488** (e.g., by a multiple of four), and send the multiplied carrier signals to the fourth amplifier **484***d*.

(213) The fourth amplifier **484***d* may be configured to receive the multiplied carrier signals from the third frequency mixer **480***c*, adjust an amplitude of the multiplied carrier signals such that the

amplified carrier signals can drive the fourth frequency mixer **480***d*, and send the amplified carrier signals to the fourth frequency mixer **480***d*.

(214) The fourth frequency mixer **480***d* may be configured to receive the amplified carrier signals from the fourth amplifier **484***d*, multiply the amplified carrier signals (e.g., by a multiple of two), and send the remultiplied carrier signals to the fifth amplifier **484***e*.

(215) The fifth amplifier **484***e* may be configured to receive the remultiplied carrier signals from the fourth frequency mixer **480***d*, adjust an amplitude of the remultiplied carrier signals such that the reamplified carrier signals can drive the quadrature coupler **494**, and send the reamplified carrier signals to the quadrature coupler **494**.

(216) The sixth amplifier **484***f* may be configured to receive the I-BB baseband signals **404***a* from the I-BB client-side input **400***a*, adjust an amplitude of the I-BB baseband signals **404***a* such that the amplified I-BB input signals can drive the fifth frequency mixer **480***e*, and send the amplified I-BB signals to the fifth frequency mixer **480***e*.

(217) The seventh amplifier **484***g* may be configured to receive the Q-BB baseband signals **404***b* from the Q-BB client-side input **400***b*, adjust an amplitude of the Q-BB baseband signals **404***b* such that the amplified Q-BB baseband signals **404***b* can drive the sixth frequency mixer **480***f*, and the amplified Q-BB signals to the sixth frequency mixer **480***f*.

(218) The quadrature coupler **494** may be configured to receive the reamplified carrier signals from the fifth amplifier **484***e*, split the reamplified carrier signals into first carrier signals and second carrier signals, send the first carrier signals to the fifth frequency mixer **480***e*, and send the second carrier signals to the sixth frequency mixer **480***f*, wherein the first carrier signals and the second carrier signals are out of phase by 90°.

(219) The fifth frequency mixer **480***e* may be configured to receive the amplified I-BB signals from the sixth amplifier **484***f*, receive the first carrier signals from the quadrature coupler **494**, up-convert the amplified I-BB signals with the first carrier signals to produce I antenna feed signals having the I component of the client data encoded therein and having the predetermined frequency of the carrier signals **488**, and send the I antenna feed signals to the power combiner **498**.

(220) The sixth frequency mixer **480***f* may be configured to receive the amplified Q-BB signals from the seventh amplifier **484***g*, receive the second carrier signals from the quadrature coupler **494**, up-convert the amplified Q-BB signals with the second carrier signals to produce Q antenna feed signals signals having the Q component of the client data encoded therein and having the predetermined frequency of the carrier signals **488**, and send the Q antenna feed signals to the power combiner **498**.

(221) The power combiner **498** may be configured to receive the I antenna feed signals from the fifth frequency mixer **480***e*, receive the Q antenna feed signals from the sixth frequency mixer **480***f*, combine the I antenna feed signals and the Q antenna feed signals to produce the antenna feed signals **412**, and send the antenna feed signals **412** to the RF interface **464**. In some implementations, the RF interface **464** may be electrically connected to one of the first antennas **416** and configured to send the antenna feed signals **412** to the first antenna **416**. In other implementations, however, one of the first antennas **416** may be included in place of the RF interface **464**.

(222) Referring now to FIG. **5**A, shown therein is a block diagram of an exemplary implementation of the first receiver **216***a* (hereinafter, the "first receiver **216***a*") shown in FIG. **2**. However, it should be understood that the description of any particular one of the receivers **216** may be applicable to any of the receivers **216** described herein. The first receiver **216***a* (and, therefore, each of the receiver **216**) generally comprises one or more second antennas **516** configured to detect the radiated signals **420** received from the first hollow waveguide **208***a* and generate one or more antenna output signals **512** (hereinafter, the "antenna output signals **512**") based on the radiated signals **420**, receiver circuitry **508** configured to receive the antenna output signals **512** from the second antennas **516** and generate the baseband signals **404** based on the antenna output signals

512, and a client-side output **500** configured to receive the baseband signals **404** from the receiver circuitry **508** and transmit the baseband signals **404** to one or more external component (e.g., a control module **224**).

(223) In some implementations, the antenna output signals **512** are received from the second antennas **516** on one or more transmission lines (not shown) (hereinafter, the “transmission lines”), wherein each of the transmission lines has two or more conductors (not shown) (hereinafter, the “conductors”). In some implementations, the transmission lines have a first transmission loss and the first hollow waveguide **208***a* has a second transmission loss that is less than the first transmission loss. In some implementations, the second transmission loss is in a range between 0.001 and 20.00 dB/m/Tb/s.

(224) In some implementations, as shown in FIG. **5**A, each of the second antennas **516**, the receiver circuitry **508**, and the client-side output **500** may be disposed on a substrate **524**. However, in other implementations, one or more of the second antennas **516**, the receiver circuitry **508**, and the client-side output **500** may be disposed on a first substrate (not shown), and one or more of the second antennas **516**, the receiver circuitry **508**, and the client-side output **500** may not be disposed on the first substrate. For example, the one or more of the second antennas **516**, the receiver circuitry **508**, and the client-side output **500** may be disposed on a second substrate (not shown). In such implementations, the first substrate and the second substrate may be in a stacked arrangement.

(225) In some implementations, the substrate **524** may have a plurality of layers (not shown). In such implementations, one or more of the second antennas **516**, the receiver circuitry **508**, and the client-side output **500** may be disposed on a first layer (not shown), and one or more of the second antennas **516**, the receiver circuitry **508**, and the client-side output **500** may be disposed on a second layer (not shown).

(226) In some implementations, one or more of the second antennas **516**, the receiver circuitry **508**, and the client-side output **500** may be integrated into a monolithic semiconductor die (not shown). In some implementations, one or more of the second antennas **516**, the receiver circuitry **508**, and the client-side output **500** may implemented using one or more of CMOS technology, SiGe semiconductor technology, and III-V compound semiconductor technology.

(227) In some implementations, the radiated signals **420** include a first complementary radiated signal (not shown) having a first polarization and a second complementary radiated signal (not shown) having a second polarization different from the first polarization. In such implementations, the second antennas **516** may be configured to generate the antenna output signals **512** based on the radiated signals **420** including the first complementary radiated signal and the second complementary radiated signal. The first polarization and the second polarization may be orthogonal to each other.

(228) In some implementations, the radiated signals **420** may be formed by a first complementary radiated signal (not shown) having a first polarization and a second complementary radiated signal (not shown) having a second polarization different from the first polarization interacting in the first hollow waveguide **208***a*. In such implementations, the radiated signals **420** may have a third polarization different from the first polarization and the second polarization. In such implementations, the second antennas **516** may be configured generate the antenna output signals **512** based on the radiated signals **420** formed by the first complementary radiated signal and the second complementary radiated signal.

(229) Referring now to FIG. **5**B, in some implementations, the client-side output **500** is configured to receive a serial baseband signal (i.e., the baseband signals **404**) from the receiver circuitry **508**. In such implementations, the first receiver **216***a* (and, therefore, any of the receivers **216**) may further comprise a second deserializer **526** configured to receive the baseband signals **404** from the client-side output **500**, split the serial baseband signal into the parallel baseband signals **428**, and transmit the parallel baseband signals **428** to one or more external component (e.g., a control module **224**). In some such implementations, splitting the serial baseband signal into the parallel

baseband signals **428** utilizes at least one of PDM, TDM, and WDM.

(230) Referring now to FIG. **5**C, in some implementations, the client-side output **500** is configured to receive the parallel baseband signals **428** from the receiver circuitry **508**. In such implementations, the first receiver **216***a* (and, therefore, any of the receivers **216**) may further comprise a second serializer **532** configured to receive the parallel baseband signals **428** from the client-side output **500** and combine the parallel baseband signals **428** into the serial baseband signal (i.e., the baseband signals **404**). In some such implementations, combining the parallel baseband signals **428** into the baseband signals **404** utilizes at least one of PDM, TDM, and WDM.

(231) Referring now to FIG. **5**D, shown therein is an exemplary implementation of the receiver circuitry **508** shown in FIGS. **5**A-**5**C. In some implementations, the receiver circuitry **508** comprises one or more LOs **536** (hereinafter, the "LO **536**") configured to generate one or more reference signals **540** (hereinafter, the "reference signals **540**") having a baseband frequency less than the transmission frequency, one or more down-conversion circuits **552** (hereinafter, the "down-converter **552**") configured to receive the antenna output signals **512** from the second antennas **516** and the reference signals **540** from the LO **536** and down-convert the antenna output signals **512** (i.e., lower a frequency of the antenna output signals **512** from the transmission frequency to the baseband frequency) using the reference signals **540** to generate one or more modulated signals **548** (hereinafter, the "modulated signals **548**"), and one or more demodulation circuits **544** (hereinafter, the "demodulator **544**") configured to receive the modulated signals **548** from the down-converter **552** and demodulate the modulated signals **548** to generate the baseband signals **404**.

(232) Referring now to FIG. **5**E, in implementations in which the second antennas **516** are configured to receive the radiated signals **420** formed by a first complementary radiated signal (not shown) having a first polarization and a second complementary radiated signal (not shown) having a second polarization different from the first polarization interacting in the first hollow waveguide **208***a*, the receiver circuitry **508** may be configured to receive the antenna output signals **512** from the second antennas **516**. In such implementations, the demodulator **544** may be configured to receive the modulated signals **548** from the down-converter **552** and demodulate the modulated signals **548** to generate the parallel baseband signals **428**.

(233) In some implementations, the receiver circuitry **508** may further comprise a splitter **556** configured to receive the antenna output signals **512** from the second antennas **516** and split the antenna output signals **512** into a plurality of parallel antenna output signals **560** (hereinafter, the "parallel antenna output signals **560**"). However, in other implementations, the second antennas **516** may be configured to detect the first complementary radiated signal and the second complementary radiated signal based on the radiated signals **420** received from the first hollow waveguide **208***a* and generate the antenna output signals **512** based on the first complementary radiated signal and the second complementary radiated signal.

(234) In some implementations, detecting the first complementary radiated signal and the second complementary radiated signal based on the radiated signals **520** received from the first hollow waveguide **208***a* utilizes at least one of PDM, TDM, and WDM.

(235) Referring now to FIG. **5**F, shown therein is a block diagram of another exemplary implementation of the first receiver **216***a* shown in FIG. **2**. In the implementation shown in FIG. **5**F, the first receiver **216***a* comprises an RF interface **564** configured to receive the antenna output signals **512**, the receiver circuitry **508** configured to generate the baseband signals **404** based on the antenna output signals **512**, the client-side output **500** configured to transmit the baseband signals **404** to one or more external component (e.g., a control module **224**), and a digital enhancement and control unit **568** configured to provide digital control and/or processing capabilities for one or more of the components of the first receiver **216***a*.

(236) In the implementation shown, the receiver circuitry **508** comprises one or more demodulation block **544***a* (hereinafter, the "demodulation block **544***a*"), a frequency synthesizer **572** comprising a

PLL **576** and a first LO **536***a*, a second LO **536***b*, a first frequency mixer **580***a*, a second frequency mixer **580***b*, a first amplifier **584***a*, a second amplifier **584***b*, and a third amplifier **584***c*.

(237) The RF interface **564** may be configured to send the antenna output signals **512** to the first amplifier **584***a*. In some implementations, the RF interface **564** may be configured to receive the antenna output signals **512** from one of the second antennas **516**. In other implementations, one of the second antennas **516** may be included in place of the RF interface **564**.

(238) The first amplifier **584***a* may be configured to receive the antenna output signals **512** from the RF interface **564**, adjust an amplitude of the antenna output signals **512** such that the amplified transmission signals can drive the first frequency mixer **580***a*, and send the amplified transmission signals to the first frequency mixer **580***a*.

(239) The frequency synthesizer **572** (i.e., the first LO **536***a* and the PLL **576**) may be configured to generate first carrier signals having a continuous waveform (e.g., a sinusoidal waveform) having a predetermined frequency (e.g., within the THz frequency band **104** or, in some implementations, in a range between 300 GHz and 10 THz). In some implementations, the predetermined frequency of the first carrier signals is in a range between 30 GHz and 300 GHz. In some such implementations, the predetermined frequency of the first carrier signals is 240 GHz. In other implementations, the predetermined frequency of the first carrier signals is in a range between 300 GHz and 3 THz. The first LO **536***a* may be further configured to send the first carrier signals to the second amplifier **584***b*.

(240) The second amplifier **584***b* may be configured to receive the first carrier signals from the first LO **536***a*, adjust an amplitude of the first carrier signals to generate amplified carrier signals that can drive the first frequency mixer **580***a*, and send the amplified carrier signals to the first frequency mixer **580***a*.

(241) The first frequency mixer **580***a* may be configured to receive the antenna output signals **512** from the first amplifier **584***a*, receive the amplified carrier signals from the second amplifier **584***b*, down-convert the antenna output signals **512** with the amplified carrier signals to produce modulated signals having the client data encoded therein and having the BB frequency, and send the modulated signals to the third amplifier **584***c*.

(242) The third amplifier **584***c* may be configured to receive the modulated signals from the first frequency mixer **580***a*, adjust an amplitude of the modulated signals such that the amplified modulated signals can drive the second frequency mixer **580***b*, and send the amplified modulated signals to the second frequency mixer **580***b*.

(243) The second LO **536***b* may be configured to generate second carrier signals having a continuous waveform (e.g., a sinusoidal waveform) having a predetermined frequency (i.e., the BB frequency). In some implementations, the predetermined frequency of the second carrier signals (i.e., the BB frequency) is in a range between 8 GHz and 10 GHz. The second LO **536***b* may be further configured to send the second carrier signals to the second frequency mixer **580***b*.

(244) The second frequency mixer **580***b* may be configured to receive the amplified modulated signals from the third amplifier **584***c*, receive the second carrier signals from the second LO **536***b*, down-convert the amplified modulated signals with the second carrier signals to produce encoded signals having the client data encoded therein and having the predetermined frequency of the second carrier signals (i.e., the BB frequency), and send the encoded signals to the demodulation block **544***a*.

(245) The demodulation block **544***a* may be configured to receive the encoded signals from the second frequency mixer **580***b* and decode the encoded signals in a format suitable for transmission to one or more external component (e.g., a control module **224**) to generate the baseband signals **404**.

(246) In some implementations, the demodulation block **544***a* may include one or more analog-to-digital converter (ADC), one or more Serializer/Deserializer (SerDes), one or more rectifying detector **800** (shown in FIG. **8**), and/or circuitry operable to decode the encoded output signals

from a modulation format, such as AM, ASK, PSK, QAM, or QAM16, or variations thereof, for example, to produce the baseband signals **404** with the client data encoded therein. In some implementations, the demodulation block **544***a* may include circuitry operable to perform forward error correction (FEC). The demodulation block **544***a* may be further configured to send the baseband signals **404** to the client-side output **500**. In some implementations, the demodulation block **544***a* is configured to simply receive the encoded signals from the second frequency mixer **580***b* and send the encoded signals as the baseband signals **404** to the client-side output **500**.

(247) In some implementations, the client-side output **500** is a pair of output interfaces. In some such implementations, the client-side output **500** is an LVDS link configured to transmit LVDS signals, and the baseband signals **404** are LVDS signals with the client data encoded therein.

(248) Referring now to FIG. **5**G, shown therein is a block diagram of another exemplary implementation of the first receiver **216***a* shown in FIG. **2**. In the implementation shown in FIG. 5G, the first receiver **216***a* comprises the RF interface **564** configured to receive the antenna output signals **512**, an LO input **500***c* configured to receive carrier signals **588** from an external LO, the receiver circuitry **508** configured to generate Q-BB baseband signals **404***b* and I-BB baseband signals **404***a* based on the antenna output signals **512** and the carrier signals **588**, and a Q-BB client-side output **500***a* and an I-BB client-side output **500***b* configured to transmit the Q-BB baseband signals **404***b* and the I-BB baseband signals **404***a*, respectively.

(249) In the implementation shown, the receiver circuitry **508***a* comprises a third frequency mixer **580***c*, a fourth frequency mixer **580***d*, a fifth frequency mixer **580***e*, a sixth frequency mixer **580***f*, a fourth amplifier **584***d*, a fifth amplifier **584***e*, a sixth amplifier **584***f*, a seventh amplifier **584***g*, an eighth amplifier **584***h*, a ninth amplifier **584***i*, a tenth amplifier **584***j*, an eleventh amplifier **584***k*, a twelfth amplifier **584***l*, a Balun **592**, a quadrature coupler (e.g., branchline coupler) **594**, and a power divider (e.g., Wilkinson power divider) **598**.

(250) The fourth amplifier **584***d* may be configured to receive the antenna output signals **512** from the RF interface **564**, adjust an amplitude of the antenna output signals **512** such that the amplified transmission signals can drive the power divider **598**, and send the amplified transmission signals to the power divider **598**. In some implementations, the fourth amplifier **584***d* is a low-noise amplifier (LNA).

(251) The power divider **598** may be configured to receive the amplified transmission signals from the fourth amplifier **584***d*, split the amplified transmission signals into I antenna output signals having the I component of the client data encoded therein and Q antenna output signals having the Q component of the client data encoded therein, send the Q antenna output signals to the third frequency mixer **580***c*, and send the I antenna output signals to the fourth frequency mixer **580***d*.

(252) The LO input **500***c* may be configured to receive carrier signals **588** from an external LO, the carrier signals **588** having a continuous waveform (e.g., a sinusoidal waveform) having a predetermined frequency. The LO input **500***c* may be further configured to send the carrier signals **588** to the Balun **592**.

(253) The Balun **592** may be configured to isolate and/or maintain impedance differences between balanced transmission lines and unbalanced transmission lines. The Balun **492** may be further configured to send the carrier signals **588** to the sixth frequency mixer **580***f*.

(254) The sixth frequency mixer **580***f* may be configured to receive the carrier signals **588** from the Balun **592**, multiply the carrier signals **588** (e.g., by a multiple of four), and send the multiplied carrier signals to the twelfth amplifier **5841**.

(255) The twelfth amplifier **5841** may be configured receive the multiplied carrier signals from the sixth frequency mixer **580***f*, adjust an amplitude of the multiplied carrier signals to generate amplified carrier signals that can drive the fifth frequency mixer **580***e*, and send the amplified carrier signals to the fifth frequency mixer **580***e*.

(256) The fifth frequency mixer **580***e* may be configured to receive the amplified carrier signals from the twelfth amplifier **584***l*, multiply the amplified carrier signals (e.g., by a multiple of two),

and send the remultiplied carrier signals to the eleventh amplifier **584***k*.
(257) The eleventh amplifier **584***k* may be configured to receive the remultiplied carrier signals from the fifth frequency mixer **580***e*, adjust an amplitude of the remultiplied carrier signals to generate reamplified carrier signals that can drive the quadrature coupler **594**, and send the reamplified carrier signals to the quadrature coupler **594**.
(258) The quadrature coupler **594** may be configured to receive the reamplified carrier signals from the eleventh amplifier **584***k*, split the reamplified carrier signals into first carrier signals and second carrier signals, send the first carrier signals to the third frequency mixer **580***c*, and send the second carrier signals to the fourth frequency mixer **580***d*, wherein the first carrier signals and the second carrier signals are out of phase by 90°.
(259) The third frequency mixer **580***c* may be configured to receive the Q antenna output signals from the power divider **598**, receive the first carrier signals from the quadrature coupler (e.g., branchline coupler) **566**, down-convert the Q antenna output signals with the first carrier signals to generate Q-BB intermediate signals having the Q component of the client data encoded therein and having the BB frequency, and send the Q-BB intermediate signals to the fifth amplifier **584***e*.
(260) The fifth amplifier **584***e*, the sixth amplifier **584***f*, and the seventh amplifier **584***g* may be configured to receive the Q-BB intermediate signals from the third frequency mixer **580***c*, down-convert the Q-BB intermediate signals to generate the Q-BB baseband signals **404***b*, and send the Q-BB baseband signals **404***b* to the Q-BB client-side output **500***a*. In some implementations, the fifth amplifier **584***e* is a transimpedance amplifier (TIA), and the sixth amplifier **584***f* is a variable-gain amplifier (VGA).
(261) The fourth frequency mixer **580***d* may be configured to receive the I antenna output signals from the power divider **598**, receive the second carrier signals from the quadrature coupler **594**, down-convert the I antenna output signals with the second carrier signals to produce I-BB intermediate signals having the I component of the client data encoded therein and having the BB frequency, and send the I-BB intermediate signals to the eighth amplifier **584***h*.
(262) The eighth amplifier **584***h*, the ninth amplifier **584***i*, and the tenth amplifier **584***j* may be configured to receive the I-BB intermediate signals from the fourth frequency mixer **580***d*, down-convert the I-BB intermediate signals to generate the I-BB baseband signals **404***a*, and send the I-BB baseband signals **404***a* to the I-BB client-side output **500***b*. In some implementations, the eighth amplifier **584***h* is a TIA, and the ninth amplifier **584***i* is VGA.
(263) Referring now to FIG. **6**A, shown therein is a block diagram of an exemplary implementation of the first transceiver **220***a* (hereinafter, the "first transceiver **220***a*") shown in FIG. **2**. However, it should be understood that the description of any particular one of the transceivers **220** may be applicable to any of the transceivers **220** described herein. The first transceiver **220***a* (and, therefore, each of the transceivers **220**) generally comprises a third transmitter **212***c* and a third receiver **216***c*.
(264) The third transmitter **212***c* generally comprises a client-side input **600***a* configured to receive one or more first baseband signals **604***a* (hereinafter, the "first baseband signals **604***a*") having first client data encoded therein from one or more external component (e.g., a control module **224**), transmitter circuitry **608***a* configured to receive the first baseband signals **604***a* from the client-side input **600***a* and generate one or more antenna feed signals **612***a* (hereinafter, the "antenna feed signals **612**") based on the first baseband signals **604***a*, and one or more first antennas **616***a* (hereinafter, the "first antennas **616**") configured to receive the antenna feed signals **612***a* from the transmitter circuitry **608***a*, generate one or more first radiated signals **420***a* (hereinafter, the "first radiated signals **420***a*") based on the antenna feed signals **612***a*, and couple the first radiated signals **420***a* into the fourth hollow waveguide **208***d*.
(265) The third receiver **216***c* generally comprises one or more second antennas **616***b* (hereinafter, the "antennas **616***b*") configured to detect one or more second radiated signals **620***b* (hereinafter, the "second radiated signals **620***b*") received from the third hollow waveguide **208***c* and generate

one or more antenna output signals **612***b* (hereinafter, the "antenna output signals **612***b*") based on the second radiated signals **620***b*, receiver circuitry **608***b* configured to receive the antenna output signals **612***b* from the second antennas **616***b* and generate the second baseband signals **604***b* based on the antenna output signals **612***b*, and a client-side output **600***b* configured to receive the second baseband signals **604***b* from the receiver circuitry **608***b* and transmit the second baseband signals **604***b* to one or more external component (e.g., a control module **224**).

(266) Each of the components of the first transceiver **220***a* (and, therefore, each of the transceivers **220**) may be the same or similar to one or more of the components of the first transmitter **212***a* and the first receiver **216***a* as described herein.

(267) Referring now to FIG. **6**B, shown therein is a block diagram of another exemplary implementation of the first transceiver **220***a* shown in FIG. **2**. In the implementation shown in FIG. **6**B, the first transceiver **220***a* comprises the client-side input **600***a* configured to receive the first baseband signals **604***a* from one or more external component (e.g., a control module **224**), the transmitter circuitry **608***a* configured to generate the antenna feed signals **612***a* based on the input signals **640***a*, a first RF interface **664***a* configured to transmit the antenna feed signals **612***a*, a second RF interface **664***b* configured to receive the antenna output signals **612***b*, the receiver circuitry **608***b* configured to generate the second baseband signals **604***b* based on the antenna output signals **612***b*, the client-side output **600***b* configured to transmit the second baseband signals **604***b* to one or more external component, and a digital enhancement and control unit **668** configured to provide digital control and/or processing capabilities for one or more of the components of the first transceiver **220***a*.

(268) In some implementations, the first transceiver **220***a* comprises the first RF interface **664***a*, but lacks the second RF interface **664***b*. In such implementations, the first RF interface **664***a* may be configured to transmit antenna feed signals **612***a* and receive antenna output signals **612***b*. In some implementations, the first transceiver **220***a* may have a number of RF interfaces that is greater than two.

(269) In the implementation shown, the transmitter circuitry **608***a* comprises a frequency synthesizer **672** comprising a PLL **676**, a first LO **636***a*, and a signal distribution block (e.g., splitter) **698**, one or more modulation block **644***a* (hereinafter, the "modulation block **644***a*"), a second LO **636***b*, a first frequency mixer **680***a*, a third frequency mixer **680***c*, a first amplifier **684***a*, a third amplifier **684***c*, and a fifth amplifier **684***e*.

(270) In the implementation shown, the receiver circuitry **608***b* comprises the frequency synthesizer **672** comprising the PLL **676**, the first LO **636***a*, and the signal distribution **698**, the modulation block **644***a*, a third LO **636***c*, a second frequency mixer **680***b*, a fourth frequency mixer **680***d*, a second amplifier **684***b*, a fourth amplifier **684***d*, and a sixth amplifier **684***f*.

(271) In some implementation shown in FIG. **6**B, each of the components of the first transceiver **220***a* are disposed on a single substrate **624**, which may be a portion of a semiconductor wafer.

(272) The modulation block **644***a* may be configured to: (1) receive the first baseband signals **604***a* from the client-side input **600***a*, encode the first baseband signals **604***a* in a format suitable for modulation onto a carrier signal, and send the encoded input signals the third frequency mixer **680***c*; and (2) receive the encoded output signals from the fourth frequency mixer **680***d*, decode the encoded output signals in a format suitable for transmission to one or more external component (e.g., a control module **224**), and send the second baseband signals **604***b* to the client-side output **600***b*.

(273) In some implementations, the modulation block **644***a* may include one or more DAC, one or more ADC, one or more Serializer/Deserializer (SerDes), one or more folded modulator **700** (shown in FIG. **7**), one or more rectifying detector **800** (shown in FIG. **8**) and/or circuitry operable to encode the first baseband signals **604***a* in a modulation format, such as AM, ASK, PSK, QAM, or QAM16, or variations thereof, for example, and decode encoded output signals from the modulation format to produce second baseband signals **604***b* having the client data encoded therein.

In some implementations, the modulation block **644***a* may include circuitry operable to perform forward error correction (FEC).

(274) The frequency synthesizer **672** may be configured to generate first carrier signals having a continuous waveform (e.g., a sinusoidal waveform) having a predetermined frequency (e.g., within the THz frequency band **104** or in some implementations, a range between 300 GHz and 10 THz). In some implementations, the predetermined frequency of the first carrier signals is in a range between 30 GHz and 300 GHz. In some such implementations, the predetermined frequency of the first carrier signals is 240 GHz. In other implementations, the predetermined frequency of the first carrier signals is in a range between 300 GHz and 3 THz. The frequency synthesizer **672** may be further configured to send the first carrier signals to the signal distribution block **698**.

(275) The signal distribution block **698** may be configured to receive the first carrier signals from the first LO **636***a* and distribute the first carrier signals to the third amplifier **684***c* and the fourth amplifier **684***d*.

(276) Referring now to the transmitter circuitry **608***a*, in some implementations, the client-side input **600***a* is a pair of input interfaces. In some such implementations, the client-side input **600***a* is an LVDS link configured to receive LVDS signals, and the first baseband signals **604***a* are LVDS signals having the client data encoded therein. The client-side input **600***a* may be further configured to send the first baseband signals **604***a* to the modulation block **644***a*.

(277) The second LO **636***b* may be configured to generate second carrier signals having a continuous waveform (e.g., a sinusoidal waveform) having a predetermined frequency (i.e., the BB frequency). In some implementations, the predetermined frequency of the second carrier signals (i.e., the BB frequency) is in a range between 8 GHz and 10 GHz. The second LO **636***b* may be further configured to send the second carrier signals to the third frequency mixer **680***c*.

(278) The third frequency mixer **680***c* may be configured to receive the encoded input signals from the modulation block **644***a*, receive the second carrier signals from the second LO **636***b*, up-convert the encoded input signals with the second carrier signals to produce first modulated signals having the client data encoded therein and having the predetermined frequency of the second carrier signals (i.e., the BB frequency), and send the first modulated signals to the fifth amplifier **684***e*.

(279) The fifth amplifier **684***e* may be configured to receive the first modulated signals from the third frequency mixer **680***c*, adjust an amplitude of the first modulated signals such that the amplified first modulated signals can drive the first frequency mixer **680***a*, and send the amplified first modulated signals to the first frequency mixer **680***a*.

(280) The third amplifier **684***c* may be configured to receive the first carrier signals from the signal distribution block **698**, adjust an amplitude of the first carrier signals to generate amplified carrier signals that can drive the first frequency mixer **680***a*, and send the amplified carrier signals to the first frequency mixer **680***a*.

(281) The first frequency mixer **680***a* may be configured to receive the amplified carrier signals from the third amplifier **684***c*, receive the amplified first modulated signals from the fifth amplifier **684***e*, up-convert the amplified first modulated signals with the amplified carrier signals to produce second modulated signals having the data encoded therein and having the predetermined frequency of the amplified carrier signals (i.e., within the THz frequency band **104** or, in some implementations, in a range between 300 GHz and 10 THz), and send the second modulated signals to the first amplifier **684***a*.

(282) The first amplifier **684***a* may be configured to receive the second modulated signals from the first frequency mixer **680***a*, adjust an amplitude of the second modulated signals such that the amplified second modulated signals can be transmitted by the first RF interface **664***a*, and send the amplified second modulated signals to the first RF interface **664***a*.

(283) The first RF interface **664***a* may be configured to receive the amplified second modulated signals from the first amplifier **684***a* and send the amplified second modulated signals as antenna feed signals **612***a* (i.e., having the data encoded therein) having a frequency within a predetermined

frequency range (e.g., the THz frequency band **104** or, in some implementations, in a range between 300 GHz and 10 THz). In some implementations, the first RF interface **664***a* may be connected to one of the antennas **616** and configured to send the antenna feed signals **612***a* to the antenna **616**. In other implementations, however, one of the antennas **616** may be included in place of the first RF interface **664***a*.

(284) Referring now to the receiver circuitry **608***b*, the second RF interface **664***b* may be configured to receive the antenna output signals **612***b* (i.e., having client data encoded therein) within a predetermined frequency range (e.g., the THz frequency band **104** or, in some implementations, in a range between 300 GHz and 10 THz) and send the antenna output signals **612***b* to the second amplifier **684***b*. As described in further detail below, the second RF interface **664***b* may be configured to receive the antenna output signals **612***b* from one of the antennas **616**. In other implementations, however, one of the antennas **616** may be included in place of the second RF interface **664***b*.

(285) The second amplifier **684***b* may be configured to receive the antenna output signals **612***b* from the second RF interface **664***b*, adjust an amplitude of the antenna output signals **612***b* to generate amplified second transmission signals that can drive the second frequency mixer **680***b*, and send the amplified second transmission signals to the second frequency mixer **680***b*.

(286) The fourth amplifier **684***d* may be configured to receive the first carrier signals from the signal distribution block **698**, adjust an amplitude of the first carrier signals to generate amplified carrier signals that can drive the second frequency mixer **680***b*, and send the amplified carrier signals to the second frequency mixer **680***b*.

(287) The second frequency mixer **680***b* may be configured to receive the amplified second transmission signals from the second amplifier **684***b*, receive the amplified carrier signals from the fourth amplifier **684***d*, down-convert the amplified second transmission signals with the amplified carrier signals to produce third modulated signals having the data encoded therein and having the IF or the BB frequency, and send the third modulated signals to the sixth amplifier **684***f*.

(288) The sixth amplifier **684***f* may be configured to receive the third modulated signals from the second frequency mixer **680***b*, adjust an amplitude of the third modulated signals such that the amplified third modulated signals can drive the fourth frequency mixer **680***d*, and send the amplified third modulated signals to the fourth frequency mixer **680***d*.

(289) The third LO **636***c* may be configured to generate reference signals having a continuous waveform (e.g., a sinusoidal waveform) having a predetermined frequency (i.e., a BB frequency). In some implementations, the predetermined frequency of the reference signals (i.e., the BB frequency) is in a range between 8 GHz and 10 GHz. The third LO **636***c* may be further configured to send the reference signals to the fourth frequency mixer **680***d*.

(290) The fourth frequency mixer **680***d* may be configured to receive the amplified third modulated signals from the sixth amplifier **684***f*, receive the reference signals from the third LO **636***c*, down-convert the amplified third modulated signals with the reference signals to produce encoded output signals having the client data encoded therein and having the predetermined frequency of the reference signals (i.e., the BB frequency), and send the encoded output signals to the modulation block **644***a*.

(291) The client-side output **600***b* may be configured to transmit the second baseband signals **604***b* having the client data encoded therein to one or more external component (e.g., a control module **224**). In some implementations, the client-side output **600***b* is a pair of output interfaces. In some such implementations, the client-side output **600***b* is an LVDS link configured to transmit LVDS signals, and the second baseband signals **604***b* are LVDS signals having the client data encoded therein.

(292) Referring now to FIG. **7**, shown therein is a schematic diagram of an exemplary implementation of a folded modulator **700** constructed in accordance the present disclosure. The folded modulator **700** may be configured to perform broadband direct modulation to generate the

encoded signals and to minimize distortion while doing so. The folded modulator **700** may employ a cascade architecture (e.g., a cascaded circuit drive that is “stacked” or “folded”) in order to produce a linear or near-linear modulated output (i.e., the encoded signals). In implementations in which the folded modulator **700** employs a cascade architecture, the size of the stack may be directly proportional to the bandwidth.

(293) Referring now to FIG. **8**, shown therein is a schematic diagram of an exemplary implementation of a rectifying detector **800** constructed in accordance the present disclosure. The rectifying detector **800** may be configured to perform direct detection of incoming signals (i.e., the encoded signals). The rectifying detector **800** may be further configured to detect an envelope of the encoded signals or one or more amplitude transition of the encoded signals to generate the output signals.

(294) Referring now to FIG. **9**A, shown therein is a side view of an exemplary implementation of an antenna **900** coupled with a fifth hollow waveguide **208***e* constructed in accordance with the present disclosure. However, it should be understood that the description referring to any particular one of the antennas **416**, **516**, **616**, **900** may refer to any of the antennas **416**, **516**, **616**, **900** described herein. As shown in FIG. **8**A, the antenna **900** generally comprises a ground plane **904**, a radiator **908** mounted on the ground plane **904**, and a coaxial feedline **912** electrically connected to the radiator **908**. In some implementations, the antenna **900** may lack the ground plane **904**. In some implementations, the antenna **900** further comprises a casing (not shown) enclosing the radiator **908**. The antenna **900** may be a vertical antenna (i.e., an antenna extending orthogonally from a substrate) or a horizontal antenna (i.e., an antenna extending laterally from a substrate).

(295) The radiator **908** may be configured to transmit and detect radiated signals configured for coherent detection. In the implementation shown, the radiator **908** is a helical radiator configured to transmit and detect radiated signals having a circular polarization. In this implementation, the radiator **908** has a length I.sub.radiator, a diameter d.sub.radiator, and a spacing s.sub.radiator between adjacent turns of the radiator **908**. The radiator **908** is preferably disposed at a distance d.sub.gap from the fifth hollow waveguide **208***e*.

(296) The radiator **908** may be wound in a predetermined direction, such as clockwise (i.e., a left-hand wind) or counter-clockwise (i.e., a right-hand wind). While the radiator **908** of the antenna **900** is depicted in FIG. **9**A as having a right-hand wind or a counter-clockwise rotational direction, it should be understood that the radiator **908** of the antenna **900** may be provided with a left-hand wind or a clockwise rotational direction.

(297) In some implementations, signals for transmission may be sent to the antenna **900** via the coaxial feedline **912**. In other implementations, received RF signals may be sent from the antenna **900** via the coaxial feedline **912**.

(298) In some implementations, the length I.sub.radiator of the radiator **908** may be proportional to the wavelength of the signals being transmitted and/or received. In some implementations, the length I.sub.radiator of the radiator **908** is in a range between 10 microns and 10 mm. In some implementations, the diameter d.sub.radiator of the radiator **908** may be proportional to the wavelength of the signals being transmitted and/or received. In some implementations, the diameter d.sub.radiator of the radiator **908** is in a range between 10 microns and 10 mm. In some implementations, the spacing S.sub.radiator between adjacent turns of the radiator **908** may be in a range between 1 micron and 1 mm.

(299) The predetermined distance d.sub.gap at which the antenna **900** is spaced from the hollow waveguide **208** may vary depending upon the carrier frequency of the RF signal being transmitted by the antenna **900**. In some implementations, the predetermined distance d.sub.gap at which the antenna **900** is spaced from the hollow waveguide **208** is in a range between 3 μm and 3 mm. In one implementation, the predetermined distance d.sub.gap at which the antenna **900** is spaced from the hollow waveguide **208** is 1 mm. In some implementations, the antenna **900** may be directly connected to the fifth hollow waveguide **208***e*.

(300) Referring now to FIG. **9**B, shown therein is a top plan view of another exemplary implementation of the antenna **900** coupled with the fifth hollow waveguide **208***e* constructed in accordance with the present disclosure. The antenna **900** is similar in construction and function as the antenna **900**, with the exception that the antenna **900** includes a first radiator **908***a* formed of a conductive material having a plurality of coplanar windings. In one implementation, the first radiator **908***a* is in the form of a spiral. The first radiator **908***a* may be wound in a predetermined direction, such as clockwise (i.e., a left-hand wind) or counter-clockwise (i.e., a right-hand wind). While the first radiator **908***a* of the antenna **900** is depicted in FIG. **9**B as having a right-hand wind or a counter-clockwise rotational direction, it should be understood that the first radiator **908***a* of the antenna **900** may be provided with a left-hand wind or a clockwise rotational direction.
(301) Other implementations of the antenna **900** include implementation as a gain horn antenna, a Cassegrain antenna, an omnidirectional antenna, a horn lens antenna, a spot focus antenna, a waveguide probe antenna, a scalar feed horn antenna, a wide-angle scalar feed horn antenna, a trihedral antenna, and a conical horn antenna.
(302) Referring now to FIG. **10**, shown therein is another exemplary implementation of the antenna **900**. As shown in FIG. **10**, the antenna **900** may be implemented as a bifilar helix antenna. The bifilar helix antenna **900** generally comprises a ground plane **904***a* having a first differential pad **1100***a* and a second differential pad **1100***b* and a second radiator **908***b* mounted on the ground plane **904***a*. In some implementations, the bifilar helix antenna **900** may lack the ground plane **904***a*. The second radiator **908***b* is generally in the shape of a double helix and may have a first feed point **1104***a* electrically connected to the first differential pad **1100***a* and a second feed point **1104***b* electrically connected to the second differential pad **1100***b*. A first coaxial feedline **1108***a* and a second coaxial feedline **1108***b* may be electrically connected to the first differential pad **1100***a* and the second differential pad **1100***b*, respectively.
(303) In some implementations, the second radiator **908***b* may be configured to transmit and detect differential radiated signals. That is, in the transmit direction, the second radiator **908***b* may receive a first complementary antenna feed signal from the first feed point **1104***a* and a second complementary antenna feed signal from the second feed point **1104***b* and transmit the radiated signals based on the first complementary antenna feed signal and the second complementary antenna feed signal. Further, in the receive direction, the second radiator **908***b* may receive the radiated signals and provide the first complementary antenna output signal to the first feed point **1104***a* and the second complementary antenna output signal to the second feed point **1104***b*. In such implementations, the first complementary antenna output signal and the second complementary antenna output signal may be equal in magnitude but opposite in phase (i.e., out of phase by 180°).
(304) The second radiator **908***b* may be wound in a predetermined direction, such as clockwise or counter-clockwise. While the second radiator **908***b* of the bifilar helix antenna **900** is depicted in FIG. **9** as having a left-hand wind or a clockwise rotational direction, it should be understood that the second radiator **908***b* of the bifilar helix antenna **900** may be provided with a right-hand wind or a counter-clockwise rotational direction.
(305) The second radiator **908***b* may comprise a first radiator portion **1112** and a second radiator portion **1114**. The first radiator portion **1112** has a first end formed by the first feed point **1104***a* and a second end **1116** spaced a distance from the first feed point **1104***a*. The first radiator portion **1112** is in the form of a spiral (i.e., a helix shape). The second radiator portion **1114** has a third end formed by the second feed point **1104***b* and a fourth end **1118** spaced a distance from the second feed point **1104***b*. The second radiator portion **1114** is in the form of a spiral (i.e., a helix shape). The second end **1116** of the first radiator portion **1112** is connected to the fourth end **1118** of the second radiator portion **1114**.
(306) Referring now to FIGS. **11** and **12**, shown therein is another exemplary implementation of the bifilar helix antenna **900** shown in FIG. **10**. As shown in FIGS. **11** and **12**, in some implementations, a conductive cone **1200** may be provided surrounding the bifilar helix antenna

900 (i.e., such that the bifilar helix antenna **900** is enclosed within the conductive cone **1200**). The second radiator **908***b* may be wound in a predetermined direction, such as clockwise or counter-clockwise. While the second radiator **908***b* of the bifilar helix antenna **900** enclosed within the conductive cone **1200** is depicted in FIGS. **11** and **12** as having a left-hand wind or a clockwise rotational direction, it should be understood that the second radiator **908***b* of the bifilar helix antenna **900** enclosed within the conductive cone **1200** may be provided with a right-hand wind or a counter-clockwise rotational direction.

(307) The conductive cone **1200** may have a first end **1204***a*, a second end **1204***b* opposite the first end **1204***a*, and a sidewall **1208** extending between the first end **1204***a* and the second end **1204***b*. The sidewall **1208** may define a first opening **1212***a* at the first end **1204***a* and a second opening **1212***b* at the second end **1204***b*. As shown in FIGS. **11** and **12**, the first end **1204***a* of the conductive cone **1200** is generally provided with a diameter d.sub.4 shorter than a diameter d.sub.5 of the second end **1204***b* of the conductive cone **1200**.

(308) The bifilar helix antenna **900** enclosed within the conductive cone **1200** may be configured to transmit circularly polarized signals with a relatively high gain (e.g., more than 6 decibels relative to isotropic (dBi), such as 10 dBi, 12 dBi, 14 dBi, 15 dBi, 16 dBi, 18 dBi, or 20 dBi, for example). In the implementation shown in FIGS. **11** and **12**, the bifilar helix antenna **900** enclosed within the conductive cone **1200** may function as an efficient, wide-bandwidth polarizer. That is, the bifilar helix antenna **900** enclosed within the conductive cone **1200** may be configured to transmit circularly polarized RF signals with a high radiation efficiency (e.g., greater than 50%, such as 60%, 70%, 75%, 80%, 85%, 90%, or 95%, for example). Losses in radiation efficiency are generally due to losses in conductors or substrates. Further, the bifilar helix antenna **900** enclosed within the conductive cone **1200** may be configured to transmit circularly polarized signals with a wide bandwidth (e.g., greater than 10% of center frequency, such as 12%, 14%, 15%, 16%, 18%, 20%, 22%, 24%, or 25%, for example).

(309) The diameter of the bifilar helix antenna **900** may be less than the wavelength of the signals transmitted by the bifilar helix antenna **900**. In some implementations, the conductive cone **1200** may be constructed of a conductive material, such as aluminum, copper, silver, gold, other conductive metals, combinations thereof, and/or the like.

(310) It will be understood by persons having ordinary skill in the art that circularly polarized signals transmitted by a radiator **908** of a first particular one of the antennas **900** may be received only by a radiator **908** of a second particular one of the antennas **900** having the same rotational direction. That is, for example, the radiator **908** shown in FIG. **8**A and the first radiator **908***a* shown in FIG. **8**B are depicted as having a right-hand wind or a counter-clockwise rotational direction. As a result, circularly polarized RF signals transmitted by the radiator **908** shown in FIG. **9**A or the first radiator **908***a* shown in FIG. **9**B would have a right-hand circular polarization (RHCP). On the other hand, the second radiator **908***b* shown in FIGS. **10-12** is depicted as having a left-hand wind or a clockwise rotational direction. As a result, circularly polarized RF signals transmitted by the second radiator **908***b* shown in FIGS. **10-12** would have a left-hand circular polarization (LHCP).

(311) Because circularly polarized signals transmitted by a radiator **908** of a first particular one of the antennas **900** may be received only by a radiator **908** of a second particular one of the antennas **900** having the same rotational direction, circularly polarized RF signals transmitted by the radiator **908** as depicted in FIG. **9**A or the first radiator **908***a* as depicted in FIG. **9**B (i.e., RHCP RF signals) could not be received by the second radiator **908***b* as depicted in FIGS. **10-12**. Similarly, circularly polarized signals transmitted by the second radiator **908***b* as depicted in FIGS. **10-12** (i.e., LHCP RF signals) could not be received by the radiator **908** as depicted in FIG. **9**A or the first radiator **908***a* as depicted in FIG. **9**B. However, circularly polarized signals transmitted by the radiator **908** as depicted in FIG. **8**A (i.e., RHCP RF signals) could be received by the first radiator **908***a* as depicted in FIG. **9**B, and circularly polarized signals transmitted by the second radiator **908***b* as depicted in FIG. **10** (i.e., LHCP RF signals) could be received by the second radiator **908***b* as

depicted in FIGS. **11** and **12**.

(312) Referring now to FIG. **13**, shown therein is a diagrammatic view of an electric field **1300** produced by the bifilar helix antenna **900** enclosed within the conductive cone **1200** shown in FIGS. **11** and **12**. As illustrated in FIG. **13**, the bifilar helix antenna **900** enclosed within the conductive cone **1200** may be operable to produce the electric field **1300** such that a near-field region of the electric field **1300** and a far-field region of the electric field **1300** are established with a greater directivity than would be provided by conventional antennas. Further, the bifilar helix antenna **900** enclosed within the conductive cone **1200** may be operable to produce the electric field **1300** in a manner that does not interfere with the circular polarization of the circularly polarized radiated signals transmitted by the second radiator **908***b*.

(313) Referring now to FIG. **14**, shown therein is a diagrammatic view of a radiation pattern **1400** of the bifilar helix antenna **900** enclosed within the conductive cone **1200** shown in FIGS. **11** and **12**. The radiation pattern **1400** may correspond to a transmission signal having a frequency of 2,000 GHz and a phase of 0°. As shown in FIG. **14**, a first curve **1404** demonstrates an LHCP gain of the bifilar helix antenna **900** enclosed within the conductive cone **1200**, while a second curve **1408** demonstrates a total directivity of the bifilar helix antenna **900** enclosed within the conductive cone **1200**. A difference between the first curve **1404** and the second curve **1408** may indicate metal and polarization losses. As illustrated in FIG. **14** and as described above in relation to FIG. **13**, the bifilar helix antenna **900** enclosed within the conductive cone **1200** may be operable to produce the electric field **1300** such that a near-field region **1304** of the electric field **1300** and a far-field region **1308** of the electric field **1300** are established with a greater directivity than would be provided by conventional antennas.

(314) Referring now to FIGS. **15** and **16**, shown therein are side views of exemplary implementations of a non-uniform bifilar helix antenna **1500** (hereinafter, the “non-uniform antenna **1500**”) constructed in accordance with the present disclosure. Providing the antenna with a non-uniform design is effective because the size of the helix determines the frequency of operation. By varying characteristic dimensions of the helix, a wider band of frequencies may be effectively radiated.

(315) Similar to the bifilar helix antenna **900** described above, the non-uniform antenna **1500** may comprise the ground plane **904***a* having the first differential pad **1100***a* and the second differential pad **1100***b* and a non-uniform third radiator **908***c* mounted on the ground plane **904***a*. The third radiator **908***c* may have a plurality of turns **1504***a-n* including at least a first turn **1504***a* and a second turn **1504***b*. For purposes of clarity, only the first turn **1504***a* and the second turn **1504***b* are labeled with a reference character. The first turn **1504***a* may have a first characteristic dimension, while the second turn **1504***b* may have a second characteristic dimension different from the first characteristic dimension. The first turn **1504***a* may be adjacent to the second turn **1504***b* or non-adjacent to (i.e., spaced from) the second turn **1504***b*.

(316) In the implementation shown in FIG. **15**, the first turn **1504***a* has a first pitch p.sub.1, the second turn **1504***b* has a second pitch p.sub.2, and the first pitch p.sub.1 is less than the second pitch p.sub.2, the implementation shown in FIG. **15**, the first turn **1504***a* has the first pitch p.sub.1, the second turn **1504***b* has the second pitch p.sub.2, and the first pitch p.sub.1 is greater than the second pitch p.sub.2.

(317) In some implementations, the non-uniform antenna **1500** may lack the ground plane **904***a*. The third radiator **908***c* is generally in the shape of a double helix and may have the first feed point **1104***a* electrically connected to the first differential pad **1100***a* and the second feed point **1104***b* electrically connected to the second differential pad **1100***b*. The first coaxial feedline **1108***a* and the second coaxial feedline **1108***b* may be electrically connected to the first differential pad **1100***a* and the second differential pad **1100***b*, respectively.

(318) In some implementations, the third radiator **908***c* may be configured to emit and receive differential signals. That is, in the transmit direction, the third radiator **908***c* may receive a first

complementary signal from the first feed point **1104***a* and a second complementary signal from the second feed point **1104***b* and transmit the transmission signal. Further, in the receive direction, the third radiator **908***c* may receive the transmission signal and provide the first complementary signal to the first feed point **1104***a* and the second complementary signal to the second feed point **1104***b*. In such implementations, the first complementary signal and the second complementary signal may be equal in magnitude but opposite in phase (i.e., out of phase by 180°).

(319) The third radiator **908***c* may be wound in a predetermined direction, such as clockwise or counter-clockwise. While the third radiator **908***c* of the non-uniform antenna **1500** is depicted in FIGS. **15** and **16** as having a right-hand wind or a counter-clockwise rotational direction, it should be understood that the third radiator **908***c* of the non-uniform antenna **1500** may be provided with a left-hand wind or a clockwise rotational direction.

(320) The third radiator **908***c* may comprise the first radiator portion **1112** and the second radiator portion **1114**. The first radiator portion **1112** has the first end formed by the first feed point **1104***a* and the second end **1116** spaced a distance from the first feed point **1104***a*. The first radiator portion **1112** is in the form of a spiral (i.e., a helix shape). The second radiator portion **1114** has the third end formed by the second feed point **1104***b* and the fourth end **1118** spaced a distance from the second feed point **1104***b*. The second radiator portion **1114** is in the form of a spiral (i.e., a helix shape). While the second end **1116** and the fourth end **1118** are shown as being disconnected from each other, it should be understood that, in some implementations, the second end **1116** of the first radiator portion **1112** is connected to the fourth end **1118** of the second radiator portion **1114**.

(321) The non-uniform antenna **1500** provides a wider frequency response in comparison to uniform antennas existing in the prior art and the uniform bifilar helix antennas discussed herein. A mathematical equation for the helical shape of the non-uniform radiator **908***c* of the non-uniform antenna **1500** in three-dimensional space is shown in Table 1 below and in a graph **1700** shown in FIG. **17**, while the polarization discrimination of a uniform antenna across the frequency range between 0.80 THz and 1.40 THz is shown in a graph **1800** shown in FIG. **18**. As shown in FIGS. **17** and **18**, the polarization discrimination may be determined by subtracting the left-hand circular polarization directivity (i.e., DirLHCP) from the right-hand circular polarization directivity (i.e., DirRHCP). As shown in Table 1 and FIG. **16**, the right-hand circular polarization directivity (i.e., DirRHCP) of the non-uniform antenna **1500** may be relatively constant (i.e., 11.5 dBi±1 dBi) in the frequency range between 0.80 THz and 1.40 THz. Furthermore, as shown in FIG. **17**, the polarization discrimination (i.e., DirRHCP-DirLHCP) of the non-uniform antenna **1500** remains above 25 dB across the frequency range between 0.80 THz and 1.40 THz. Conversely, as shown in FIG. **18**, the polarization discrimination (i.e., DirRHCP-DirLHCP) of a uniform antenna dips below 25 dB at the band edges and slightly below 25 dB in the midband range.

(322) TABLE-US-00001 TABLE 1 Mathematical Equation for a Helical Shape of the Non-Uniform Radiator 908c of the Non-Uniform Antenna 1500 in Three-Dimensional Space X(t) 41 * cos(t) [μm] Y(t) 41 * sin(t) [μm] Z(t) 0.293 * t * (t + 25) [μm] start(t) 0 end(t) 25.13

(323) Referring now to FIGS. **18** and **20**, shown therein are side views of more exemplary implementations of the non-uniform antenna **1500** shown in FIGS. **15** and **16**. For purposes of clarity, the differential pads **1100** and the feed points **1104** are not labeled with a reference character in FIGS. **18** and **19**. In the implementations shown in FIGS. **19** and **20**, the first characteristic dimension and the second characteristic dimension are not pitches, but diameters. In the implementation shown in FIG. **19**, the first turn **1504***a* has a first diameter d.sub.1, the second turn **1504***b* has a second diameter d.sub.2, and the first diameter d.sub.1 is less than the second diameter d.sub.2. In the implementation shown in FIG. **20**, the first turn **1504***a* has the first diameter d.sub.1, the second turn **1504***b* has the second diameter d.sub.2, and the first diameter d.sub.1 is greater than the second diameter d.sub.2.

(324) Varying the diameters d.sub.a-n of the turns **1504** of the third radiator **908***c* rather than the pitches p.sub.a-n of the turns **1504** of the third radiator **908***c* may be advantageous in different

bands or with different ground plane dimensions, wire dimensions, etc.

(325) It should be understood that the third radiator **908**_c_ and/or the non-uniform antenna **1500** may be included in place of any of the respective radiators **908** and/or antennas **900** described herein. Further, it should be understood that, while the second turn **1504**_b_ is shown as being directly adjacent to the first turn **1504**_a_, there may be one or more turns in between the first turn **1504**_a_ and the second turn **1504**_b_. Finally, it should be understood that, while the first turn **1504**_a_ is shown as being directly adjacent to the ground plane **904**_a_, there may be one or more turns in between the ground plane **904**_a_ and the first turn **1504**_a_.

(326) Referring now to FIGS. **21** and **22**A-**22**C, shown therein is a differential waveguide probe antenna **2100** constructed in accordance with the present disclosure. The differential waveguide probe antenna **2100** is configured to generate and transmit the transmission signal. Conversely, the differential waveguide probe antenna **2100** is further configured to receive the transmission signal. The differential waveguide probe antenna **2100** comprises a pair of waveguide probes **2104** including a first waveguide probe **2104**_a_ and a second waveguide probe **2104**_b_.

(327) In some implementations, the differential waveguide probe antenna **2100** may further comprise an intermediary waveguide **2108** configured to propagate the transmission signal. In such implementations, the differential waveguide probe antenna **2100** may be further configured to generate and transmit the transmission signal into the intermediary waveguide **2108**. Conversely, in such implementations, the differential waveguide probe antenna **2100** may be further configured to receive the transmission signal from the intermediary waveguide **2108**.

(328) The intermediary waveguide **2108** may have a first end **2112**_a_, a second end **2112**_b_ (the first end **2112**_a_ and the second end **2112**_b_, collectively, the “ends **2112**”) opposite the first end **2112**_a_, and a surface **2116** extending between the ends **2112**. In some implementations, a back reflector **2118** may abut the first end **2112**_a_. The surface **2116** may be constructed of a metal and may have a diameter d.sub.a less than two wavelengths of the transmission signal at 10 THz (or a maximum frequency in the frequency band occupied by the transport network **200**) (i.e., 60 μm) and greater than one-half wavelength at 300 GHz (or a minimum frequency in the frequency band occupied by the transport network **200**) (i.e., 0.5 mm). The intermediary waveguide **2108** may be constructed as such in order to ensure that one or more intended waveguide modes are established. That is, were the intermediary waveguide **2108** to be constructed at a smaller size, the one or more intended waveguide modes may not be able to propagate, and were the intermediary waveguide **2108** to be constructed at a larger size, one or more unintended waveguide modes may be excited. In some implementations, the one or more intended waveguide modes of the intermediary waveguide **2108** sufficiently matches the one or more intended waveguide modes of the hollow waveguide **208** such that a coupling loss between the intermediary waveguide **2108** and the hollow waveguide **208** is minimized (e.g., the coupling loss is in a range between 0.1 dB and 5.0 dB).

(329) The waveguide probes **2104** may be positioned on opposite sides of the surface **2116** of the intermediary waveguide **2108** and may extend into the intermediary waveguide **2108** toward each other, but may be spaced a distance from each other. The waveguide probes **2104** may thus establish a strong electrical field in line with the one or more intended waveguide mode. Each of the waveguide probes **2104** may be excited with the transmission signal. In some implementations, each of the waveguide probes **2104** may be excited with the transmission signal at an equal strength and/or an opposite phase. That is, the waveguide probes **2104** may be configured to receive the transmission signal as a differential signal having a first complementary signal and a second complementary signal and generate and transmit the transmission signal in the electromagnetic wave form. Conversely, the waveguide probes **2104** may be further configured to receive the transmission signal and provide the transmission signal as a differential signal having a first complementary signal and a second complementary signal.

(330) In some implementations, the intermediary waveguide **2108** may have a flared end at the second end **2112**_b_ configured to facilitate a mode transition between the intermediary waveguide

2108 and the hollow waveguide **208**. In such implementations, the surface **2116** at the flared end may have a diameter d.sub.b greater than the diameter d.sub.a. In some such implementations, the flared end may be formed integrally with the intermediary waveguide **2108**. However, in other such implementations, the flared end may be constructed as a horn **2120** separate from but coupled to the intermediary waveguide **2108**. The horn **2120** may have a first end **2124***a* abutting the second end **2112***b* of the intermediary waveguide **2108**, a second end **2124***b* (the first end **2124***a* and the second end **2124***b*, collectively, the "ends **2124**") opposite the first end **2124***a*, and a curved surface **2128** extending between the ends **2124**. The curved surface **2128** at the first end **2124***a* may have a diameter d.sub.c equal to the diameter d.sub.a. The differential waveguide probe antenna **2100** may be configured to transmit the transmission signal with a wide (i.e., greater than 50%) bandwidth into the hollow waveguide **208** at least in part because an energy contribution from each of the waveguide probes **2104** effectively cancels out the higher-order, unintended waveguide modes of the other waveguide probe **2104**. A polarization discrimination of the differential waveguide probe antenna **2100** across a frequency range between 0.60 THz and 1.80 THz is shown in a graph **2500** shown in FIG. **22**D.

(331) Referring now to FIGS. **23**, **24**A, and **24**B, shown therein is an exemplary implementation of a differential tapered antenna **2600** constructed in accordance with the present disclosure. The differential tapered antenna **2600** is configured to generate and transmit the transmission signal in the electromagnetic wave form—and, conversely, receive the transmission signal in the electromagnetic wave form—and comprises a pair of conductors including a first conductor **2604***a* and a second conductor **2604***b* (collectively, the "conductors **2604**") spaced a distance d.sub.d from the first conductor **2604***a*.

(332) The differential tapered antenna **2600** may be similar in some respects to a tapered slot antenna and in some respects to a ridged horn antenna. However, the differential tapered antenna **2600** differs from such antennas due to the differential tapered antenna **2600** having a differential launch and being coupled into the hollow waveguide **208** such that the transmission signal has multiple waveguide modes.

(333) In the implementation shown in FIGS. **23**, **24**A, and **24**B, the intermediary waveguide **2108** has a first planar, yet longitudinally directed curved surface **2608***a* and a second planar, yet longitudinally directed curved surface **2608***b* (collectively, the "curved surfaces **2608**") bordering a space **2612**. In the implementation shown in FIGS. **23**, **24**A, and **24**B, the conductors **2604** collectively define the curved surfaces **2608** of the intermediary waveguide **2108** and form the space **2612** between the conductors **2604**. As described above, the intermediary waveguide **2108** is configured to propagate the transmission signal in the electromagnetic wave form. In such implementations, the differential tapered antenna **2600** may be further configured to generate and transmit the transmission signal into the intermediary waveguide **2108** and receive the transmission signal from the intermediary waveguide **2108**.

(334) In some implementations, the distance d.sub.d between the first conductor **2604***a* and the second conductor **2604***b* at the first end **2112***a* of the intermediary waveguide **2108** is less than two wavelengths of the transmission signal at 10 THz (or the maximum frequency in the frequency band occupied by the transport network **200**) and greater than one-half wavelength at 300 GHz (or the minimum frequency in the frequency band occupied by the transport network **200**). The distance d.sub.d may be selected to establish a single waveguide mode for the frequency of the transmission signal. In some implementations, a distance d.sub.e between the conductors **2604** at the second end **2112***b* of the intermediary waveguide **2108** is greater than the distance d.sub.d. This tapered shape may establish a continuously scaled geometry which enables an ultra-wide (i.e., greater than 50%) bandwidth. As energy launches down the conductors **2604**, the one or more intended waveguide mode is established between the conductors **2604** and subsequently launches into the intermediary waveguide **2108**.

(335) In some implementations, each of the conductors **2604** may be fed with the transmission

signal at an equal strength and/or an opposite phase. That is, the conductors **2604** may be configured to receive the transmission signal as a differential signal having a first complementary signal and a second complementary signal and generate and transmit the transmission signal in the electromagnetic wave form. Conversely, the conductors **2604** may be further configured to receive the transmission signal and provide the transmission signal as a differential signal having a first complementary signal and a second complementary signal.

(336) A thickness and a width of the transmission lines at the feed point may be selected to establish a characteristic impedance matched to the receiver and/or driver. Persons having ordinary skill in the art will understand how to perform such calculations. As shown in FIG. **24**B, the differential tapered antenna **2600** may further comprise one or more ground connection, such as a first ground connection **2800***a* and a second ground connection **2800***b*. A polarization discrimination of the differential tapered antenna **2600** across a frequency range between 0.50 THz and 2.00 THz is shown in a graph **2900** shown in FIG. **24**C.

(337) Referring now to FIG. **25**, shown therein is an exemplary implementation of a differential microstrip patch antenna **3000** constructed in accordance with the present disclosure. The differential microstrip patch antenna **3000** is configured to generate and transmit the transmission signal in the electromagnetic wave form—and, conversely, receive the transmission signal in the electromagnetic wave form—and comprises a pair of microstrip patch antennas including a first microstrip patch antenna **3004***a* and a second microstrip patch antenna **3004***b* (collectively, the "microstrip patch antennas **3004**") spaced a distance d.sub.f from the first microstrip patch antenna **3004***a*.

(338) In some implementations, the differential microstrip patch antenna **3000** may further comprise the horn **2120** having the first end **2124***a* proximal to the microstrip patch antennas **3004**, the second end **2124***b* distal to the microstrip patch antennas **3004**, and the curved surface **2128** extending between the ends **2124**. The curved surface **2028** at the first end **2124***a* may have the diameter d.sub.c, and the curved surface **2028** at the second end **2124***b* may have the diameter d.sub.b greater than the diameter d.sub.c.

(339) In some implementations, each of the microstrip patch antennas **3004** may be fed with the transmission signal at an equal strength and/or an opposite phase. That is, the microstrip patch antennas **3004** may be configured to receive the transmission signal as a differential signal having a first complementary signal and a second complementary signal and generate and transmit the transmission signal in the electromagnetic wave form. Conversely, the microstrip patch antennas **3004** may be further configured to receive the transmission signal and provide the transmission signal as a differential signal having a first complementary signal and a second complementary signal.

(340) The differential waveguide probe antenna **2100**, the differential tapered antenna **2600**, and the differential microstrip patch antenna **3000** are configured to generate the transmission signal in a linearly polarized form.

(341) Referring now to FIGS. **26** and **27**A-**27**C, shown therein is a diagrammatic view of an exemplary implementation of a single-ended waveguide probe antenna **3008** constructed in accordance with the present disclosure. In some implementations, the single-ended waveguide probe antenna **3008** may lack the second waveguide probe **2104***b*, thereby only comprising the first waveguide probe **2104***a*. Further, in some implementations, the surface **2116** of the intermediary waveguide **2108** may define an opening **3012** through which the first waveguide probe **2104***a* extends. As referenced above, in some implementations, the first end **2112***a* of the intermediary waveguide **2108** may serve as a back reflector.

(342) Referring now to FIGS. **28**, **29**A-**29**C, and **30**A-**30**C, shown therein are diagrammatic views of exemplary implementations of a slot antenna **5800** constructed in accordance with the present disclosure. As shown in FIGS. **28**, **29**A-**29**C, and **30**A-**30**C, the slot antenna **5800** may include the ground plane **904** disposed between the intermediary waveguide **2108** and the back reflectors **2118**.

In some implementations, the ground plane **904** may define one or more slots **3016** (e.g., a first slot **3016***a* shown in FIGS. **28** and **29**A-C and a second slot **3016***b* shown in FIGS. **30**A-**30**C) (hereinafter, the “slots **3016**”).

(343) Referring now to FIG. **31**, shown therein is a diagrammatic view of an exemplary implementation of a transport network **3100** (hereinafter, the “network **3100**”) constructed in accordance with the present disclosure. The network **3100** generally comprises a first network element **3102***a*, a second network element **3102***b*, and a hollow waveguide **3104** communicatively coupled to the first network element **3102***a* and the second network element **3102***b*.

(344) While the network **3100** is described herein as comprising the first network element **3102***a* transmitting signals and the second network element **3102***b* receiving such signals, it should be understood that the network **3100** may be bidirectional; that is, the network **3100** may further comprise the second network element **3102***b* transmitting signals and the first network element **3102***a* receiving such signals. Accordingly, in some such implementations, the hollow waveguide **3104** may be bidirectional (i.e., configured to simultaneously propagate signals in both directions); however, in other such implementations, the hollow waveguide **3104** comprises a first hollow waveguide (not shown) configured to propagate signals in a first direction (e.g., from the first network element **3102***a* to the second network element **3102***b*), and a second hollow waveguide (not shown) configured to propagate signals in a second direction opposite the first direction (e.g., from the second network element **3102***b* to the first network element **3102***a*).

(345) The first network element **3102***a* generally comprises one or more transmitter **3106** (hereinafter, the “transmitter **3106**” or, collectively, the “transmitters **3106**”) and a transmitter antenna array **3108**. The transmitter **3106** may include transmitter circuitry configured to generate a plurality of channel signals **3112**, such as a first channel signal **3112***a* and a second channel signal **3112***b* shown in FIG. **30**. The transmitter antenna array **3108** may comprise a plurality of transmitter antennas **3116**, such as a first transmitter antenna **3116***a* and a second transmitter antenna **3116***b* shown in FIG. **30**.

(346) The channel signals **3112** may have input data encoded with a modulation format and a carrier frequency in a range between 300 GHz and 10 THz. That is, the first channel signal **3112***a* and the second channel signal **3112***b* may have first input data encoded with a first modulation format and a first carrier frequency in the range between 300 GHz and 10 THz. The first channel signal **3112***a* and the second channel signal **3112***b* are preferably identical signals carrying the same data with the same modulation format and at a same frequency, with the exception that the first channel signal **3112***a* and second channel signal **3112***b* may be phase shifted relative to one another to change a polarization angle of an electromagnetic wave generated by the transmitter antenna array **3108** as discussed below. The second channel signal **3112***b* may have second input data (e.g., the same as the first input data) encoded with a second modulation format and a second carrier frequency in the range between 300 GHz and 10 THz.

(347) Each of the modulation formats described herein may be selected from a group consisting of: intensity-modulation (IM)/direct-detection (DD) (IM/DD); non-return-to-zero modulation (NRZ); pulse-amplitude-modulation-n (PAMn); IM-PAMn; m-quadrature-amplitude-modulation (mQAM); and single-sideband modulation (SSB). In implementations wherein one or more of the modulation formats is PAMn or IM-PAMn, n may be a power of 2 (e.g., 2, 4, 8, 16, 32, 64, etc.). Similarly, in implementations wherein one or more of the modulation formats is mQAM, m may be a power of 2 greater than or equal to 4 (e.g., 4, 8, 16, 32, 64, etc.).

(348) In some implementations, the first modulation format and the second modulation format are the same modulation format. In some implementations, the first carrier frequency and the second carrier frequency have the same carrier frequency in the range between 300 GHz and 10 THz.

(349) The first transmitter antenna **3116***a* and the second transmitter antenna **3116***b* of the transmitter antenna array **3108** may be configured to receive the channel signals **3112** and transmit a plurality of wireless signals **3120**, such as a first wireless signal **3120***a* and a second wireless

signal **3120***b* shown in FIG. **30**. That is, the first transmitter antenna **3116***a* may be configured to receive the first channel signal **3112***a* having the first input data encoded with the first modulation format and the first carrier frequency and transmit the first wireless signal **3120***a* having the first input data encoded with the first modulation format and the first carrier frequency, while the second transmitter antenna **3116***b* may be configured to receive the second channel signal **3112***b* having the second input data encoded with the second modulation format and the second carrier frequency and transmit a second wireless signal **3120***b* having the second input data encoded with the second modulation format and the second carrier frequency.

(350) The first transmitter antenna **3116***a* may be configured to induce a first circular polarization into the first wireless signal **3120***a*. In some implementations, the first circular polarization is a left-hand circular polarization (LHCP). However, in other implementations, the first circular polarization may be a right-hand circular polarization (RHCP). Similarly, the second transmitter antenna **3116***b* may be configured to induce a second circular polarization into the second wireless signal **3120***b*, wherein the second circular polarization is orthogonal to the first circular polarization. Thus, in implementations where the first circular polarization is an LHCP, the second circular polarization is an RHCP. However, in implementations in which the first circular polarization is an RHCP, the second circular polarization is an LHCP.

(351) The first transmitter antenna **3116***a* and the second transmitter antenna **3116***b* are positioned adjacent to each other such that the first wireless signal **3120***a* and the second wireless signal **3120***b* interact to form a linearly polarized wireless signal **3124** having a linear polarization. In some implementations, the linear polarization is a horizontal linear polarization (HLP). In other implementations, the linear polarization is a vertical linear polarization (VLP). Persons having ordinary skill in the art will understand that an HLP or a VLP may have a polarization angle such that the polarization is not perfectly horizontal nor perfectly vertical.

(352) It should be understood that a circular polarization is generally composed of linear polarizations with a 90° phase shift, as shown in Equations (1) and (2) below:

(353) $E_{LHCP} = \frac{1}{\sqrt{2}}(E_x - jE_y)$ (1) $E_{RHCP} = \frac{1}{\sqrt{2}}(E_x + jE_y)$ (2)

Combining the two orthogonal circular polarizations results in a first linear polarization—an HLP in this example—as shown in Equation (3) below:

(354) $E_{LHCP} + E_{RHCP} = \frac{1}{\sqrt{2}}(E_x - jE_y) + \frac{1}{\sqrt{2}}(E_x + jE_y)$ (3)

$E_{LHCP} + E_{RHCP} = \frac{1}{\sqrt{2}}E_x - \frac{1}{\sqrt{2}}jE_y + \frac{1}{\sqrt{2}}E_x + \frac{1}{\sqrt{2}}jE_y E_{LHCP} + E_{RHCP} = \frac{2}{\sqrt{2}}E_x$

$E_{LHCP} + E_{RHCP} = \sqrt{2}E_x$

Due to the law of conservation of energy, the canceled j terms (i.e., −jE.sub.y and jE.sub.y) do not result in a loss of energy. Forming the linearly polarized wireless signal **3124** from multiple circularly polarized wireless signals provides high broadband polarization diversity, perhaps due to the polarization non-idealities of each individual antenna being canceled out during the operation.

(355) Further, combining the two orthogonal circular polarizations after applying a phase shift of 180° to the one of the circular polarizations—the RHCP in this example—results in a second linear polarization orthogonal to the first linear polarization—a VLP in this example—as shown in Equation (4) below:

(356) $E_{LHCP} + E_{RHCP\angle 180^\circ} = \frac{1}{\sqrt{2}}(E_x - jE_y) + \frac{1}{\sqrt{2}}(-E_x - jE_y)$ (4)

$E_{LHCP} + E_{RHCP} = \frac{1}{\sqrt{2}}E_x - \frac{1}{\sqrt{2}}jE_y - \frac{1}{\sqrt{2}}E_x - \frac{1}{\sqrt{2}}jE_y E_{LHCP} + E_{RHCP} = -\frac{2}{\sqrt{2}}E_y$

$E_{LHCP} + E_{RHCP} = -\sqrt{2}E_y$

However, applying a phase shift of 180° to the one of the circular polarizations as such may result in a decrease in the quality of the excited field.

(357) Alternatively, combining the two orthogonal circular polarizations after applying a physical phase shift of 90° to each of the circular polarizations—such as by physically rotating the first transmitter antenna **3116***a* and the second transmitter antenna **3116***b*—results in a third linear

polarization orthogonal to the first linear polarization—a VLP in this example—as shown in Equation (5) below:

(358) $E_{\text{LHCP .Math. }90^\circ} + E_{\text{RHCP .Math. }90^\circ} = \frac{1}{\sqrt{2}}(E_y + jE_x) + \frac{1}{\sqrt{2}}(E_y - jE_x)$ (5)

$$E_{LHCP} + E_{RHCP} = \frac{1}{\sqrt{2}}E_y + \frac{1}{\sqrt{2}}jE_x + \frac{1}{\sqrt{2}}E_y - \frac{1}{\sqrt{2}}jE_x E_{LHCP} + E_{RHCP} = \frac{2}{\sqrt{2}}E_y$$

$$E_{LHCP} + E_{RHCP} = \sqrt{2}E_y$$

Finally, it will be understood by persons having ordinary skill in the art that applying a phase shift to each of the circular polarizations in a range between 0° and 180° may result in the polarization having a polarization angle such that the polarization is not perfectly horizontal nor perfectly vertical.

(359) The second network element **3102***b* generally comprises a receiver antenna array **3128** and one or more receiver **3132** (hereinafter, the "receiver **3132**" or, collectively, the "receivers **3132**"). The receiver antenna array **3128** may comprise a plurality of receiver antennas **3136**, such as a first receiver antenna **3136***a* and a second receiver antenna **3136***b* shown in FIG. **30**. The receiver **3132** may include receiver circuitry configured to extract the input data from the channel signals **3112**. That is, the receiver circuitry may be configured to extract the first input data from the first channel signal **3112***a* and the second input data from the second channel signal **3112***b*, which as discussed above is preferably the same input data.

(360) The first receiver antenna **3136***a* and the second receiver antenna **3136***b* of the receiver antenna array **3128** may be configured to receive the wireless signals **3120** and generate the channel signals **3112**. That is, the first receiver antenna **3136***a* may be configured to receive the first wireless signal **3120***a* having the first input data encoded with the first modulation format and the first carrier frequency and generate the first channel signal **3112***a* having the first input data encoded with the first modulation format and the first carrier frequency, while the second receiver antenna **3136***b* may be configured to receive the second wireless signal **3120***b* having the second input data encoded with the second modulation format and the second carrier frequency and generate the second channel signal **3112***b* having the second input data encoded with the second modulation format and the second carrier frequency.

(361) The first receiver antenna **3136***a* may be configured to receive wireless signals having the first circular polarization. Similarly, the second receiver antenna **3136***b* may be configured to receive wireless signals having the second circular polarization.

(362) Referring now to FIG. **32**A, shown therein is a diagrammatic view of an exemplary implementation of the transmitter **3106** shown in FIG. **31**. As described above, the transmitter **3106** may include transmitter circuitry configured to generate the channel signals **3112**, such as the first channel signal **3112***a* and the second channel signal **3112***b* shown in FIG. **32**A. Accordingly, the transmitter **3106** may comprise a plurality of channel signal generators **3200** configured to generate the channel signals **3112**, such as a first channel signal generator **3200***a* configured to generate the first channel signal **3112***a* and a second channel signal generator **3200***b* configured to generate the second channel signal **3112***b* shown in FIG. **32**A.

(363) Misalignment of the transmitter antenna array **3108** with the receiver antenna array **3128** greater than a certain amount (e.g., 3.2°) may cause a decrease in power in the intended polarization by cos(θ).sup.2 and an increase in power in an unintended polarization by sin(θ).sup.2, where theta is the misalignment angle thereby causing a decrease in polarization diversity. To address this challenge, in some implementations, the transmitter **3106** may further comprise a phase-shift circuit **3204** configured to induce a phase shift in the first channel signal **3112***a* relative to the second channel signal **3112***b* to induce a polarization angle in the linearly polarized wireless signal **3124**. As shown in FIG. **31**, in such implementations, the phase-shift circuit **3204** may be further configured to receive a polarization signal **3208** and induce the phase shift in the first channel signal **3112***a* relative to the second channel signal **3112***b* based at least in part upon the polarization signal **3208**. Inducing the phase shift in the first channel signal **3112***a* relative to the

second channel signal **3112***b* based at least in part upon the polarization signal **3208** may have the effect of aligning the transmitter antenna array **3108** of the first network element **3102***a* with the receiver antenna array **3128** of the second network element **3102***b* so as to maximize received power at the receiver **3132**.
(364) Referring now to FIG. **32**B, shown therein is a diagrammatic view of an exemplary implementation of the receiver **3132** shown in FIG. **31**. As described above, the receiver circuitry may be configured to extract the first input data from the first channel signal **3112***a* and the second input data from the second channel signal **3112***b*. Accordingly, the receiver **3132** may comprise a plurality of channel signal generators **3212** configured to generate the channel signals **3112**, such as a first channel signal generator **3212***a* configured to generate the first channel signal **3112***a* and a second channel signal generator **3212***b* configured to generate the second channel signal **3112***b* shown in FIG. **32**B.
(365) As described above, misalignment of the transmitter antenna array **3108** with the receiver antenna array **3128** greater than a certain amount (e.g., 3.2°) may cause a decrease in power in the intended polarization by cos(θ).sup.2 and an increase in power in an unintended polarization by sin(θ).sup.2, thereby causing a decrease in polarization diversity. To address this challenge, in some implementations, the receiver **3132** may further comprise aa polarization signal generator **3216** configured to generate the polarization signal **3208** based on a polarization angle between the first channel signal **3112***a* and the second channel signal **3112***b*. The polarization signal generator **3216** may include power measurement circuitry for measuring the power of the first channel signal **3112***a* and the second channel signal **3112***b.*
(366) To calibrate the polarization angle induced by the phase-shift circuit **3204**, a series of first channel signals **3112***a* and second channel signals **3112***b* having known phase shifts relative to one another can be supplied to the transmitter antenna array **3108** and subsequently received by the receiver antenna array **3128**. The first channel signals **3112***a* and the second channel signals **3112***b* are received by the receiver **3132** and the power, for example, is analyzed by the polarization signal generator **3216** to generate the polarization signals **3208**. The polarization signal generator **3216** may include a receiver processor **3220** executing logic operable to search for the strongest signal and use this information regarding the strongest signal to generate the polarization signals **3208** and a receiver communication unit **3224** that transmits the polarization signals **3208** to the phase-shift circuit **3204**, which may include a transmitter communication unit **3228** that receives the polarization signals **3208** from the polarization signal generator **3216**. The polarization signals **3208** are correlated with particular ones of the first channel signals **3112***a* and the second channel signals **3112***b* having the known phase shifts and may be indicative of the power received by receiver **3132**. The phase-shift circuit **3204** may include a transmitter processor **3232** executing logic operable to analyze the polarization signals **3208** and select the phase shift for the first channel signal **3112***a* and the second channel signal **3112***b* that delivers the most power to the receiver **3132**. Then, the selected phase shift is used by the phase-shift circuit **3204** thereafter for forming the first channel signal **3112***a* and the second channel signal **3112***b*. This calibration procedure can be accomplished on a periodic basis, such as hourly, daily, and/or the like.
(367) Referring now to FIG. **33**, shown therein is a diagrammatic view of an exemplary implementation of the transmitter antenna array **3108** shown in FIG. **31**. It should be understood that the receiver antenna array **3128** may be similar in construction and function to the transmitter antenna array **3108**, except that the receiver antennas **3136** of the receiver antenna array **3128** may be wound in in an opposite direction to their respective counterparts (i.e., the transmitter antennas **3116**) of the transmitter antenna array **3108**.
(368) As described above, the transmitter antenna array **3108** may comprise the transmitter antennas **3116**, which may include the first transmitter antenna **3116***a* and the second transmitter antenna **3116***b* shown in FIG. **33**. The transmitter antennas **3116** of the transmitter antenna array **3108** are shown in FIG. **33** as being arranged in a 1×2 grid pattern.

(369) In some implementations, a distance d between the first transmitter antenna **3116***a* and the second transmitter antenna **3116***b* may be equal to 1.5 times the wavelength λ. However, in other implementations, the distance d between the first transmitter antenna **3116***a* and the second transmitter antenna **3116***b* may be a number greater than or less than 1.5 times the wavelength λ. For example, in some implementations, the distance d between the first transmitter antenna **3116***a* and the second transmitter antenna **3116***b* may be less than the wavelength λ. However, in other implementations, the distance d between the first transmitter antenna **3116***a* and the second transmitter antenna **3116***b* may be a multiple of the wavelength λ.
(370) Referring now to FIG. **34**, shown therein is a diagrammatic view of another exemplary implementation of a transport network **3100***a* (hereinafter, the "network **3100***a*") constructed in accordance with the present disclosure. The network **3100***a* generally comprises a third network element **3102***c*, a fourth network element **3102***d*, and the hollow waveguide **3104** communicatively coupled to the third network element **3102***c* and the fourth network element **3102***d*.
(371) Unlike the first network element **3102***a* shown in FIG. **31**, the third network element **3102***c* generally comprises a plurality of the transmitters **3106**, such as a first transmitter **3106***a* and a second transmitter **3106***b* shown in FIG. **34**, and a transmitter antenna array **3108***a*. Similarly, unlike the second network element **3102***b* shown in FIG. **31**, the fourth network element **3102***d* generally comprises a receiver antenna array **3128***a* and a plurality of the receivers **3132**, such as a first receiver **3132***a* and a second receiver **3132***b* shown in FIG. **34**.
(372) The transmitter antenna array **3108***a* may be similar to the transmitter antenna array **3108** shown in FIG. **31**, except that the first transmitter antenna **3116***a* and the second transmitter antenna **3116***b* form a first transmitter antenna pair, and the transmitter antenna array **3108***a* further comprises a third transmitter antenna **3116***c* and a fourth transmitter antenna **3116***d* forming a second transmitter antenna pair. While the transmitter antenna array **3108***a* is described herein as comprising four of the transmitter antennas **3116**, it should be understood that the transmitter antenna array **3108***a* may comprise a number of the transmitter antennas **3116** greater or less than four.
(373) The first transmitter **3106***a* may be similar to the transmitter **3106** shown in FIG. **31**, while the second transmitter **3106***b* may be similar to the transmitter **3106** shown in FIG. **30** except that the channel signals **3112** transmitted by the second transmitter **3106***b* may include the third channel signal **3112***c* and the fourth channel signal **3112***d*.
(374) As described above, the channel signals **3112** may have the input data encoded with a particular modulation format and a particular carrier frequency in the range between 300 GHz and 10 THz. That is, as described above, the first channel signal **3112***a* may have the first input data encoded with the first modulation format and the first carrier frequency in the range between 300 GHz and 10 THz, and the second channel signal **3112***b* may have the second input data encoded with the second modulation format and the second carrier frequency in the range between 300 GHz and 10 THz. Similarly, the third channel signal **3112***c* may have third input data encoded with a third modulation format and a third carrier frequency in the range between 300 GHz and 10 THz, and the fourth channel signal **3112***d* may have fourth input data encoded with a fourth modulation format and a fourth carrier frequency in the range between 300 GHz and 10 THz.
(375) In some implementations, two or more of the first modulation format, the second modulation format, the third modulation format, and the fourth modulation format are the same modulation format. For example, the first modulation format and the second modulation format can be the same modulation format. Further, the third modulation format and the fourth modulation format can be the same modulation format.
(376) In some implementations, two or more of the first carrier frequency, the second carrier frequency, the third carrier frequency, and the fourth carrier frequency are the same carrier frequency in the range between 300 GHz and 10 THz. For example, the first carrier frequency and the second carrier frequency may be the same carrier frequency; and the third carrier frequency and

the fourth carrier frequency may be the same carrier frequency. In some implementations, two or more of the first carrier frequency and the second carrier frequency may be different from the third carrier frequency, and the fourth carrier frequency.

(377) As described above, the transmitter antennas **3116** of the transmitter antenna array **3108***a* may be configured to receive the channel signals **3112** and transmit the wireless signals **3120**, such as the first wireless signal **3120***a* and the second wireless signal **3120***b*, as well as a third wireless signal **3120***c* and a fourth wireless signal **3120***d*, shown in FIG. **34**. That is, the third transmitter antenna **3116***c* may be configured to receive the third channel signal **3112***c* having the third input data encoded with the third modulation format and the third carrier frequency and transmit the third wireless signal **3120***c* having the third input data encoded with the third modulation format and the third carrier frequency, and the fourth transmitter antenna **3116***d* may be configured to receive the fourth channel signal **3112***d* having the fourth input data encoded with the fourth modulation format and the fourth carrier frequency and transmit the fourth wireless signal **3120***d* having the fourth input data encoded with the fourth modulation format and the fourth carrier frequency.

(378) As described above, the first transmitter antenna **3116***a* may be configured to induce the first circular polarization into the first wireless signal **3120***a*, and the second transmitter antenna **3116***b* may be configured to induce the second circular polarization into the second wireless signal **3120***b*, wherein the second circular polarization is orthogonal to the first circular polarization. Similarly, the third transmitter antenna **3116***c* may be configured to induce a third circular polarization into the third wireless signal **3120***c*, and the fourth transmitter antenna **3116***d* may be configured to induce a fourth circular polarization into the fourth wireless signal **3120***d*, wherein the fourth circular polarization is orthogonal to the third circular polarization.

(379) The first wireless signal **3120***a* and the second wireless signal **3120***b* may interact to form a first linearly polarized wireless signal **3124***a* having the first linear polarization. Similarly, the third wireless signal **3120***c* and the fourth wireless signal **3120***d* may interact to form a second linearly polarized wireless signal **3124***b* having a second linear polarization, wherein the second linear polarization is orthogonal to the first linear polarization. The first linearly polarized wireless signal **3124***a* may be similar to the linearly polarized wireless signal **3124** shown in FIG. **30**. Similarly, the second linearly polarized wireless signal **3124***b* may be similar to the linearly polarized wireless signal **3124** shown in FIG. **31**.

(380) In some implementations, the first linearly polarized wireless signal **3124***a* has a first polarization angle, and the second linearly polarized wireless signal **3124***b* has a second polarization angle, wherein the first polarization and angle and the second polarization angle are offset by a number of degrees within a range between 86.8° and 93.2°.

(381) The receiver antenna array **3128***a* may be similar to the receiver antenna array **3128** shown in FIG. **31**, except that the first receiver antenna **3136***a* and the second receiver antenna **3136***b* form a first receiver antenna pair, and the receiver antenna array **3128***a* further comprises a third receiver antenna **3136***c* and a fourth receiver antenna **3136***d* forming a second receiver antenna pair. While the receiver antenna array **3128***a* is described herein as comprising four of the receiver antennas **3136**, it should be understood that the receiver antenna array **3128***a* may comprise an even number of the receiver antennas **3136** greater or less than four.

(382) The first receiver **3132***a* may be similar to the receiver **3132** shown in FIG. **31**, while the second receiver **3132***b* may be similar to the receiver **3132** shown in FIG. **30** except that the channel signals **3112** received by the second receiver **3132***b* may include the third channel signal **3112***c* and the fourth channel signal **3112***d*.

(383) As described above, the receiver antennas **3136** of the receiver antenna array **3128***a* may be configured to receive the wireless signals **3120** and generate the channel signals **3112**. That is, the third receiver antenna **3136***c* may be configured to receive the third wireless signal **3120***c* having the third input data encoded with the third modulation format and a third carrier frequency and generate the third channel signal **3112***c* having the third input data encoded with the third

modulation format and the third carrier frequency, and the fourth receiver antenna **3136***d* may be configured to receive the fourth wireless signal **3120***d* having the fourth input data encoded with the fourth modulation format and the fourth carrier frequency and generate the fourth channel signal **3112***d* having the fourth input data encoded with the fourth modulation format and the fourth carrier frequency.

(384) As described above, the first receiver antenna **3136***a* may be configured to receive the first wireless signal **3120***a* having the first input data encoded with the first modulation format and the first carrier frequency and generate the first channel signal **3112***a* having the first input data encoded with the first modulation format and the first carrier frequency, while the second receiver antenna **3136***b* may be configured to receive the second wireless signal **3120***b* having the second input data encoded with the second modulation format and the second carrier frequency and generate the second channel signal **3112***b* having the second input data encoded with the second modulation format and the second carrier frequency. Similarly, the third receiver antenna **3136***c* may be configured to receive the third wireless signal **3120***c* having the third input data encoded with the third modulation format and the third carrier frequency and generate the third channel signal **3112***c* having the third input data encoded with the third modulation format and the third carrier frequency, while the fourth receiver antenna **3136***d* may be configured to receive the fourth wireless signal **3120***d* having the fourth input data encoded with the fourth modulation format and the fourth carrier frequency and generate the fourth channel signal **3112***d* having the fourth input data encoded with the fourth modulation format and the fourth carrier frequency.

(385) Referring now to FIG. **35**A, shown therein is a diagrammatic view of an exemplary implementation of the transmitter antenna array **3108***a* shown in FIG. **34**. It should be understood that the receiver antenna array **3128***a* may be similar to the transmitter antenna array **3108***a* including the receiver antennas **3136** being would in the same direction as the transmitter antennas **3116**. In some implementations, the receiver antennas **3136** of the receiver antenna array **3128***a* may be wound in an opposite direction to their respective counterparts (i.e., the transmitter antennas **3116**) of the transmitter antenna array **3108***a*.

(386) As described above, the transmitter antenna array **3108***a* may comprise the transmitter antennas **3116**, which may include the first transmitter antenna **3116***a*, the second transmitter antenna **3116***b*, the third transmitter antenna **3116***c*, and the fourth transmitter antenna **3116***d* shown in FIG. **35**A. The transmitter antennas **3116** of the transmitter antenna array **3108***a* are shown in FIG. **35**A as being arranged in an n×m grid pattern, where n and m are both equal to two. While the transmitter antennas **3116** of the transmitter antenna array **3108***a* shown in FIG. **35**A are arranged in a 2×2 grid pattern, it should be understood that the transmitter antennas **3116** of the transmitter antenna array **3108***a* may be arranged in any n×m grid pattern wherein n and m are both a multiple of two. In some implementations, a distance d between each of the transmitter antennas **3116** and the nearest neighbor of such transmitter antenna **3116** may be equal to 1.5λ. However, in other implementations, the distance d between each of the transmitter antennas **3116** and the nearest neighbor of such transmitter antenna **3116** may be a number greater than or less than 1.5λ.

(387) Referring now to FIG. **35**B, shown therein is a diagrammatic view of another exemplary implementation of the transmitter antenna array **3108***a* shown in FIG. **34**. However, it should be understood that the receiver antenna array **3128***a* may be similar to the transmitter antenna array **3108***a* including the receiver antennas **3136** being would in the same direction as the transmitter antennas **3116**. In some implementations, he receiver antennas **3136** of the receiver antenna array **3128***a* may be wound in in an opposite direction to their respective counterparts (i.e., the transmitter antennas **3116**) of the transmitter antenna array **3108***a*.

(388) As described above, the transmitter antenna array **3108***a* may comprise the transmitter antennas **3116**, which may include the first transmitter antenna **3116***a*, the second transmitter antenna **3116***b*, the third transmitter antenna **3116***c*, and the fourth transmitter antenna **3116***d* shown in FIG. **35**B. The transmitter antennas **3116** of the transmitter antenna array **3108***a* are shown in

FIG. **35**B as being arranged in 1×m grid pattern. While the transmitter antennas **3116** of the transmitter antenna array **3108***a* shown in FIG. **35**B are arranged in a 1×4 grid pattern, it should be understood that the transmitter antennas **3116** of the transmitter antenna array **3108***a* may be arranged in any 1×m grid pattern wherein m is a multiple of four.

(389) As described above, in some implementations, a distance d between each of the transmitter antennas **3116** and the nearest neighbor of such transmitter antenna **3116** may be equal to 1.5 times the wavelength λ. However, in other implementations, the distance d between each of the transmitter antennas **3116** and the nearest neighbor of such transmitter antenna **3116** may be a number greater than or less than 1.5 times the wavelength λ. For example, in some implementations, the distance d between each of the transmitter antennas **3116** and the nearest neighbor of such transmitter antenna **3116** may be less than the wavelength λ. However, in other implementations, the distance d between each of the transmitter antennas **3116** and the nearest neighbor of such transmitter antenna **3116** may be a multiple of the wavelength λ.

(390) Referring now to FIG. **36**, shown therein is a diagrammatic view of an exemplary implementation of a method **3600** of using the network **3100** in accordance with the present disclosure. As shown in FIG. **36**, the method **3600** generally comprises the step of: transmitting the first wireless signal **3120***a* and the second wireless signal **3120***b* simultaneously from the transmitter antenna array **3108** into the hollow waveguide **3104**, the first wireless signal **3120***a* and the second wireless signal **3120***b* having the input data (i.e., the first input data and the second input data, respectively) encoded with the modulation format (i.e., the first modulation format and the second modulation format, respectively) and having the carrier frequency (i.e., the first carrier frequency and the second carrier frequency, respectively) in the range between 300 GHz and 10 THz, the first wireless signal **3120***a* having an LHCP and the second wireless signal **3120***b* having an RHCP such that the first wireless signal **3120***a* interacts with the second wireless signal **3120***b* to form the linearly polarized wireless signal **3124** (step **3604**).

(391) In some implementations, the method **3600** further comprises, before the step of transmitting (step **3604**): generating the first wireless signal **3120***a* by applying the first channel signal **3112***a* to the first transmitter antenna **3116***a* of the transmitter antenna array **3108**; generating the second wireless signal **3120***b* by applying the second channel signal **3112***b* to the second transmitter antenna **3116***b* of the transmitter antenna array **3108**; and inducing a phase shift in the first channel signal **3112***a* relative to the second channel signal **3112***b* to induce a polarization angle in the linearly polarized wireless signal **3124**.

(392) In some such implementations, the method **3600** further comprises receiving the polarization signal **3208**, wherein inducing is defined further as inducing the phase shift in the first channel signal **3112***a* relative to the second channel signal **3112***b* based at least in part upon the polarization signal **3208**.

(393) In some implementations, the transmitter antenna array **3108** is the transmitter antenna array **3108***a*, the input data is the first input data, the modulation format is the first modulation format, the carrier frequency is the first carrier frequency, and the linearly polarized wireless signal **3124** is the first linearly polarized wireless signal **3124***a*. In such implementations, the method **3600** may further comprise: transmitting the third wireless signal **3120***c* and the fourth wireless signal **3120***d* simultaneously from the transmitter antenna array **3108***a* into the hollow waveguide **3104**, the third wireless signal **3120***c* having the third input data encoded with the third modulation format and having the third carrier frequency in the range between 300 GHz and 10 THz, and the fourth wireless signal **3120***d* having the fourth input data encoded with the fourth modulation format and having the fourth carrier frequency in the range between 300 GHz and 10 THz, the third wireless signal **3120***c* having an LHCP and the fourth wireless signal **3120***d* having an RHCP such that the third wireless signal **3120***c* interacts with the fourth wireless signal **3120***d* to form a second linearly polarized wireless signal **3124***b*.

(394) In some such implementations, the method **3600** further comprises, before the step of

transmitting the third wireless signal **3120***c*: generating the third wireless signal **3120***c* by applying the third channel signal **3112***c* to the third transmitter antenna **3116***c* of the transmitter antenna array **3108***a*; generating the fourth wireless signal **3120***d* by applying the fourth channel signal **3112***d* to the fourth transmitter antenna **3116***d* of the transmitter antenna array **3108***a*; and inducing a phase shift in the third channel signal **3112***c* relative to the fourth channel signal **3112***d* to induce a polarization angle in the second linearly polarized wireless signal **3124***b*.
(395) In some such implementations, the method **3600** further comprises receiving the polarization signal **3208**, wherein inducing is defined further as inducing the phase shift in the third channel signal **3112***c* relative to the fourth channel signal **3112***d* based at least in part upon the polarization signal **3208**.
(396) In some implementations, the first linearly polarized wireless signal **3124***a* has a first polarization angle and the second linearly polarized wireless signal **3124***b* has a second polarization angle, and the first polarization angle and the second polarization angle are offset within a range between 86.8° and 93.2°.
(397) In some implementations, the step of transmitting is defined further as: transmitting the first wireless signal **3120***a* by the first transmitter antenna **3116***a*; transmitting the second wireless signal **3120***b* by the second transmitter antenna **3116***b*; transmitting the third wireless signal **3120***c* by the third transmitter antenna **3116***c*; and transmitting the fourth wireless signal **3120***d* by the fourth transmitter antenna **3116***d*; wherein the first transmitter antenna **3116***a*, the second transmitter antenna **3116***b*, the third transmitter antenna **3116***c*, and the fourth transmitter antenna **3116***d* are arranged in an n×m grid pattern where n and m are at least two, as shown in FIG. **35**A.
(398) In other implementations, the step of transmitting is defined further as: transmitting the first wireless signal **3120***a* by the first transmitter antenna **3116***a*; transmitting the second wireless signal **3120***b* by the second transmitter antenna **3116***b*; transmitting the third wireless signal **3120***c* by the third transmitter antenna **3116***c*; and transmitting the fourth wireless signal **3120***d* by the fourth transmitter antenna **3116***d*; wherein the first transmitter antenna **3116***a*, the second transmitter antenna **3116***b*, the third transmitter antenna **3116***c*, and the fourth transmitter antenna **3116***d* are arranged in a 1×m grid pattern where m is at least four, as shown in FIG. **35**B.
(399) Referring now to FIG. **37**A, shown therein is a diagrammatic view of an exemplary implementation of a dual-polarization (dual-pol) transmitter network element **3700***a* (hereinafter, the “transmitter network element **3700***a*”) constructed in accordance with the present disclosure. As shown in FIG. **37**A, the transmitter network element **3700***a* may comprise a dual-pol hollow waveguide referred to hereinafter by way of example as a dual-pol hollow waveguide **3704** configured to simultaneously propagate signals having a first polarization and a second polarization different from the first polarization, one or more modulator **3708***a-n* (e.g., a first modulator **3708***a* and a second modulator **3708***b* shown in FIG. **37**A) (collectively, the “modulators **3708**”) configured to generate a first channel signal **3712***a* and a second channel signal **3712***b* (collectively, the “channel signals **3712**”), and one or more antenna **3716***a-n* (e.g., a first antenna **3716***a* and a second antenna **3716***b* shown in FIG. **37**A) (collectively, the “antennas **3716**”) configured to receive the first channel signal **3712***a* and the second channel signal **3712***b* and to couple the first channel signal **3712***a* into the dual-pol hollow waveguide **3704** with the first polarization and the second channel signal **3712***b* into the dual-pol hollow waveguide **3704** with the second polarization. The antennas **3716** may be similar to the antennas **900** described above.
(400) The channel signals **3712** are also referred to herein as the “transmitted channel signals **3712**” (i.e., the “first transmitted channel signal **3712***a*” and the “second transmitted channel signal **3712***b*”) when viewed from the perspective of the transmitter network element **3700***a* and the “received channel signals **3712**” (i.e., the “first received channel signal **3712***a*” and the “second received channel signal **3712***b*”) when viewed from the perspective of the receiver network element **3700***b* (shown in FIG. **37**B). However, it should be understood that the received channel signals **3712** may have the same data and the same RF frequency as the transmitted channel signals **3712**,

though the received channel signals **3712** may exhibit linear distortions caused by the dual-pol hollow waveguide **3704** and/or the antennas **3716**.

(401) In implementations where the antennas **3716** include the first antenna **3716***a* and the second antenna **3716***b*, the first antenna **3716***a* may be configured to apply the first polarization to the first transmitted channel signal **3712***a* as the first antenna **3716***a* couples the first transmitted channel signal **3712***a* into the dual-pol hollow waveguide **3704**, and the second antenna **3716***b* may be configured to apply the second polarization to the second transmitted channel signal **3712***b* as the second antenna **3716***b* couples the second transmitted channel signal **3712***b* into the dual-pol hollow waveguide **3704**.

(402) The first transmitted channel signal **3712***a* may have first data encoded in a first modulation format, and the second transmitted channel signal **3712***b* may have second data encoded in a second modulation format. In some implementations, the first modulator **3708***a* may be configured to generate the first transmitted channel signal **3712***a* and the second modulator **3708***b* may be configured to generate the second transmitted channel signal **3712***b* such that the first transmitted channel signal **3712***a* has a first channel frequency and the second transmitted channel signal **3712***b* has a second channel frequency, wherein the first channel frequency and the second channel frequency are in a range between 300 GHz and 10 THz. In such implementations, the modulators **3708** may be described as performing "direct modulation". However, in other implementations, each of the first modulator **3708***a* and the second modulator **3708***b* may comprise an intermediate frequency (IF) modulator configured to generate the first transmitted channel signal **3712***a* and the second transmitted channel signal **3712***b*, respectively, such that the first transmitted channel signal **3712***a* has a first intermediate frequency less than the first channel frequency and the second transmitted channel signal **3712***b* has a second intermediate frequency less than the second channel frequency. In such implementations, each of the first modulator **3708***a* and the second modulator **3708***b* may further comprise one or more up-converter (not shown) configured to receive the first transmitted channel signal **3712***a* and the second transmitted channel signal **3712***b*, respectively, and up-convert the first transmitted channel signal **3712***a* and the second transmitted channel signal **3712***b* such that the first transmitted channel signal **3712***a* has the first channel frequency and the second transmitted channel signal **3712***b* has the second channel frequency, wherein the first channel frequency and the second channel frequency are in the range between 300 GHz and 10 THz. In such implementations, the modulators **3708** may be described as performing "IF modulation".

(403) The first modulation format and the second modulation format may be selected from a group consisting of: intensity-modulation (IM)/direct-detection (DD) (IM/DD); non-return-to-zero modulation (NRZ); pulse-amplitude-modulation-n (PAMn); IM-PAMn; m-quadrature-amplitude-modulation (mQAM); single-sideband-modulation (SSB); quadrature-phase-shift-keying (QPSK); and differential-detection QPSK (DQPSK). In some implementations, the first modulation format is the same the second modulation format. However, in other implementations, the first modulation format is different from the second modulation format. In implementations wherein one or more of the first modulation format and the second modulation format is PAMn or IM-PAMn, n may be a power of 2 (e.g., 2, 4, 8, 16, 32, 64, etc.). Similarly, in implementations wherein one or more of the first modulation format and the second modulation format is mQAM, m may be a power of 2 greater than or equal to 4 (e.g., 4, 8, 16, 32, 64, etc.). It should be understood that 4QAM (i.e., mQAM in implementations where m is equal to 4) may be the same as QPSK.

(404) In some implementations, the modulators **3708** may be further configured to receive one or more input signal **3720** (e.g., a first input signal **3720***a*, a second input signal **3720***b*, a third input signal **3720***c*, and a fourth input signal **3720***d* shown in FIG. **37**A) (collectively, the "input signals **3720**"). In some such implementations, as shown in FIG. **37**A, the first input signal **3720***a* and the second input signal **3720***b* may form a first pair of input signals **3724***a*, and the third input signal **3720***c* and the fourth input signal **3720***d* may form a second pair of input signals **3724***b*. In such

implementations, the first pair of input signals **3724***a* may have the first data encoded in the first modulation format, and the second pair of input signals **3724***b* may have the second data encoded in the second modulation format. Further, in such implementations, the first input signal **3720***a* and the third input signal **3720***c* may be I input signals, and the second input signal **3720***b* and the fourth input signal **3720***d* may be Q input signals.

(405) The first polarization may be orthogonal to the second polarization. In some implementations, the first polarization is a left-hand circular polarization (LHCP). In such implementations, the second polarization is a right-hand circular polarization (RHCP). In other implementations, the first polarization is a horizontal linear polarization (HLP). In such implementations, the second polarization is a vertical linear polarization (VLP). Persons having ordinary skill in the art will understand that the HLP and the VLP may have a rotation such that the HLP is not perfectly horizontal and the VLP is not perfectly vertical.

(406) Referring now to FIG. **37**B, shown therein is a diagrammatic view of another exemplary implementation of a dual-pol receiver network element **3700***b* (hereinafter, the "receiver network element **3700***b*") constructed in accordance with the present disclosure. As shown in FIG. **37**B, the receiver network element **3700***b* may comprise the dual-pol hollow waveguide **3704**, the antennas **3716** (e.g., the first antenna **3716***a* and the second antenna **116***b* shown in FIG. **37**B) configured to receive the first received channel signal **3712***a* and the second received channel signal **3712***b* from the dual-pol hollow waveguide **3704**, and one or more demodulator **3728***a-n* (e.g., a first demodulator **3728***a* and a second demodulator **3728***b* shown in FIG. **37**B) (collectively, the "demodulators **3728**") configured to receive the first received channel signal **3712***a* and the second received channel signal **3712***b*.

(407) In some implementations, the demodulators **3728** may be further configured to produce one or more output signal **3732** (e.g., a first output signal **3732***a*, a second output signal **3732***b*, a third output signal **3732***c*, and a fourth output signal **3732***d* shown in FIG. **37**B) (collectively, the "output signals **3732**") based on the first received channel signal **3712***a* and the second received channel signal **3712***b* (i.e., the first output signal **3732***a* and the second output signal **3732***b* being produced based on the first received channel signal **3712***a*, and the third output signal **3732***c* and the fourth output signal **3732***d* being produced based on the second received channel signal **3712***b*).

(408) In some implementations, as shown in FIG. **37**B, the first output signal **3732***a* and the second output signal **3732***b* may form a first pair of output signals **3736***a*. Similarly, the third output signal **3732***c* and the fourth output signal **3732***d* may form a second pair of output signals **3736***b*. In such implementations, the first output signal **3732***a* and/or the third output signal **3732***c* may have in-phase (I) data, and the second output signal **3732***b* and/or the fourth output signal **3732***d* may have quadrature (Q) data. The output signals **3732** may be configured for data detection (i.e., extraction of the first data and the second data).

(409) In implementations where the antennas **3716** include the first antenna **3716***a* and the second antenna **3716***b*, the first antenna **3716***a* may be configured to receive RF signals having the first polarization, and the second antenna **3716***b* may be configured to receive RF signals having the second polarization.

(410) In some implementations, the first demodulator **3728***a* and the second demodulator **3728***b* may be configured to demodulate the first received channel signal **3712***a* and the second received channel signal **3712***b*, respectively, to produce the first pair of output signals **3736***a* (i.e., the first output signal **3732***a* and the second output signal **3732***b*) based on the first received channel signal **3712***a* and the second pair of output signals **3736***b* (i.e., the third output signal **3732***c* and the fourth output signal **3732***d*) based on the second received channel signal **3712***b* such that the first output signal **3732***a* and the second output signal **3732***b* of the first pair of output signals **3736***a* have the first channel frequency and the third output signal **3732***c* and the fourth output signal **3732***d* of the second pair of output signals **3736***b* have the second channel frequency in the range between 300 GHz and 10 THz. In such implementations, the demodulators **3728** may be described as performing

"direct demodulation". However, in other implementations, each of the first demodulator **3728***a* and the second demodulator **3728***b* may comprise one or more down-converter (not shown) configured to receive the first received channel signal **3712***a* and the second received channel signal **3712***b*, respectively, and down-convert the first received channel signal **3712***a* and the second received channel signal **3712***b* such that the first received channel signal **3712***a* has the first intermediate frequency less than the first channel frequency and the second received channel signal **3712***b* has the second intermediate frequency less than the second channel frequency. In such implementations, each of the first demodulator **3728***a* and the second demodulator **3728***b* may further comprise an IF demodulator configured to demodulate the first received channel signal **3712***a* and the second received channel signal **3712***b*, respectively, to produce the first pair of output signals **3736***a* (i.e., the first output signal **3732***a* and the second output signal **3732***b*) based on the first received channel signal **3712***a* and the second pair of output signal **3736***b* (i.e., the third output signal **3732***c* and the fourth output signal **3732***d*) based on the second received channels signal **3712***b*. In such implementations, the demodulators **3728** may be described as performing "IF demodulation".

(411) Referring now to FIG. **38**, shown therein is a diagrammatic view of a dual-pol signal **3800** comprising a plurality of wavelength-division multiplexed (WDM) signals **3804** in accordance with the present disclosure. The dual-pol signal **3800** is also referred to herein as the "transmitted dual-pol signal **3800**" when viewed from the perspective of the transmitter network element **3700***a* and the "received dual-pol signal **3800**" when viewed from the perspective of the receiver network element **3700***b* (shown in FIG. **37**B). However, it should be understood that the received dual-pol signal **3800** may have the same data and the same frequency as the transmitted dual-pol signal **3800**, though the received dual-pol signal **3800** may exhibit linear distortions caused by the dual-pol hollow waveguide **3704** and/or the antennas **3716**. Similarly, the WDM signals **3804** are also referred to herein as the "transmitted WDM signals **3804**" when viewed from the perspective of the transmitter network element **3700***a* and the "received WDM signals **3804**" when viewed from the perspective of the receiver network element **3700***b* (shown in FIG. **37**B). However, it should be understood that the received WDM signals **3804** may have the same data and the same frequency as the transmitted WDM signals **3804**, though the received WDM signals **3804** may exhibit linear distortions caused by the dual-pol hollow waveguide **3704** and/or the antennas **3716**.

(412) As will be discussed in greater detail below, in some implementations, the transmitter network element **3700***a* may be configured to transmit the transmitted dual-pol signal **3800** having the plurality of the transmitted WDM signals **3804** (e.g., a first transmitted WDM signal **3804***a* and a second transmitted WDM signal **3804***b* shown in FIG. **38**), wherein each of the transmitted WDM signals **3804** comprises a plurality of the transmitted channel signals **3712**, wherein each of the transmitted channel signals **3712** has a channel frequency in the range between 300 GHz and 10 THz. The transmitter network element **3700***a* may be configured to transmit a number of the transmitted WDM signals **3804** that is at least one (i.e., a single channel with a single polarization). Similarly, in some implementations, the receiver network element **3700***b* may be configured to receive the received dual-pol signal **3800** having the plurality of received WDM signals **3804**, wherein each of the received WDM signals **3804** comprises a plurality of the received channel signals **3712**, wherein each of the received channel signals **3712** has a channel frequency in the range between 300 GHz and 10 THz.

(413) The first WDM signal **3804***a* is shown in FIG. **38** as having a plurality of first channel signals **3712***a* (e.g., a first f.sub.1 channel signal **3712***a*-**1**, a first f.sub.2 channel signal **3712***a*-**2**, a first f.sub.3 channel signal **3712***a*-**3**, and a first f.sub.4 channel signal **3712***a*-**4** shown in FIG. **38**) (collectively, the "first channel signals **3712***a*"). Similarly, the second WDM signal **3804***b* is shown in FIG. **38** as having a plurality of second channel signals **3712***b* (e.g., a second f.sub.1 channel signal **3712***b*-**1**, a second f.sub.2 channel signal **3712***b*-**2**, a second f.sub.3 channel signal **3712***b*-**3**, and a second f.sub.4 channel signal **3712***b*-**4** shown in FIG. **38**) (collectively, the "second channel

signals **3712***b*”).
(414) As shown in FIG. **38**, the first f.sub.1 channel signal **3712***a***-1** and the second f.sub.1 channel signal **3712***b***-1** may have a first channel frequency f.sub.1, the first f.sub.2 channel signal **3712***a***-2** and the second f.sub.2 channel signal **3712***b***-2** may have a second channel frequency f.sub.2, the first f.sub.3 channel signal **3712***a***-3** and the second f.sub.3 channel signal **3712***b***-3** may have a third channel frequency f.sub.3, and the first f.sub.4 channel signal **3712***a***-4** and the second f.sub.4 channel signal **3712***b***-4** may have a fourth channel frequency f.sub.4. Each of the WDM signals **3804** may have a number of the channel signals **3712** that is at least one (i.e., a single channel with a single polarization).
(415) As shown in FIG. **38**, in some such implementations, adjacent ones of the first channel signals **3712***a* may be spaced apart 200 GHz from each other, and adjacent ones of the second channel signals **3712***b* may be spaced apart 200 GHz from each other. However, in other implementations, adjacent ones of the first channel signals **3712***a* may be spaced apart from each other in a range between 50 GHz and 400 GHz, and adjacent ones of the second channel signals **3712***b* and may be spaced apart from each other in the range between 50 GHz and 400 GHz.
(416) Referring now to FIG. **39**, shown therein is a diagrammatic view of an exemplary implementation of a dual-pol transport network **3900** constructed in accordance with the present disclosure. As shown in FIG. **39**, the dual-pol transport network **3900** may comprise a first dual-pol network element **3902***a*, a second dual-pol network element **3902***b*, and the dual-pol hollow waveguide **3704** extending between the first dual-pol network element **3902***a* and the second dual-pol network element **3902***b* configured to simultaneously propagate signals having the first polarization and the second polarization different from the first polarization.
(417) While the transport network **3900** is described herein as comprising the first dual-pol network element **3902***a* transmitting signals and the second dual-pol network element **3902***b* receiving such signals, it should be understood that the transport network **3900** may be bidirectional; that is, the transport network **3900** may further comprise the second dual-pol network element **3902***b* transmitting signals and the first dual-pol network element **3902***a* receiving such signals. Accordingly, in some such implementations, the dual-pol hollow waveguide **3704** may be bidirectional (i.e., configured to simultaneously propagate signals in both directions); however, in other such implementations, the dual-pol hollow waveguide **3704** comprises a first dual-pol hollow waveguide (not shown) configured to propagate signals in a first direction (e.g., from the first dual-pol network element **3902***a* to the second dual-pol network element **3902***b*), and a second dual-pol hollow waveguide (not shown) configured to propagate signals in a second direction opposite the first direction (e.g., from the second dual-pol network element **3902***b* to the first dual-pol network element **3902***a*). Nevertheless, the first dual-pol network element **3902***a* is also referred to herein as the “transmitter network element **3902***a*” and the second dual-pol network element **3902***b* is also referred to herein as the “receiver network element **3902***b*”.
(418) The transmitter dual-pol network element **3902***a* may comprise the plurality of modulators **3708** (e.g., a first f.sub.1 modulator **3708***a***-1**, a second f.sub.1 modulator **3708***b***-1**, a first f.sub.2 modulator **3708***a***-2**, a second f.sub.2 modulator **3708***b***-2**, a first f.sub.3 modulator **3708***a***-3**, a second f.sub.3 modulator **3708***b***-3**, a first f.sub.4 modulator **3708***a***-4**, and a second f.sub.4 modulator **3708***b***-4** shown in FIG. **39**), a first combiner **3904***a*, a second combiner **3904***b*, a third combiner **3904***c*, and one or more first dual-pol antenna **3716***c* (hereinafter, the “first dual-pol antenna **3716***c*”).
(419) The first f.sub.1 modulator **3708***a***-1**, the first f.sub.2 modulator **3708***a***-2**, the first f.sub.3 modulator **3708***a***-3**, and the first f.sub.4 modulator **3708***a***-4** may be collectively referred to as the “first modulators **3708***a*”, and the second f.sub.1 modulator **3708***b***-1**, the second f.sub.2 modulator **3708***b***-2**, the second f.sub.3 modulator **3708***b***-3**, and the second f.sub.4 modulator **3708***b***-4** may be collectively referred to as the “second modulators **3708***b*”. While four of the modulators **3708** are shown in FIG. **39**, it should be understood that the first modulators **3708***a* and the second

modulators **3708***b* may include a number of the modulators **3708** that is greater or less than four.

(420) Each of the first modulators **3708***a* may be configured to generate a particular one of the first transmitted channel signals **3712***a* having the first data encoded in the first modulation format. In some implementations, each of the first modulators **3708***a* may be configured to generate a particular one of the first transmitted channel signals **3712***a* such that each of the first transmitted channel signals **3712***a* has one of a plurality of distinct first channel frequencies in the range between 300 GHz and 10 THz. In such implementations, the first modulators **3708***a* may be described as performing "direct modulation". However, in other implementations, each of the first modulators **3708***a* may comprise an IF modulator configured to generate the particular one of the first transmitted channel signals **3712***a* such that each of the first transmitted channel signals **3712***a* has one of a plurality of distinct first intermediate frequencies less than the distinct first channel frequency of such first transmitted channel signal **3712***a*. In such implementations, each of the first modulators **3708***a* may further comprise one or more first up-converter (not shown) configured to receive the first transmitted channel signals **3712***a* and up-convert the first transmitted channel signals **3712***a* such that each of the first transmitted channel signals **3712***a* has the distinct first channel frequency of such first transmitted channel signal **3712***a*. In such implementations, the first modulators **3708***a* may be described as performing "IF modulation".

(421) In some implementations, each of the first modulators **3708***a* may be further configured to receive the first pair of input signals **3724***a*, wherein each of the first pairs of input signals **3724***a* has the first input signal **3720***a*, the second input signal **3720***b*, and the first data encoded in the first modulation format.

(422) Each of the second modulators **3708***b* may be configured to generate a particular one of the second transmitted channel signals **3712***b* having the second data encoded in the second modulation format. In some implementations, each of the second modulators **3708***b* may be configured to generate the particular one of the second transmitted channel signals **3712***b* such that each of the second transmitted channel signals **3712***b* has one of a plurality of distinct second channel frequencies in the range between 300 GHz and 10 THz. In such implementations, the second modulators **3708***b* may be described as performing "direct modulation". However, in other implementations, each of the second modulators **3708***b* may comprise an IF demodulator configured to generate the particular one of the second transmitted channel signals **3712***b* such that each of the second transmitted channel signals **3712***b* has one of a plurality of distinct second intermediate frequencies less than the distinct second channel frequency of such second transmitted channel signal **3712***b*. In such implementations, each of the second modulators **3708***b* may further comprise one or more second up-converter (not shown) configured to receive the second transmitted channel signals **3712***b* and up-convert the second transmitted channel signals **3712***b* such that each of the second transmitted channel signals **3712***b* has the distinct channel frequency of such second transmitted channel signal **3712***b*. In such implementations, the second modulators **3708***b* may be described as performing "IF modulation".

(423) The first combiner **3904***a* may be configured to receive the first transmitted channel signals **3712***a* from the first modulators **3708***a* and combine the first transmitted channel signals **3712***a* into the first transmitted WDM signal **3804***a*, and the second combiner **3904***b* may be configured to receive the second transmitted channel signals **3712***b* from the second modulators **3708***b* and combine the second transmitted channel signals **3712***b* into the second transmitted WDM signal **3804***b*. The third combiner **3904***c* may be configured to receive the first transmitted WDM signal **3804***a* and the second transmitted WDM signal **3804***b* and combine the first transmitted WDM signal **3804***a* and the second transmitted WDM signal **3804***b* into the transmitted dual-pol signal **3800**. One or more of the first combiner **3904***a*, the second combiner **3904***b*, and the third combiner **3904***c* may be multiplexers. In some implementations, one or more of the first combiner **3904***a* and the second combiner **3904***b* is a WDM combiner, and the third combiner **3904***c* is a polarization combiner.

(424) The first dual-pol antenna **3716***c* may be configured to receive the transmitted dual-pol signal **3800** from the third combiner **3904***c* and couple the transmitted dual-pol signal **3800** into the dual-pol hollow waveguide **3704** (i.e., the first transmitted WDM signal **3804***a* having the first polarization and the second transmitted WDM signal **3804***b* having the second polarization).
(425) The receiver dual-pol network element **3902***b* may comprise a plurality of demodulators **3728** (e.g., a first f.sub.1 demodulator **3728***a***-1**, a second f.sub.1 demodulator **3728***b***-1**, a first f.sub.2 demodulator **3728***a***-2**, a second f.sub.2 demodulator **3728***b***-2**, a first f.sub.3 demodulator **3728***a***-3**, a second f.sub.3 demodulator **3728***b***-3**, a first f.sub.4 demodulator **3728***a***-4**, and a second f.sub.4 demodulator **3728***b***-4** shown in FIG. **39**), a first splitter **3912***a*, a second splitter **3912***b*, a third splitter **3912***c*, and one or more second dual-pol antenna **3716***d* (hereinafter, the "second dual-pol antenna **3716***d*").
(426) The second dual-pol antenna **3716***d* may be configured to receive the received dual-pol signal **3800** (i.e., the first received WDM signal **3804***a* having the first polarization and the second received WDM signal **3804***b* having the second polarization) from the dual-pol hollow waveguide **3704**.
(427) The third splitter **3912***c* may be configured to receive the received dual-pol signal **3800** from the second dual-pol antenna **3716***d* and split the received dual-pol signal **3800** into the first received WDM signal **3804***a* and the second received WDM signal **3804***b*. The first splitter **3912***a* may be configured to receive the first received WDM signal **3804***a* from the third splitter **3912***c* and split the first received WDM signal **3804***a* into the plurality of first received channel signals **3712***a*. The second splitter **3912***b* may be configured to receive the second received WDM signal **3804***b* from the third splitter **3912***c* and split the second received WDM signal **3804***b* into the plurality of second received channel signals **3712***b*. One or more of the first splitter **3912***a*, the second splitter **3912***b*, and the third splitter **3912***c* may be demultiplexers. In some implementations, one or more of the first splitter **3912***a* and the second splitter **3912***b* is a WDM splitter, and the third splitter **3912***c* is a polarization splitter.
(428) The first f.sub.1 demodulator **3728***a***-1**, the first f.sub.2 demodulator **3728***a***-2**, the first f.sub.3 demodulator **3728***a***-3**, and the first f.sub.4 demodulator **3728***a***-4** are collectively referred to as the "first demodulators **3728***a*", and the second f.sub.1 demodulator **3728***b***-1**, the second f.sub.2 demodulator **3728***b***-2**, the second f.sub.3 demodulator **3728***b***-3**, and the second f.sub.4 demodulator **3728***b***-4** are collectively referred to as the "second demodulators **3728***b*". While four of the demodulators **3728** are shown in FIG. **39**, it should be understood that the first demodulators **3728***a* and the second demodulators **3728***b* may include a number of the demodulators **3728** that is greater or less than four.
(429) Each of the first demodulators **3728***a* may be configured to demodulate a particular one of the first received channel signals **3712***a* to produce the first pairs of output signals **3736***a* having the first data encoded in the first modulation format and configured for data detection. In some implementations, each of the first demodulators **3728***a* may be configured to demodulate the particular one of the first received channel signals **3712***a* having one of the plurality of distinct first channel frequencies in the range between 300 GHz and 10 THz. However, in other implementations, the receiver dual-pol network element **3902***b* may further comprise one or more first down-converter (not shown) configured to receive the first received channel signals **3712***a* and down-convert the first received channel signals **3712***a* such that each of the first received channel signals **3712***a* has one of the plurality of distinct first intermediate frequencies less than the distinct first channel frequency of such first received channel signal **3712***a*. In such implementations, each of the first demodulators **3728***a* may be configured to demodulate the particular one of the first received channel signals **3712***a* having one of the plurality of distinct first intermediate frequencies.
(430) Each of the second demodulators **3728***b* may be configured to demodulate a particular one of the second received channel signals **3712***b* to produce the second pair of output signals **3736***b* having the second data encoded in the second modulation format and configured for data detection.

In some implementations, each of the second demodulators **3728***b* may be configured to demodulate the particular one of the second received channel signals **3712***b* having one of the plurality of distinct second channel frequencies in the range between 300 GHz and 10 THz. However, in other implementations, the second dual-pol network element **3902***b* may further comprise one or more second down-converter (not shown) configured to receive the second received channel signals **3712***b* and down-convert the second received channel signals **3712***b* such that each of the second received channel signals **3712***b* has one of the plurality of distinct second intermediate frequencies less than the distinct second channel frequency of such second received channel signal **3712***b*. In such implementations, each of the second demodulators **3728***b* may be configured to demodulate the particular one of the second received channel signals **3712***b* having one of the plurality of distinct second intermediate frequencies.
(431) Referring now to FIG. **40**, in some implementations, the transmitter dual-pol network element **3902***a* may comprise a first antenna **3716***e* configured to transmit RF signals having the first polarization and a second antenna **3716***f* configured to transmit RF signals having the second polarization. In such implementations, the receiver dual-pol network element **3902***b* may comprise a third antenna **3716***g* configured to receive RF signals having the first polarization and a fourth antenna **3716***h* configured to receive RF signals having the second polarization. Each of the first antenna **3716***e*, the second antenna **3716***f*, the third antenna **3716***g*, and the fourth antenna **3716***h* may be single-polarization or dual-polarization antennas.
(432) Referring now to FIG. **41**, in some implementations, the transmitter dual-pol network element **3902***a* may comprise a plurality of first antennas **3716***e* (e.g., a first f.sub.1 antenna **3716***e***-1**, a first f.sub.2 antenna **3716***e***-2**, a first f.sub.3 antenna **3716***e***-3**, and a first f.sub.4 antenna **3716***e***-4**) (collectively, the "first antennas **3716***e*") and a plurality of second antennas **3716***f* (e.g., a second f.sub.1 antenna **3716***f***-1**, a second f.sub.2 antenna **3716***f***-2**, a second f.sub.3 antenna **3716***f***-3**, and a second f.sub.4 antenna **3716***f***-4**) (collectively, the "second antennas **3716***f*").
(433) Each of the first antennas **3716***e* may be configured to receive a particular one of the first transmitted channel signals **3712***a* and apply the first polarization to the particular one of the first transmitted channel signals **3712***a* as the particular one of the first transmitted channel signals **3712***a* is coupled into the dual-pol hollow waveguide **3704**, and each of the second antennas **3716***f* may be configured to receive a particular one of the second transmitted channel signals **3712***b* and apply the second polarization to the particular one of the second transmitted channel signals **3712***b* as the particular one of the second transmitted channel signals **3712***b* is coupled into the dual-pol hollow waveguide **3704**.
(434) Similarly, in such implementations, the receiver dual-pol network element **3902***b* may comprise a plurality of third antennas **3716***g* (e.g., a third f.sub.1 antenna **3716***g***-1**, a third f.sub.2 antenna **3716***g***-2**, a third f.sub.3 antenna **3716***g***-3**, and a third f.sub.4 antenna **3716***g***-4**) (collectively, the "third antennas **3716***g*") and a plurality of fourth antennas **3716***h* (e.g., a fourth f.sub.1 antenna **3716***h***-1**, a fourth f.sub.2 antenna **3716***h***-2**, a fourth f.sub.3 antenna **3716***h***-3**, and a fourth f.sub.4 antenna **3716***h***-4**) (collectively, the "fourth antennas **3716***h*").
(435) Each of the third antennas **3716***g* may be configured to receive a particular one of the first received channel signals **3712***a* having the first polarization from the dual-pol hollow waveguide **3704**, and each of the fourth antennas **3716***h* may be configured to receive a particular one of the second received channel signals **3712***b* having the second polarization from the dual-pol hollow waveguide **3704**.
(436) Referring now to FIG. **42**, shown therein is a diagrammatic view of an exemplary implementation of the first modulator **3708***a* shown in FIG. **37**A. However, it should be understood that any of the modulators **3708** described herein may be similar to the first modulator **3708***a* as shown in FIG. **42**. As shown in FIG. **42**, the first modulator **3708***a* may comprise a transmitter local oscillator (LO) **4200** configured to generate a transmitter LO signal **4204**, a transmitter phase-shifter **4208** configured to receive the transmitter LO signal **4204** and shift the phase of the

transmitter LO signal **4204** by a predetermined amount (e.g., 90°) to produce a quadrature LO signal **4212**, a first transmitter mixer **4216***a* configured to receive the transmitter LO signal **4204** and the first input signal **3720***a* and mix the transmitter LO signal **4204** with the first input signal **3720***a* to produce a first transmitter mixer output signal **4220***a*, a second transmitter mixer **4216***b* configured to receive the quadrature LO signal **4212** and the second input signal **3720***b* and mix the quadrature LO signal **4212** with the second input signal **3720***b* to produce a second transmitter mixer output signal **4220***b*, and a transmitter adder **4224** configured to receive the first transmitter mixer output signal **4220***a* and the second transmitter mixer output signal **4220***b* and combine the first transmitter mixer output signal **4220***a* and the second transmitter mixer output signal **4220***b* to produce the first transmitted channel signal **3712***a*.

(437) As described above, in some implementations, the first modulator **3708***a* may be configured to generate the first transmitted channel signal **3712***a* such that the first transmitted channel signal **3712***a* has a first channel frequency in a range between 300 GHz and 10 THz. However, in other implementations, the first modulator **3708***a* may be configured to generate the first transmitted channel signal **3712***a* such that the first transmitted channel signal **3712***a* has an intermediate frequency less than the first channel frequency. In such implementations, the first modulator **3708***a* may further comprise one or more up-converter (not shown) configured to receive the first transmitted channel signal **3712***a* and up-convert the first transmitted channel signal **3712***a* such that the first transmitted channel signal **3712***a* has the first channel frequency.

(438) Referring now to FIG. **43**, shown therein is a diagrammatic view of an exemplary implementation of the first demodulator **3728***a* shown in FIG. **37**B. However, it should be understood that any of the demodulators **3728** described herein may be similar to the first demodulator **3728***a* shown in FIG. **43**. As shown in FIG. **43**, the first demodulator **3728***a* may comprise a receiver LO **4300**—which may be a voltage-controlled oscillator (VCO)—configured to generate a receiver LO signal **4304** having an LO frequency within a predetermined range (e.g., within 1 GHz) of the first channel frequency of the RF carrier embedded in the first received channel signal **3712***a*, a receiver phase-shifter **4308** configured to receive the receiver LO signal **4304** and shift the phase of the receiver LO signal **4304** by a predetermined amount (e.g., 90°) to produce a quadrature LO signal **4312**, a first receiver mixer **4316***a* configured to receive the receiver LO signal **4304** and the first received channel signal **3712***a* and mix the receiver LO signal **4304** with the first received channel signal **3712***a* to produce a first receiver mixer output signal **4320***a*, a second receiver mixer **4316***b* configured to receive the quadrature LO signal **4312** and the first received channel signal **3712***a* and mix the quadrature LO signal **4312** with the first received channel signal **3712***a* to produce a second receiver mixer output signal **4320***b*, a first lowpass filter (LPF) **4324***a* configured to receive the first receiver mixer output signal **4320***a* and attenuate frequencies higher than a predetermined cutoff frequency to produce a first baseband signal **4322***a* (also referred to herein as “U(t)” or the “in-phase (I) channel”), a second LPF **4324***b* configured to receive the second receiver mixer output signal **4320***b* and attenuate frequencies higher than the predetermined cutoff frequency to produce a second baseband signal **4322***b* (also referred to herein as “V(t)” or the “quadrature (Q) channel”), a carrier recovery module **4332** configured to receive the first baseband signal **4322***a* and the second baseband signal **4322***b* and produce a carrier recovery control signal **4336** to cause an LO frequency and an LO phase of the receiver LO signal **4304** to match the first channel frequency and the first channel phase of the RF carrier embedded in the first received channel signal **3712***a*, a module **4340** having circuitry configured to form a pre-equalized output signal **4344** having a complex signal representation of U(t)+j*V(t), and an equalizer **4348** configured to equalize the pre-equalized output signal **4344** by applying a plurality of complex tap weights having a complex signal presentation of h.sub.i+j*h.sub.q to produce the first output signal **3732***a* configured for data detection, also in a complex form (i.e., having an I component and a Q component). The complex representation is mathematically convenient. The plurality of complex tap weights may be determined and/or adjusted based on a tap weight control

algorithm, such as Least Mean Squares (LMS), Zero Forcing (ZF), and/or the like. The operation of the equalizer **4348** may be described as convolution between the input signal (i.e., the pre-equalized output signal **4344**) and the equalizer transfer function—with a delay line finite impulse response (FIR) structure and the plurality of complex tap weights—in the time domain or the multiplication between the input signal (i.e., the pre-equalized output signal **4344**) and the equalizer transfer function—with a delay line FIR structure and the plurality of complex tap weights—in the frequency domain. Persons having ordinary skill in the art will understand that the equalizer transfer function may be determined based on the plurality of complex tap weights.

(439) It should be understood that the description above generally refers to implementations in which one or more of the receiver network elements **3700***a* (shown in FIG. **37**B), **3902***b* (shown in FIG. **39**) includes a coherent receiver. In other implementations, such as implementations in which one or more of the first modulation format and the second modulation format is DQPSK, the first demodulator **3728***a* may lack one or more of the carrier recovery module **4332** and the receiver LO **4300**; however, in such implementations, the first demodulator **3728***a* may further comprise a differential detection circuit.

(440) As described above, in some implementations, the first demodulator **3728***a* may be configured to demodulate the first received channel signal **3712***a* to produce the first output signal **3732***a* such that the first output signal **3732***a* has the first channel frequency in the range between 300 GHz and 10 THz. However, in other implementations, the first demodulator **3728***a* may further comprise one or more down-converter (not shown) configured to, prior to demodulating the first received channel signal **3712***a* to produce the first output signal **3732***a*, down-convert the first received channel signal **3712***a* such that the first received channel signal **3712***a* has an intermediate frequency less than the first channel frequency. In such implementations, the first demodulator **3728***a* may further comprise one or more down-converter (not shown) configured to receive the first received channel signal **3712***a* and down-convert the first received channel signal **3712***a* such that the first received channel signal **3712***a* has the intermediate frequency less than the first channel frequency.

(441) Referring now to FIG. **44**, shown therein is a diagrammatic view of another exemplary implementation of receiver network element **3700***b* constructed in accordance with the present disclosure. The receiver network element **3700***b* in the implementation shown in FIG. **44** generally comprises a first portion **4400***a* and a second portion **4400***b*. In some implementations, the first portion **4400***a* is an X-pol portion and the second portion **4400***b* is a Y-pol portion. In such implementations, the first received channel signal **3712***a* may be a received X-pol signal and the second received channel signal **3712***b* may be a received Y-pol signal. As shown in FIG. **44**, the receiver network element **3700***b* may comprise a first demodulator **3728***a*, a second demodulator **3728***b*, and an equalizer **4402**. As shown in FIG. **44**, the equalizer **4402** may comprise a plurality of complex equalizers **4404** (e.g., a first complex equalizer **4404***a*, a second complex equalizer **4404***b*, a third complex equalizer **4404***c*, and a fourth complex equalizer **4404***d* shown in FIG. **44**) and a plurality of adders **4406** (e.g., a first adder **4406***a* and a second adder **4406***b* shown in FIG. **44**).

(442) It should be understood that, although the first received channel signal **3712***a* (i.e., the X-pol signal) and the second received channel signal **3712***b* (i.e., the Y-pol signal) may use the same RF carrier, it may be necessary to include multiple carrier recovery modules (i.e., a first carrier recover module **4332-1** and a second carrier recovery module **4332-2** shown in FIG. **44**) in the receiver network element **3700***b*.

(443) The operation of module **4340-1** and module **4340-2** in FIG. **44** is similar to the module **4340** in FIG. **43**. The first complex equalizer **4404***a* may be configured to receive a first pre-equalized output signal **4344-1** produced by the first demodulator **3728***a* and a first complex tap weight h.sub.xx and multiply the first pre-equalized output signal **4344-1** and the first complex tap weight h.sub.xx to produce a first equalizer intermediate signal **4408***a*.

(444) The second complex equalizer **4404***b* may be configured to receive a second pre-equalized

output signal **4344-2** produced by the second demodulator **3728***b* and a second complex tap weight h.sub.yx and multiply the second pre-equalized output signal **4344-2** and the second complex tap weight h.sub.yx to produce a second equalizer intermediate signal **4408***b*.

(445) The third complex equalizer **4404***c* may be configured to receive the first pre-equalized output signal **4344-1** produced by the first demodulator **3728***a* and a third complex tap weight h.sub.xy and multiply the first pre-equalized output signal **4344-1** and the third complex tap weight h.sub.xy to produce a third equalizer intermediate signal **4408***c*.

(446) The fourth complex equalizer **4404***d* may be configured to receive the second pre-equalized output signal **4344-2** produced by the second demodulator **3728***b* and a fourth complex tap weight h.sub.yy and multiply the second pre-equalized output signal **4344-2** and the fourth complex tap weight h.sub.yy to produce a fourth equalizer intermediate signal **4408***d*.

(447) The first adder **4406***a* may be configured to receive the first equalizer intermediate signal **4408***a* produced by the first complex equalizer **4404***a* and the second equalizer intermediate signal **4408***b* produced by the second complex equalizer **4404***b* and add the first equalizer intermediate signal **4408***a* and the second equalizer intermediate signal **4408***b* to produce the first output signal **3732***a*. In some implementations, the first output signal **3732***a* is an equalized X-pol signal.

(448) The second adder **4406***b* may be configured to receive the third equalizer intermediate signal **4408***c* produced by the third complex equalizer **4404***c* and the fourth equalizer intermediate signal **4408***d* produced by the fourth complex equalizer **4404***d* and add the third equalizer intermediate signal **4408***c* and the fourth equalizer intermediate signal **4408***d* to produce the second output signal **3732***b*. In some implementations, the second output signal **3732***b* is an equalized Y-pol signal.

(449) It should be understood that, where the cross-pol discrimination of the system is outside of a predetermined range (e.g., between 16 dB and 25 dB, depending on the modulation format being used and the system link budget), the second complex tap weight h.sub.yx and the third complex tap weight h.sub.xy may be set to zero, which would cause the first portion **4400***a* and the second portion **4400***b* of the receiver network element **3700***b* in FIG. **44** to operate as two separate single-pol receiver network elements, such as is shown in FIG. **43**. In such a case, the receiver network element **3700***b* may operate more efficiently, thereby requiring less power.

(450) It should be understood that the implementation of the first demodulator **3728***a* shown in FIG. **43** and the implementation of the receiver network element **3700***b* shown in FIG. **44** are illustrative implementations provided as examples. It should be further understood that the approach described above may be referred to as an “analog approach”. Conversely, a “digital approach” may also be used instead which may include one or more ADC and a digital signal processor (DSP) configured to perform the demodulation and equalization described herein.

(451) In some implementations, the first demodulator **3728***a* and the second demodulator **3728***b* may be configured to demodulate the first channel signal **3712***a* and the second channel signal **3712***b*, respectively, to produce the first input signal **3720***a* and the second input signal **3720***b* such that the first input signal **3720***a* has the first channel frequency and the second input signal **3720***b* has the second channel frequency in the range between 300 GHz and 10 THz. However, in other implementations, the first demodulator **3728***a* may be configured to, prior to demodulating the first channel signal **3712***a* to produce the first input signal **3720***a*, down-convert the first channel signal **3712***a* such that the first channel signal **3712***a* has an intermediate frequency less than the first channel frequency. In such implementations, the first demodulator **3728***a* may further comprise one or more down-converter (not shown) configured to receive the first channel signal **3712***a* and down-convert the first channel signal **3712***a* such that the first channel signal **3712***a* has the intermediate frequency less than the first channel frequency. Similarly, in such implementations, the second demodulator **3728***b* may be configured to, prior to demodulating the second channel signal **3712***b* to produce the second input signal **3720***b*, down-convert the second channel signal **3712***b* such that the second channel signal **3712***b* has an intermediate frequency less than the

second channel frequency. In such implementations, the second demodulator **3728***b* may further comprise one or more down-converter (not shown) configured to receive the second channel signal **3712***b* and down-convert the second channel signal **3712***b* such that the second channel signal **3712***b* has the intermediate frequency less than the second channel frequency.

(452) Referring now to FIG. **45**, shown therein is a diagrammatic view of an exemplary implementation of a method **4500** of use in accordance with the present disclosure. As shown in FIG. **45**, the method **4500** generally comprises the step of: coupling, by one or more antenna **900**, **3716**, a first wavelength division multiplexed (WDM) signal **3804***a* into a hollow waveguide, e.g., the dual-pol hollow waveguide **3704** with a first polarization, and a second WDM signal **3804***b* into the hollow waveguide, e.g., the dual-pol hollow waveguide **3704** with a second polarization so as to simultaneously propagate RF signals having the first polarization and the second polarization through the dual-pol hollow waveguide **3704**, the first WDM signal **3804***a* having a first channel frequency in a range between 300 Gigahertz (GHz) and 10 Terahertz (THz), and the second WDM signal **3804***b* having a second channel frequency in a range between 300 GHz and 10 THz (step **4504**).

(453) In some implementations, coupling the first WDM signal **3804***a* and the second WDM signal **3804***b* into the dual-pol hollow waveguide **3704** (step **4504**) includes coupling the first WDM signal **3804***a* and the second WDM signal **3804***b* having a modulation format selected from a group consisting of: intensity-modulation (IM)/direct-detection (DD) (IM/DD); non-return-to-zero modulation (NRZ); pulse-amplitude-modulation-n (PAMn); IM-PAMn; m-quadrature-amplitude-modulation (mQAM); quadrature-phase-shift-keying (QPSK); differential-detection QPSK (DQPSK); and single-sideband modulation (SSB).

(454) In some implementations, coupling the first WDM signal **3804***a* and the second WDM signal **3804***b* into the dual-pol hollow waveguide **3704** (step **4504**) includes coupling the first WDM signal **3804***a* to a first antenna **3716***e* configured to apply the first polarization and coupling the second WDM signal **3804***b* to a second antenna **3716***f* configured to apply the second polarization, the first antenna **3716***e* being separate from the second antenna **3716***f*.

(455) In some implementations, the first polarization is a left-hand circular polarization (LHCP), and the second polarization is a right-hand circular polarization (RHCP). In some implementations, the first polarization is a horizontal linear polarization (HLP), and the second polarization is a vertical linear polarization (VLP).

(456) In some implementations, wherein coupling the first WDM signal **3804***a* and the second WDM signal **3804***b* into the dual-pol hollow waveguide **3704** (step **4504**) includes coupling the first WDM signal **3804***a* and the second WDM signal **3804***b* to a dual-pol antenna (e.g., the first dual-pol antenna **3716***c*) configured to simultaneously transmit RF signals having the first polarization and the second polarization into the dual-pol hollow waveguide **3704**.

(457) In some implementations, the method **4500** further comprises the step of combining a plurality of first channel signals **3712***a* to form the first WDM signal **3804***a*, the first channel signals having a plurality of channel frequencies in the range between 300 GHz and 10 THz, and wherein at least some of the first channel signals are encoded with data. In some such implementations, adjacent ones of the first channel signals **3712***a* are spaced in a range from 50 GHz to 400 GHz.

(458) Referring now to FIG. **46**A, shown therein is a diagrammatic view of another exemplary implementation of a network element **4600** constructed in accordance with the present disclosure. As shown in FIG. **46**A, the network element **4600** generally comprises one or more demodulator **4604** (hereinafter, the “demodulator **4604**”) and one or more modulator **4608** (hereinafter, the “modulator **4608**”) that are coupled together as shown in FIG. **46**A with one or more bus or electrical circuit.

(459) The demodulator **4604** may be configured to receive one or more input signal **4612** (hereinafter, the “input signals **4612**”), such as a first input signal **4612***a* and a second input signal

4612*b* shown in FIG. **46**A, and extract a series of phase signals **4616** (hereinafter, the "phase signals **4616**") and a series of amplitude signals **4620** (hereinafter, the "amplitude signals **4620**") from the input signals **4612**. The demodulator **4604** may be configured to decompose the input signals **4612** into individual bitstreams and produce the phase signals **4616** and the amplitude signals **4620** based on the individual bitstreams. The demodulator **4604** may be thus configured to extract a first phase signal **4616***a* and a first amplitude signal **4620***a* from the first input signal **4612***a*. Similarly, the demodulator **4604** may be configured to extract a second phase signal **4616***b* and a second amplitude signal **4620***b* from the second input signal **4612***b*.

(460) The input signals **4612** may have input data encoded therein. For example, the first input signal **4612***a* may have first input data encoded therein, and the second input signal **4612***b* may have second input data encoded therein. As described in more detail below, the first input data and the second input data may be encoded in the first input signal **4612***a* and the second input signal **4612***b*, respectively, in a first modulation format which can be a pulse-amplitude modulated (PAMn) format.

(461) The modulator **4608** may be configured to receive the phase signals **4616** and the amplitude signals **4620** and modulate the phase signals **4616** and the amplitude signals **4620** indicative of the first and second input data onto an output signal **4624** such that the output signal **4624** has the first and second input data encoded in a second modulation format. The output signal **4624** may have a carrier frequency in the THz frequency band **104**. In some implementations, the carrier frequency is in a range between 500 GHz and 10 THz. As described in more detail below, the modulator **4608** may be further configured to receive or generate a local oscillator (LO) signal **5012** being an electrical signal in the 500 GHz to 10 THz range (shown in FIGS. **50**A and **50**B), onto which the phase signals **4616** and the amplitude signals **4620** are modulated to produce the output signal **4624**. Further, as described in more detail below, the second modulation format may be different from the first modulation format.

(462) In some implementations, the first modulation format is a pulse-amplitude-modulation-n (PAMn) format, and the second modulation format is an m-quadrature-amplitude-modulation (mQAM) format. In some such implementations, the first modulation format is a pulse-amplitude-modulation-4 (PAM4) format and the second modulation format is a 16-quadrature-amplitude-modulation (16QAM) format. However, in other implementations, the first modulation format and the second modulation format may be modulation formats other than PAMn, PAM4, mQAM, or 16QAM.

(463) Referring now to FIG. **46**B, shown therein is a diagrammatic view of another exemplary implementation of the network element **4600** constructed in accordance with the present disclosure. As shown in FIG. **46**B, in some implementations, the network element **4600** further comprises an antenna **4628** configured to receive the output signal **4624** and couple the output signal **4624** into a hollow waveguide **4632**. In some implementations, the hollow waveguide **4632** is a fiber (either hollow or solid) configured to propagate electromagnetic waves in the THz frequency band **104**. The antenna **4628** is coupled to the modulator **4608** with one or more signal path, which may be a bus or electrical circuit.

(464) Referring now to FIG. **47**A, shown therein is a diagrammatic view of an exemplary implementation of a demodulator **4604** constructed in accordance with the present disclosure. As shown in FIG. **47**A, the demodulator **4604** may comprise a first splitter **4700***a* and a second splitter **4700***b*, one or more phase demodulator **4704** (hereinafter, the "phase demodulators **4704**"), and one or more amplitude demodulator **4708** (hereinafter, the "amplitude demodulators **4708**") that are coupled together as shown in FIG. **47**A with one or more bus or electrical circuit.

(465) The first splitter **4700***a* and the second splitter **4700***b* may be configured to receive the first input signal **4612***a* and the second input signal **4612***b*, respectively, split the first input signal **4612***a* and the second input signal **4612***b*, respectively, into at least two pre-demodulation signals **4712** (hereinafter, the "pre-demodulation signals **4712**"). For example, the first splitter **4700***a* may be

configured to receive the first input signal **4612***a* and split the first input signal **4612***a* into a first pre-demodulation signal **4712***a* and a second pre-demodulation signal **4712***b* shown in FIG. **47**A, and the second splitter **4700***b* may be configured to receive the second input signal **4612***b* and split the second input signal **4612***b* into a third pre-demodulation signal **4712***c* and a fourth pre-demodulation signal **4712***d* shown in FIG. **47**A.

(466) The phase demodulators **4704** may include a first phase demodulator **4704***a* and a second phase demodulator **4704***b*. The first phase demodulator **4704***a* may be configured to extract a series of first phase signals **4616***a* from the first pre-demodulation signal **4712***a*, and the second phase demodulator **4704***b* may be configured to extract a series of second phase signals **4616***b* from the third pre-demodulation signal **4712***c* such that the first phase signals **4616***a* are synchronized with the second phase signals **4616***b* and can be used to represent the input data encoded into the input signals **4612**.

(467) The amplitude demodulators **4708** may include a first amplitude demodulator **4708***a* and a second amplitude demodulator **4708***b*. The first amplitude demodulator **4708***a* may be configured to extract the first amplitude signal **4620***a* from the second pre-demodulation signal **4712***b*, and the second amplitude demodulator **4708***b* may be configured to extract the second amplitude signal **4620***b* from the fourth pre-demodulation signal **4712***d*.

(468) Referring now to FIG. **47**B, shown therein is a diagrammatic view of another exemplary implementation of a demodulator **4604***a* constructed in accordance with the present disclosure. As shown in FIG. **47**B, the demodulator **4604***a* may comprise a clock-and-data-recovery circuit (CDR) **4716** configured to extract the first phase signal **4616***a* and the first amplitude signal **4620***a* from the first input signal **4612***a*, and the second phase signal **4616***b* and the second amplitude signal **4620***b* from the second input signal **4612***b*. That is, the CDR circuit **4716**, like the demodulator **4604** described above, may be configured to decompose the input signals **4612** into individual bitstreams and produce the phase signals **4616** and the amplitude signals **4620** based on the individual bitstreams. As described herein, the CDR circuit **4716** may be similar to a conventional CDR circuit in that the CDR circuit **4716** of the present disclosure receives the input signals **4612** (e.g., PAM4 signals); however, unlike the conventional CDR circuit which may provide signals having the same modulation format (e.g., PAM4 signals) as outputs, the CDR circuit **4716** of the present disclosure may provide the phase signals **4616** and the amplitude signals **4620** as outputs.

(469) Referring now to FIG. **48**, shown therein is a diagrammatic view of an exemplary implementation of a first phase demodulator **4704***a* constructed in accordance with the present disclosure. However, it should be understood that any one of the phase demodulators **4704** described herein may be similar in form and function to the first phase demodulator **4704***a* shown in FIG. **48**. As shown in FIG. **48**, the first phase demodulator **4704***a* may comprise an amplifier **4800**, a first alternating current (AC) coupler **4804***a*, and a first comparator **4808***a* that are coupled together as shown in FIG. **48** with one or more bus or electrical circuit.

(470) The amplifier **4800** may be configured to receive the first pre-demodulation signal **4712***a* (in electrical form) and limit an amplitude of the first pre-demodulation signal **4712***a* to produce an amplitude-limited signal **4812** (in electrical form). In some implementations, the amplifier **4800** is a limiting amplifier.

(471) The first AC coupler **4804***a* may be configured to receive the amplitude-limited signal **4812** and block passage of direct current (DC) signals while allowing passage of AC signals, thereby removing any DC offset from the amplitude-limited signal **4812** to produce a first threshold-centered signal **4816***a*, wherein the first threshold-centered signal **4816***a* is centered around a predetermined threshold voltage. In some implementations, the predetermined threshold voltage is zero.

(472) The first comparator **4808***a* may be configured to receive the first threshold-centered signal **4816***a* and determine a polarity (i.e., positive or negative) of the first threshold-centered signal **4816***a* to produce the first phase signal **4616***a* (in electrical form). In some implementations, the

first comparator **4808***a* is a sign-check comparator.

(473) Referring now to FIG. **49**, shown therein is a diagrammatic view of an exemplary implementation of a first amplitude demodulator **4708***a* constructed in accordance with the present disclosure. However, it should be understood that any one of the amplitude demodulators **4708** described herein may be similar in form and function to the first amplitude demodulator **4708***a* shown in FIG. **49**. As shown in FIG. **49**, the first amplitude demodulator **4708***a* may comprise a magnitude extraction circuit **4900**, a second AC coupler **4804***b*, and a second comparator **4808***b* that are coupled together as shown in FIG. **49** with one or more bus or electrical circuit.

(474) The magnitude extraction circuit **4900** may be configured to receive the second pre-demodulation signal **4712***b* (in electrical form) and determine an amplitude of the second pre-demodulation signal **4712***b* to produce a rectified signal **4904** (in electrical form). In some implementations, the magnitude extraction circuit **4900** is a rectifier. In other implementations, the magnitude extraction circuit **4900** may be a squaring circuit, for example.

(475) The second AC coupler **4804***b* may be configured to receive the rectified signal **4904** (in electrical form) and block passage of DC signals while allowing passage of AC signals, thereby removing any DC offset from the rectified signal **4904** to produce a second threshold-centered signal **4816***b* (in electrical form), wherein the second threshold-centered signal **4816***b* is centered around a predetermined threshold voltage. In some implementations, the predetermined threshold voltage is zero.

(476) The second comparator **4808***b* may be configured to receive the second threshold-centered signal **4816***b* and determine a polarity (i.e., positive or negative) of the second threshold-centered signal **4816***b* to produce the first amplitude signal **4620***a* (in electrical form). That is, if the polarity of the second threshold-centered signal **4816***b* is positive, the first amplitude signal **4620***a* may have a nonzero value (e.g., 1), and if the polarity of the second threshold-centered signal **4816***b* is negative, the first amplitude signal **4620***a* may have a zero value (i.e., 0). In some implementations, the second comparator **4808***b* is a sign-check comparator.

(477) Referring now to FIG. **50**A, shown therein is a diagrammatic view of an exemplary implementation of a modulator **4608** constructed in accordance with the present disclosure. As shown in FIG. **50**A, the modulator **4608** may comprise a third splitter **4700***c*, a first phase modulator **5000***a*, a second phase modulator **5000***b*, a first amplitude modulator **5004***a*, a second amplitude modulator **5004***b*, and a combiner **5008** that are coupled together as shown in FIG. **50** with one or more bus or electrical circuit.

(478) The third splitter **4700***c* may be configured to receive an LO signal **5012** generated by an LO generator (not shown) external to the modulator **4608** and split the LO signal **5012** into one or more unmodulated carrier signal **5016** (in electrical form) (hereinafter, the "unmodulated carrier signals **5016**"). That is, the third splitter **4700***c* may be configured to receive the LO signal **5012** and split the LO signal **5012** into a first unmodulated carrier signal **5016***a* (in electrical form) and a second unmodulated carrier signal **5016***b* (in electrical form) shown in FIG. **50**A. In some implementations, the first unmodulated carrier signal **5016***a* may represent an I component of the output signal **4624**, and the second unmodulated carrier signal **5016***b* may represent a Q component of the output signal **4624**.

(479) The first phase modulator **5000***a* may be configured to receive the first unmodulated carrier signal **5016***a* (i.e., the I component of the output signal **4624**) and the first phase signal **4616***a* and modulate the first phase signal **4616***a* onto the first unmodulated carrier signal **5016***a* to produce a first phase-modulated carrier signal **5020***a*. The first amplitude modulator **5004***a* may be configured to receive the first phase-modulated carrier signal **5020***a* and the first amplitude signal **4620***a* and modulate the first amplitude signal **4620***a* onto the first phase-modulated carrier signal **5020***a* to produce a first phase-amplitude-modulated carrier signal **5024***a*.

(480) The second phase modulator **5000***b* may be configured to receive the second unmodulated carrier signal **5016***b* (i.e., the Q component of the output signal **4624**) and the second phase signal

4616*b* and modulate the second phase signal **4616***b* onto the second unmodulated carrier signal **5016***b* to produce a second phase-modulated carrier signal **5020***b*. The second amplitude modulator **5004***b* may be configured to receive the second phase-modulated carrier signal **5020***b* and the second amplitude signal **4620***b* and modulate the second amplitude signal **4620***b* onto the second phase-modulated carrier signal **5020***b* to produce a second phase-amplitude-modulated carrier signal **5024***b*.

(481) The combiner **5008** may be configured to receive the first phase-amplitude-modulated carrier signal **5024***a* and the second phase-amplitude-modulated carrier signal **5024***b* and combine the first phase-amplitude-modulated carrier signal **5024***a* and the second phase-amplitude-modulated carrier signal **5024***b* to produce the output signal **4624** such that the output signal **4624** is encoded in the second modulation format.

(482) In some implementations, the LO signal **5012** has an LO frequency equal to the carrier frequency (i.e., a frequency in the range between 500 GHz and 10 THz). However, in other implementations, the LO signal **5012** has an LO frequency less than the carrier frequency. In such implementations, the combiner **5008** is configured to receive the first phase-amplitude-modulated carrier signal **5024***a* and the second phase-amplitude-modulated carrier signal **5024***b* and combine the first phase-amplitude-modulated carrier signal **5024***a* and the second phase-amplitude-modulated carrier signal **5024***b* to produce an intermediate signal (not shown) having the LO frequency, and the network element **4600** further comprises an up-converter (not shown) configured to receive the intermediate signal and up-convert the intermediate signal to produce the output signal **4624** (in electrical form) having the carrier frequency. In still other implementations, the LO signal **5012** has an LO frequency greater than the carrier frequency. In such implementations, the combiner **5008** is configured to receive the first phase-amplitude-modulated carrier signal **5024***a* and the second phase-amplitude-modulated carrier signal **5024***b* and combine the first phase-amplitude-modulated carrier signal **5024***a* and the second phase-amplitude-modulated carrier signal **5024***b* to produce an intermediate signal (not shown) having the LO frequency, and the network element **4600** further comprises a downconverter (not shown) configured to receive the intermediate signal and down-convert the intermediate signal to produce the output signal **4624** (in electrical form) having the carrier frequency.

(483) Referring now to FIG. **50**B, shown therein is a diagrammatic view of another exemplary implementation of a modulator **4608***a* constructed in accordance with the present disclosure. As shown in FIG. **50**B, in some implementations, the third splitter **4700***c* may be electrically coupled to an LO generator **5028** and receive the LO signal **5012** (in electrical form) from the LO generator **5028** that is internal to the modulator **4608***a*.

(484) In some implementations, one or more of the first phase modulator **5000***a* and the second phase modulator **5000***b* comprises a crossbar switch **5200** (shown in FIG. **52**) configured to select one of a 0° signal and a 180° signal. In some implementations, one or more of the first amplitude modulator **5004***a* and the second amplitude modulator **5004***b* comprises a switched attenuator (e.g., a PI-type attenuator **5300***a* shown in FIG. **53**A, a T-type attenuator **5300***b* shown in FIG. **53**B, and a bridged T-type attenuator **5300***c* shown in FIG. **53**C) configured to produce signals having one of a first amplitude level and a second amplitude level. In other implementations, one or more of the first amplitude modulator **5004***a* and the second amplitude modulator **5004***b* comprises one of a switched amplifier and a variable gain amplifier.

(485) In some implementations, the first amplitude level is 1 V, and the second amplitude level is 3 V. However, in other implementations, the first amplitude level is a number of volts greater or less than 1, and the second amplitude level is a number of volts greater or less than 3. In such implementations, the first amplitude level may be a fraction (e.g., ¼, ⅓, or ½) of the second amplitude level.

(486) Referring now to FIG. **51**, shown therein is a diagrammatic view of an exemplary implementation of a method **5100** for performing direct modulation from the first modulation

format to the second modulation format in the THz frequency band **104**. As shown in FIG. **51**, the method **5100** may comprise the steps of: receiving, by the demodulator **4604**, the first input signal **4612***a* and the second input signal **4612***b*, the first input signal **4612***a* having the first input data, the second input signal **4612***b* having the second input data, the first input data and the second input data encoded in the first modulation format (step **5104**); extracting, by the demodulator **4604**, the first phase signal **4616***a* and the first amplitude signal **4620***a* from the first input signal **4612***a* and the second phase signal **4616***b* and the second amplitude signal **4620***b* from the second input signal **4612***b* (step **5108**); modulating, by the modulator **4608**, the first phase signal **4616***a*, the first amplitude signal **4620***a*, the second phase signal **4616***b*, and the second amplitude signal **4620***b* onto the output signal **4624** such that the output signal **4624** is encoded in the second modulation format, the output signal **4624** having the carrier frequency in the range between 500 GHz and 2 THz (step **5112**); converting, by the antenna **4628**, the output signal **4624** from an electrical signal to an electromagnetic wave (step **5114**); and coupling, by the antenna **4628**, the electromagnetic wave into the hollow waveguide **4632** (step **5116**).

(487) In some implementations, receiving the first input signal **4612***a* and the second input signal **4612***b* (step **5104**) is further defined as receiving, by the demodulator **4604**, the first input signal **4612***a* and the second input signal **4612***b*, the first input signal **4612***a* having the first input data, the second input signal **4612***b* having the second input data, the first input data and the second input data encoded in the first modulation format, wherein the first modulation format is the PAMn (e.g., PAM4) format.

(488) In some implementations, extracting the first phase signal **4616***a* and the first amplitude signal **4620***a* from the first input signal **4612***a* and the second phase signal **4616***b* and the second amplitude signal **4620***b* from the second input signal **4612***b* (step **5108**) is further defined as extracting, by the demodulator **4604**, the first phase signal **4616***a* and the first amplitude signal **4620***a* from the first input signal **4612***a* and the second phase signal **4616***b* and the second amplitude signal **4620***b* from the second input signal **4612***b*, wherein the demodulator **4604** includes the CDR circuit **4716**.

(489) In some implementations, extracting the first phase signal **4616***a* and the first amplitude signal **4620***a* from the first input signal **4612***a* and the second phase signal **4616***b* and the second amplitude signal **4620***b* from the second input signal **4612***b* (step **5108**) further comprises: splitting, by the first splitter **4700***a*, the first input signal **4612***a* into the first pre-demodulation signal **4712***a* and the second pre-demodulation signal **4712***b*; splitting, by the second splitter **4700***b*, the second input signal **4612***b* into the third pre-demodulation signal **4712***c* and the fourth pre-demodulation signal **4712***d*; extracting, by the first phase demodulator **4704***a*, the first phase signal **4616***a* from the first pre-demodulation signal **4712***a*; extracting, by the first amplitude demodulator **4708***a*, the first amplitude signal **4620***a* from the second pre-demodulation signal **4712***b*; extracting, by the second phase demodulator **4704***b*, the second phase signal **4616***b* from the third pre-demodulation signal **4712***c*; extracting, by the second amplitude demodulator **4708***b*, the second amplitude signal **4620***b* from the fourth pre-demodulation signal **4712***d*.

(490) In some implementations, extracting the first phase signal **4616***a* and the first amplitude signal **4620***a* from the first input signal **4612***a* and the second phase signal **4616***b* and the second amplitude signal **4620***b* from the second input signal **4612***b* (step **5108**) is further defined as: extracting, by the first phase demodulator **4704***a*, the first phase signal **4616***a* from the first pre-demodulation signal **4712***a* by passing the first pre-demodulation signal **4712***a* to the amplifier **4800** having an output connected to an input of the first comparator **4808***a*; extracting, by the first amplitude demodulator **4708***a*, the first amplitude signal **4620***a* from the second pre-demodulation signal **4712***b* by passing the second pre-demodulation signal **4712***b* to the magnitude extraction circuit **4900** having an output connected to an input of the second comparator **4808***b*; extracting, by the second phase demodulator **4704***b*, the second phase signal **4616***b* from the third pre-demodulation signal **4712***c* by passing the third pre-demodulation signal **4712***c* to the amplifier

4800 having an output to an input of the first comparator **4808***a*; and extracting, by the second amplitude demodulator **4708***b*, the second amplitude signal **4620***b* from the fourth pre-demodulation signal **4712***d* by passing the fourth pre-demodulation signal **4712***d* to the magnitude extraction circuit **4900** having an output connected to an input of the second comparator **4808***b*.
(491) In some implementations, modulating the first phase signal **4616***a*, the first amplitude signal **4620***a*, the second phase signal **4616***b*, and the second amplitude signal **4620***b* onto the output signal **4624** (step **5112**) is further defined as modulating, by the modulator **4608**, the first phase signal **4616***a*, the first amplitude signal **4620***a*, the second phase signal **4616***b*, and the second amplitude signal **4620***b* onto the output signal **4624** such that the output signal **4624** is encoded in the second modulation format, the output signal **4624** having the carrier frequency in the range between 500 GHz and 2 THz, wherein the second modulation format is the mQAM (e.g., 16QAM) format.
(492) In some implementations, modulating the first phase signal **4616***a*, the first amplitude signal **4620***a*, the second phase signal **4616***b*, and the second amplitude signal **4620***b* onto the output signal **4624** (step **5112**) further comprises: splitting, by the third splitter **4700***c*, the LO signal **5012** into the first unmodulated carrier signal **5016***a* and the second unmodulated carrier signal **5016***b*; modulating, by the first phase modulator **5000***a*, the first phase signal **4616***a* onto the first unmodulated carrier signal **5016***a*; modulating, by the first amplitude modulator **5004***a*, the first amplitude signal **4620***a* onto the first unmodulated carrier signal **5016***a* (i.e., the first phase-modulated carrier signal **5020***a*); modulating, by the second phase modulator **5000***b*, the second phase signal **4616***b* onto the second unmodulated carrier signal **5016***b*; modulating, by the second amplitude modulator **5004***b*, the second amplitude signal **4620***b* onto the second unmodulated carrier signal **5016***b* (i.e., the second phase-modulated carrier signal **5020***b*); and combining, by the combiner **5008**, the first unmodulated carrier signal **5016***a* (i.e., the first phase-amplitude-modulated carrier signal **5024***a*) and the second unmodulated carrier signal **5016***b* (i.e., the second phase-amplitude-modulated carrier signal **5024***b*) into the output signal **4624** such that the output signal **4624** is encoded in the second modulation format.
(493) In some implementations, modulating the first phase signal **4616***a*, the first amplitude signal **4620***a*, the second phase signal **4616***b*, and the second amplitude signal **4620***b* onto the output signal **4624** (step **5112**) further comprises: splitting, by the third splitter **4700***c*, the LO signal **5012** into the first unmodulated carrier signal **5016***a* and the second unmodulated carrier signal **5016***b*; modulating, by the first amplitude modulator **5004***a*, the first amplitude signal **4620***a* onto the first unmodulated carrier signal **5016***a*; modulating, by the first phase modulator **5000***a*, the first phase signal **4616***a* onto the first unmodulated carrier signal **5016***a*; modulating, by the second amplitude modulator **5004***b*, the second amplitude signal **4620***b* onto the second unmodulated carrier signal **5016***b*; modulating, by the second phase modulator **5000***b*, the second phase signal **4616***b* onto the second unmodulated carrier signal **5016***b*; and combining, by the combiner **5008**, the first unmodulated carrier signal **5016***a* and the second unmodulated carrier signal **5016***b* into the output signal **4624** such that the output signal **4624** is encoded in the second modulation format.
(494) In some implementations, modulating the first phase signal **4616***a*, the first amplitude signal **4620***a*, the second phase signal **4616***b*, and the second amplitude signal **4620***b* onto the output signal **4624** (step **5112**) is further defined as: modulating, by the first phase modulator **5000***a*, the first phase signal **4616***a* onto the first unmodulated carrier signal **5016***a*, wherein the first phase modulator **5000***a* is a first crossbar switch; modulating, by the first amplitude modulator **5004***a*, the first amplitude signal **4620***a* onto the first unmodulated carrier signal **5016***a* (i.e., the first phase-modulated carrier signal **5020***a*), wherein the first amplitude modulator **5004***a* is a first switched attenuator; and modulating, by the second phase modulator **5000***b*, the second phase signal **4616***b* onto the second unmodulated carrier signal **5016***b*, wherein the second phase modulator **5000***b* is a second crossbar switch; and modulating, by the second amplitude modulator **5004***b*, the second amplitude signal **4620***b* onto the second unmodulated carrier signal **5016***b* (i.e., the second phase-

modulated carrier signal **5020***b*), wherein the second amplitude modulator **5004***b* is a second switched attenuator.

(495) Referring now to FIG. **54**, shown therein is another exemplary implementation of a transceiver **5400***a* constructed in accordance with the present disclosure.

(496) Referring now to FIG. **55**, shown therein is another exemplary implementation of a transceiver **5400***b* constructed in accordance with the present disclosure.

(497) Referring now to FIG. **56**, shown therein is another exemplary implementation of a transceiver **5400***c* constructed in accordance with the present disclosure.

(498) Referring now to FIG. **57**, shown therein is another exemplary implementation of a transmitter **5700***a* constructed in accordance with the present disclosure.

(499) Referring now to FIG. **58**, shown therein is another exemplary implementation of a receiver **5800** constructed in accordance with the present disclosure.

(500) Referring now to FIG. **59**, shown therein is another exemplary implementation of a transmitter **5700***b* constructed in accordance with the present disclosure.

(501) Referring now to FIG. **60**, shown therein is another exemplary implementation of a transmitter **5700***c* constructed in accordance with the present disclosure.

(502) Referring now to FIGS. **61-63**, shown therein are exemplary implementations of a differential circuit constructed in accordance with the present disclosure, including a first differential circuit **6100***a* (shown in FIG. **61**), a second differential circuit **6100***b* (shown in FIG. **62**), and a third differential circuit **6100***c* (shown in FIG. **63**).

(503) Referring now to FIG. **64**, shown therein is an exemplary implementation of an antenna array **6400** constructed in accordance with the present disclosure.

(504) Referring now to FIG. **65**, shown therein is a perspective view of an exemplary implementation of an antenna **6500** constructed and used in accordance with the present disclosure. As shown in FIG. **65**, the antenna **6500** comprises an electromagnetic absorber **6502** disposed around one or more radiators **6504** (e.g., a first radiator **6504***a*, a second radiator **6504***b*, a third radiator **6504***c*, and a fourth radiator **6504***d*). The one or more radiators **6504** may be constructed in accordance with the radiator **908**, as detailed above. It should be understood that while four radiators **6504** are illustrated in FIG. **65**, the antenna **6500** may include greater than, or less than, four radiators **6504**, such as one radiator **6504** or eight radiators **6504** (for example).

(505) In one implementation, one or more of the radiators **6504** may be mounted to respective ground planes **904***a-d*. For example, the first radiator **6504***a* may be mounted to a first ground plane **904***a*, the second radiator **6504***b* may be mounted to a second ground plane **904***b*, the third radiator **6504***c* may be mounted to a third ground plane **904***c*, and the fourth radiator **6504***d* may be mounted to a fourth ground plane **904***d* (not shown).

(506) In some implementations, one or more of the radiators **6504** may be disposed within the hollow waveguide **208** (not shown in FIG. **65**). In other implementations, one or more of the radiators **6504** may be disposed apart from, and coaxially to the hollow waveguide **208**. In some implementations, one or more of the radiators **6504** may be coupled to a fiber-coupled RF transmitter (such as the first transmitter **212***a*) while others of the radiators **6504** may be coupled to a fiber-coupled RF receiver (such as the first receiver **216***a*).

(507) In one implementation, the electromagnetic absorber **6502** is not disposed between the radiator **6504** (e.g., first radiator **6504***a*, the second radiator **6504***b*, the third radiator **6504***c*, the fourth radiator **6504***d*) and the hollow waveguide **208**. The electromagnetic absorber **6502**, in some implementations, may include a distal surface **6508**, an opposed proximal surface **6509**, and one or more opening **6510** formed in the distal surface **6508** and extending toward the opposed proximal surface **6509**. In the implementation shown in FIG. **11**, four openings **6510** are shown by way of example, with one of the radiators **6504** being positioned within each of the four openings **6510**.

(508) In some non-limiting implementations, only one of the radiators **6504** is positioned within a particular one of the openings **6510**. The electromagnetic absorber **6502** has a plurality of internal

surfaces **6511** defining the openings **6510**. Each of the internal surfaces **6511** surrounds one of the radiators **6504** that is positioned within the respective opening **6510**. In the example shown, the electromagnetic absorber **6502** is devoid of a cover covering any of the openings **6510** so that electromagnetic waves generated by the radiators **6504***a-d* pass directly into the hollow waveguide **208**. In implementations incorporating the cover over one or more of the openings **6510**, the cover may be selected from a material that is transparent (or mostly transparent) to the electromagnetic wave. For example, the cover may comprise a plastic material. The cover may cause less than 10% reflected power of the electromagnetic wave. The opposed proximal surface **6509** may be positioned adjacent to the ground planes **90***a-d*. In some implementations, the opposed proximal surface **6509** contacts the ground planes **90***a-d*.

(509) In one implementation, each of the openings **6510** may have a cross-sectional shape similar in shape to the radiators **6504**. In some implementations, the cross-sectional shape of the openings **6510** may be disposed apart from the radiator **6504** by an opening distance based on a wavelength of the electromagnetic wave and/or a style of the radiator **6504** or antenna **900**. For example, the opening distance, e.g., a distance between the radiator **6504** and the interior surface **6511** may be at least % of the wavelength of the electromagnetic wave.

(510) In one implementation, the electromagnetic absorber **6502** may be disposed adjacent to the hollow waveguide **208**. For example, the distal surface **6508** of the electromagnetic absorber **6502** may have a diameter, a, defining a cross-section dimension. The distal surface **6508** may be in contact with the hollow waveguide **208**. In other implementations, the electromagnetic absorber **6502** may be disposed against, e.g., touching or in-contact with, the hollow waveguide **208**. In yet other implementations, the electromagnetic absorber **6502** may have a peripheral surface **6512** disposed within the hollow waveguide **208** and adjacent to, or in contact with, the inner surface **312** of the hollow waveguide **208**. The diameter, a, defining a cross-section dimension, may be in a range of at least 4 wavelengths to 50 wavelengths of the electromagnetic wave having data encoded within a carrier frequency in a range of 300 GHz to 10 THz, the electromagnetic wave having a wavelength. In some implementations, the peripheral surface **6512** has a cylindrical shape. However, it should be understood that the peripheral surface **6512** can be provided with another shape, such as series of planar and adjacently disposed sections, so as to provide a rectangular, hexagon, or octagon shaped cross-section, for example. In some implementations, the peripheral surface **6512** may have a non-uniform shape or a fanciful shape.

(511) In one implementation, the electromagnetic absorber **6502** may be constructed of an EM-absorbing material selected to absorb, dampen, and/or otherwise limit reflection of an electromagnetic wave (e.g., the electromagnetic wave having the transmission signals). In one implementation, the EM-absorbing material may be constructed of a porous and/or lossy material. In some implementations, the EM-absorbing material is constructed of a semi-porous material having a plurality of randomly positioned and sized openings having a size on the order of a wavelength of the electromagnetic wave, i.e., between 1/100.sup.th the electromagnetic wave wavelength to about 2 times the electromagnetic wave wavelength, and preferably about ¼.sup.th the electromagnetic wave wavelength. In some implementations, the EM-absorbing material has a texture similar to that of steel wool. In some implementations, e.g., as shown in FIG. **67** and discussed in detail below, the EM-absorbing material may be constructed as part of the ground plane **904**. In one implementation, the EM-absorbing material may, for example, include a poorly-conducting material (i.e., a material with low electrical conductivity), such as a carbon material or a compound containing carbon. In other implementations, a different poorly-conducting material may be selected other than carbon.

(512) In one implementation, the EM-absorbing material may be constructed of a foam (e.g., a solid, continuous-phase material). The foam may be, for example, open-cell foam, closed-cell foam, or a combination thereof. The foam may be carbon-doped or carbon-loaded, that is, the foam may have carbon absorbed/adsorbed into, and disposed within, the foam. In some implementations,

the foam is a polyurethane foam. In one implementation, the EM-absorbing material is a colloidal suspension having carbon particles suspended in a continuous phase material.

(513) Referring now to FIG. **66**, shown therein is a cross-section view of another exemplary implementation of an electromagnetic absorber **6600** constructed in accordance with the present disclosure. As shown the electromagnetic absorber **6600** may be disposed around one or more radiators **6504**, such as the first radiator **6504***a* and the second radiator **6504***b*, and disposed within the hollow waveguide **208** (shown as a sixth hollow waveguide **208***f*). As detailed above, in some implementations, the first radiator **6504***a* and the second radiator **6504***b* may be attached to one or more ground plane **904**, as shown in FIG. **65**. In one implementation, the electromagnetic absorber **6600** may be constructed in accordance with the electromagnetic absorber **6502** detailed above, e.g., of the EM-absorbing material.

(514) In one implementation, the sixth hollow waveguide **208***f* may have the inner surface **312** defining a cavity **6604** and having a diameter, d, defining a cross-section dimension. The sixth hollow waveguide **208***f* may be constructed in accordance with any of the hollow waveguides **208***a-n* described above in more detail; however, the sixth hollow waveguide **208***f*, shown in FIG. **66**, is illustrated as a hollow-core fiber optic cable having the conductive layer **316** surrounding the dielectric layer **308**. In other implementations, the sixth hollow waveguide **208***f* may be a metallic, non-optic waveguide.

(515) The electromagnetic absorber **6600** may have a peripheral surface **6608** in contact with at least a portion of the inner surface **312** of the sixth hollow waveguide **208***f*, i.e., the hollow-core fiber optic cable. In some implementations, the electromagnetic absorber **6600** has a diameter, a, defining a cross-section dimension less than or equal to the diameter, d, of the sixth hollow waveguide **208***f*, such that the electromagnetic absorber **6600** may extend, or fit, into the cavity **6604** of the sixth hollow waveguide **208***f* so as to not interfere with the radiators **6504** receiving energy from the electromagnetic wave.

(516) In one implementation, the sixth hollow waveguide **208***f* further includes a tapering section **6612** having a first end **6614** and a second end **6616**. The first end **6614** may have an interior diameter, t, and the second end **6616** may have the diameter, d, such that within the tapering section **6612**, the diameter of the sixth hollow waveguide **208***f* changes from the diameter, d, to the interior diameter, t. As shown, the interior diameter, t, may be less than the diameter, d.

(517) In one implementation, the electromagnetic absorber **6600** may extend within the sixth hollow waveguide **208***f*. In some implementations, the electromagnetic absorber **6600** extends beyond the tapering section **6612** of the sixth hollow waveguide **208***f*. In other implementations, the electromagnetic absorber **6600** only extends within the tapering section **6612** of the sixth hollow waveguide **208***f*. In one implementation, as shown in FIG. **66**, the electromagnetic absorber **6600** may extend within a first portion **6620** of the tapering section **6612** of the sixth hollow waveguide **208***f*.

(518) In some implementations, the electromagnetic absorber **6600** within the sixth hollow waveguide **208***f* may be provided with a thickness **6622**. The thickness **6622** may be uniform within the tapering section **6612**, such as within the first portion **6620** of the tapering section **6612**. In other implementations, the electromagnetic absorber **6600** within the sixth hollow waveguide **208***f* may be provided with a varying thickness **6622**, such that from a distal surface **6624** of the electromagnetic absorber **6600** to an interior end **6626** of the electromagnetic absorber **6600**, the thickness **6622** tapers, for example, to a feather-edge, as illustrated by tapering absorber surface **6628** (which is shown in phantom). The tapering absorber surface **6628** may taper at differing rates from the distal surface **6624** to the interior end **6626** of the electromagnetic absorber **6600**.

(519) Referring now to FIG. **67**, shown therein is a cross-section diagram of an exemplary implementation of an electromagnetic absorber **6700** constructed in accordance with the present disclosure. As shown, the electromagnetic absorber **6700** may be integrated into a fifth ground plane **904***e*. In this implementation, the electromagnetic absorber **6700** may comprise a plurality of

vias **6712** having a via diameter **6704** and a depth **6708**. The plurality of vias **6712** may extend from a first surface **6714***a* towards a second surface **6714***b* of the fifth ground plane **904***e* to the depth **6708**. In some implementations, the depth **6708** may extend through at least one layer **6716** of the fifth ground plane **904***e*, such as a first layer **6716***a*. While the vias **6712** are described as having the via diameter **6704**, the vias **6712** may have a cross-section of any suitable shape, such as an oval, square, circle, and the like, or any fanciful shape. In such implementations, the via diameter **6704** may be, for example, a cross-sectional dimension.

(520) In some implementations, one or more via **6712** of the plurality of vias **6712** of the electromagnetic absorber **6700** may extend through the first layer **6716***a* while others of the plurality of vias **6712** may extend through the first layer **6716***a* and a second layer **6716***b*. As will be understood by a skilled artisan some of the vias **6712** may be characterized as blind vias meaning the vias **6712** pass through the first layer **6716***a* and not the second layer **6716***b*, or a though via meaning the vias **6712** pass through both the first layer **6716***a* and the second layer **6716***b*.

(521) In some implementations, the depth **6708** of the plurality of vias **6712** may be selected based on a wavelength of the electromagnetic wave. For example, the depth **6708** may be about one wavelength. In other implementations, the depth **6708** may be between 1/10.sup.th of a wavelength and 10 wavelengths. In some implementations, the plurality of vias **6712** may extend through multiples of the layers **6716** to reach the depth **6708** and in some implementations does not extend through all of the layers **6716**. In some implementations, a first set of the plurality of vias **6712** may be constructed such that the depth **6708** is a first depth and a second set of the plurality of vias **6712** may be constructed such that the depth **6708** is a second depth different from the first depth, thus forming multiple semi-porous ground planes with an array of vias **6712** extending between one of more of the semi-porous ground planes. In one implementation, the vias **6712** of the first set and the second set may be randomly disposed within the first surface **6714***a* of the fifth ground plane **904***e*. In other implementations, the first set and the second set may be disposed in a pattern on the fifth ground plane **904***e* selected to minimize reflection of the electromagnetic wave. In some implementations, the depth **6708** of one or more via of the plurality of vias **6712** may be randomly selected to have values between about 10% of the wavelength and about 1000% of the wavelength.

(522) In some implementations, the plurality of vias **6712** are separated from each other by a distance **6720**. The distance **6720** may be selected based on the wavelength of the electromagnetic wave. For example, the distance **6720** may be about one wavelength. In other implementations, the distance **6720** may be between about 1/10.sup.th of a wavelength and one wavelength.

(523) In one implementation, each via **6712** of the plurality of vias **6712** may be defined by a via surface **6724** extending from the first surface **6714***a* to the second surface **6714***b*, i.e., a through via. In one implementation, the surface **6724** of the vias **6712** may be constructed of a material, e.g., comprising copper, gold, and/or carbon. In some implementations, one or more via **6712** may extend through the first surface **6714***a* and the second surface **6714***b*. In some implementations, the surface **6724** of the vias **6712** may be constructed of an electrically conductive material such as copper or gold coated with an electrically lossy material such as carbon to assist in absorbing the electromagnetic wave. In some implementations, the via surface **6724** may be textured so as to assist in absorbing the electromagnetic wave. In some implementations, the material may be an EM-absorbing material (as discussed above).

(524) In some implementations, the plurality of vias **6712** may be constructed by removing material from the fifth ground plane **904***e*. For example, during manufacturing, material may be removed from the first surface **6714***a* to the depth **6708** and with the via diameter **6704**. In other implementations, the plurality of vias **6712** may be constructed by extending protrusions **6726** from the second surface **6714***b* such that the protrusions **6726** have the surface **6724**, a height equal to the depth **6708**, and are spaced from one another by a distance equal to the via diameter **6704**.

(525) In some implementations, each via **6712** of the plurality of vias **6712** may have the via diameter **6704** and an opening width **6730**. In some implementations, the via diameter **6704** may be a width of the via nearest the second surface **6714***b*. The via diameter **6704** may be the same as, or different from, the opening width **6730**. In some implementations, a first set of the plurality of vias **6712** may be constructed such that the via diameter **6704** and the opening width **6730** are the same, a second set of the plurality of vias **6712** may be constructed such that the via diameter **6704** is smaller than the opening width **6730**, and a third set of the plurality of vias **6712** may be constructed such that the via diameter **6704** is greater than the opening width **6730**. The vias **6712** of the first set, the second set, and the third set may be randomly disposed within the first surface **6714***a* of the fifth ground plane **904***e*. In some implementations, the via diameter **6704** and the opening width **6730** of one or more via of the plurality of vias **6712** may be randomly selected to have values between about 10% of the wavelength and about 110% of the wavelength.
(526) Referring now to FIG. **68**, shown therein is a cross-section diagram of an exemplary implementation of an electromagnetic absorber **6800** constructed in accordance with the present disclosure. As shown, the electromagnetic absorber **6800** is a spray-on coating constructed as a low-THz electromagnetic absorber. In one implementation, the electromagnetic absorber **6800** may be constructed of materials in accordance with the electromagnetic absorber **6502** detailed above, e.g., of the EM-absorbing material.
(527) In one implementation, the spray-on coating may be a polyurethane foam loaded with carbon that when sprayed on a substrate, such as the ground plane **904**, adheres to the ground plane **904** and forms an uneven, or non-uniform, coating, such as, of carbon. The uneven coating may comprise carbon particles **6804** of varying sizes resulting in a non-uniform coating having voids, or dimples **6808** having a cross-section dimension approximately sized to the wavelength of the electromagnetic wave (e.g., about 300 μm). The uneven coating may have a thickness **6812** of at least % of a wavelength. In some implementations, the uneven coating may have a thickness **6812** of between about one wavelength of the electromagnetic wave and about 10 wavelengths of the electromagnetic wave.
(528) Referring now to FIG. **69**, shown therein is a diagram of an exemplary implementation of an electromagnetic absorber **6900** constructed in accordance with the present disclosure. As shown, the electromagnetic absorber **6900** may be a fabric **6904** coated with an EM-absorbing material such as carbon. The fabric **6904** may be coated, for example, by use of a spray-on carbon coating having a binder to cause the carbon to adhere to the fabric **6904**. In some implementations, the fabric **6904** may include a fabric doped with carbon.
(529) In one implementation, the fabric **6904** may be formed of a plurality of strands **6908** (e.g., weft **6908***a* and warp **6908***b*) coated (or doped) with carbon particles or another poorly-conducting EM-absorbing material. In some implementations, the strands **6908** of the fabric **6904** may be carbon-doped prior to forming the fabric **6904**, while in other implementations, the strands **6908** may be doped after the fabric **6904** has been constructed.
(530) In some implementations, the fabric **6904** may be formed of a solid, continuous-phase material doped with carbon and having one or more voids **6920** disposed therethrough and defined by remaining fabric **6904**. In some implementations, carbon particles may be sprayed through the voids **6920** of the continuous phase material.
(531) Referring now to FIG. **70**, shown therein is a flow diagram of an exemplary implementation of a process **7000** constructed in accordance with the present disclosure. The process **7000** generally comprises the steps of: disposing the electromagnetic absorber around a radiator of an antenna (step **7004**); and coupling the hollow waveguide to the antenna (step **7008**).
(532) In one implementation, disposing the electromagnetic absorber around a radiator of an antenna (step **7004**) includes disposing the electromagnetic absorber (e.g., any of electromagnetic absorber **6502**, electromagnetic absorber **6600**, electromagnetic absorber **6700**, electromagnetic absorber **6800**, and electromagnetic absorber **6900**) surrounding the radiator. In one

implementation, the electromagnetic absorber does not touch the radiator(s).
(533) In one implementation, disposing the electromagnetic absorber around a radiator of an antenna (step **7004**) may include disposing more than one electromagnetic absorber around the radiator of the antenna.
(534) In one implementation, disposing the electromagnetic absorber around a radiator of an antenna (step **7004**) includes positioning the electromagnetic absorber within the cavity **6604** of the hollow waveguide **208**. In some implementations, positioning the electromagnetic absorber within the cavity **6604** includes positioning the electromagnetic absorber so as to not interfere with the radiator **6504** receiving the energy of the electromagnetic wave.
(535) In one implementation, coupling the hollow waveguide to the antenna (step **7008**) includes positioning the radiator(s) **6504** within the cavity **6604** of the hollow waveguide **208** (or the sixth hollow waveguide **208*f***). In some implementations, positioning the radiator(s) **6504** within the cavity **6604** of the hollow waveguide **208** further includes positioning the peripheral surface **6512** within the hollow waveguide **208** and adjacent to, or in contact with, the inner surface **312** of the hollow waveguide **208**. In one implementation, coupling the hollow waveguide to the antenna (step **7008**) includes positioning the radiator(s) **6504** at least partially within the cavity **6604** of the hollow waveguide **208** (or the sixth hollow waveguide **208*f***).
(536) In one implementation, coupling the hollow waveguide to the antenna (step **7008**) includes positioning the radiator(s) **6504** within the tapering section **6612** of the sixth hollow waveguide **208*f***. In some implementations, positioning the radiator(s) **6504** within the tapering section **6612** includes disposing the peripheral surface **6608** of the electromagnetic absorber **6600** against at least the first portion **6620** of the tapering section **6612** of the sixth hollow waveguide **208*f***.
(537) In one implementation, coupling the hollow waveguide to the antenna (step **7008**) includes positioning the hollow waveguide **208** in contact with the electromagnetic absorber, e.g., against the distal surface **6508** of the electromagnetic absorber **6502** (or other ones of the electromagnetic absorbers).
(538) Referring now to FIG. **71**, shown therein is a process flow diagram of an exemplary implementation of a method **7100** constructed in accordance with the present disclosure. The method **7100** generally comprises the steps of: coupling an antenna and an electromagnetic wave via a hollow waveguide (step **7104**); and positioning an electromagnetic absorber around the antenna (step **7108**). In one implementation, coupling an antenna and an electromagnetic wave via a hollow waveguide (step **7104**) includes coupling a first antenna and a second antenna with the electromagnetic wave.
(539) In one implementation, coupling an antenna and an electromagnetic wave via a hollow waveguide (step **7104**) includes coupling the antenna and the electromagnetic wave via the hollow waveguide being at least one of: a solid-core optical fiber, a hollow-core fiber, and a metallic, non-optic waveguide. Coupling the antenna and the solid-core waveguide may include disposing the electromagnetic absorber surrounding the radiators of the antenna against the solid-core fiber. Coupling the antenna and the hollow-core fiber may include positioning a radiator of the antenna within the cavity **6604** of the hollow-core fiber (e.g., the sixth hollow waveguide **208*f***).
(540) In one implementation, positioning an electromagnetic absorber around the antenna (step **7108**) includes disposing the electromagnetic absorber around the radiators of the antenna. In some implementations, disposing the electromagnetic absorber around the radiators of the antenna further includes disposing the electromagnetic absorber against, or in contact with, the inner surface **312** defining the cavity **6604** of the sixth hollow waveguide **208*f***.
(541) In some implementations, disposing the electromagnetic absorber around the radiators of the antenna further includes disposing the electromagnetic absorber against, or in contact with, (at least the first portion **6620** of) the inner surface **312** of the tapering section **6612** of the sixth hollow waveguide **208*f***.
(542) In one implementation, positioning an electromagnetic absorber around the antenna (step

7108) includes positioning the electromagnetic absorber (constructed of an EM-absorbing material) adjacent to the ground plane **904**. The electromagnetic absorber may be, for example, an absorbing carbon-material sprayed-on the ground plane **904** to form a non-uniform layer of carbon disposed on the ground plane (as described above in reference to FIG. **68**).

(543) In one implementation, positioning an electromagnetic absorber around the antenna (step **7108**) includes providing the plurality of vias **6712** within the ground plane **904**. The plurality of vias **6712** may be disposed a distance **6720** of at least one wavelength of the carrier frequency of the electromagnetic wave from one another. In some implementations, the plurality of vias **6712** have via diameter **6704** of at least one wavelength of the carrier frequency. The vias **6712** may be provided with any suitable cross-section geometry, such as a circle, square, oval, and the like, or with any fanciful shape. In some implementations, one or more of the plurality of vias **6712** may be constructed as through-vias within the ground plane, e.g., vias **6712** extending from the first surface **6714***a* through the second surface **6714***b*. When the ground plane **904** comprises more than one layer **6716**, one or more via **6712** of the plurality of vias **6712** may extend through one or more layer **6716** of the ground plane **904**.

(544) Referring now to FIG. **72**, shown therein is a process flow diagram of an exemplary implementation of a construction process **7200** constructed in accordance with the present disclosure. The construction process **7200** generally comprises the steps of: selecting an absorber substrate (step **7204**); providing a conducting material within the absorber substrate to create an absorber precursor (step **7208**); curing the absorber precursor into an EM-absorbing material (step **7212**); and affixing the EM-absorbing material to an antenna (step **7216**).

(545) In one implementation, selecting an absorber substrate (step **7204**) includes selecting one or more of: a foam (e.g., a solid, continuous-phase material), a fabric (e.g., a woven fabric or a non-woven fabric), and a spray coating. In some implementations, the absorber substrate selected may be selected as component parts. For example, selection of the foam may include selection of at least two component parts of a foam (for example, an isocyanate and a polyol) that, when combined, cause a foam to form. Similarly, selection of the fabric may include selection of component parts of the fabric such as the weft and warp for a woven fabric, or chemical compound precursors for the non-woven fabric, and selection of the spray coating may include selection of an accelerant, a binder, and a solvent.

(546) In one implementation, providing a conducting material with the absorber substrate to create an absorber precursor (step **7208**) may include absorbing, adsorbing, mixing, dissolving, suspending, coating, attaching, incorporating, doping, and/or otherwise including the conducting material within the absorber substrate to create an absorber precursor. For example, providing the conducting material with the absorber substrate may include spraying or coating the foam with the conducting material, spraying, or coating the fabric with the conducting material such that the conducting material is disposed within voids between the waft and warp for woven fabric(s) or within the one or more voids formed in non-woven fabric(s), and incorporating the conducting material within the spray coating.

(547) In one implementation, providing the conducting material with the absorber substrate to create the absorber precursor (step **7208**) may include absorbing, adsorbing, mixing, dissolving, suspending, coating, attaching, incorporating, and/or otherwise including the conducting material being one or more of: carbon, fullerenes, carbon nano-particles, a carbon compound, a semi-metal, a metalloid, and/or the like, or combinations thereof. In one implementation, providing the conducting material with the absorber substrate to create the absorber precursor (step **7208**) may include disposing such conducting materials with the absorber substrate in randomized position and/or orientation.

(548) In one implementation, providing the conducting material with the absorber substrate to create the absorber precursor (step **7208**) may include absorbing, adsorbing, mixing, dissolving, suspending, coating, attaching, incorporating, and/or otherwise including the conducting material

within one or more component part of the absorber substrate. For example, the conducting material may be incorporated into one or more of the component parts of the foam, the component parts of the spray coating, and/or the component parts of the fabric to form the absorber precursor. In this way, when the absorber precursors (e.g., the component parts of the absorber substrate having the conducting material) are combined or assembled to form the absorber substrate, the conducting materials are integrated/incorporated into the absorber substrate.

(549) In one implementation, curing the absorber precursor into an EM-absorbing material (step **7212**) may include allowing the absorber precursor to cure or set as the component parts of the absorber substrate are bonded to form the EM-absorbing material. In some implementations, curing the absorber precursor may be optional. In other implementations, curing the absorber precursor into the EM-absorbing material (step **7212**) may be performed after affixing the EM-absorbing material to a substrate, such as the fifth ground plane **904***e*, adjacent to and preferably surrounding the antenna (step **7216**). For example, when providing the conducting material as part of the spray coating, the spray coating may not be allowed to cure until after the spray coating has been affixed, or otherwise applied, to the fifth ground plane **904***e*, for example, which is adjacent to the antenna (e.g., in step **7216**). Additionally, in some implementations, curing the absorber precursor into the EM-absorbing material (step **7212**) may be performed after affixing the EM-absorbing material to the fifth ground plane **904***e*, for example, which is adjacent to the antenna (step **7216**) in order to further form a bond between the antenna and the EM-absorbing material.

(550) In one implementation, affixing the EM-absorbing material to an antenna (step **7216**) may include disposing the EM-absorbing material around one or more radiator **6504** of the antenna (e.g., the antenna **6500**). In some implementations, prior to affixing the EM-absorbing material to the antenna (step **7216**), the one or more radiator of the antenna may be (at least, temporarily) shielded to limit un-intended application of the EM-absorbing material directly to the radiator.

(551) In one implementation, affixing the EM-absorbing material to the antenna (step **7216**) may include applying the absorber precursor to the antenna. For example, when the absorber precursor is the spray coating doped with the conducting material, the absorber precursor may be disposed between the radiator and the ground plane by spraying the absorber precursor onto the ground plane **904**.

(552) In one implementation, affixing the EM-absorbing material to the antenna (step **7216**) may include applying the absorber precursor being fabric adjacent to the antenna. In some implementations, affixing the EM-absorbing material adjacent to the antenna (step **7216**) may include weaving the fabric around the one or more radiators **6504**. In other implementations, affixing the EM-absorbing material adjacent to the antenna (step **7216**) may include providing a slit in the fabric such that the one or more radiators **6504** may be positioned through the slit. In yet other implementations, affixing the EM-absorbing material adjacent to the antenna (step **7216**) may include providing a first fabric on a first side of the radiator **6504** of the antenna and a second fabric on a second side of the radiator **6504** of the antenna. The first fabric and the second fabric may overlap each other at a seam formed therebetween. The first fabric and the second fabric may be formed to include the same conducting materials or different conducting materials. In some implementations, the first fabric may be a woven fabric, while the second fabric may be a non-woven fabric. In some implementations, affixing the EM-absorbing material adjacent to the antenna (step **7216**) may include disposing the fabric adjacent to the antenna prior to curing the absorber precursor such that the absorber precursor cures while in contact a substrate adjacent to the antenna to bond the EM-absorbing material to the substrate (such as to the fifth ground plane **904***e*).

(553) In one implementation, affixing the EM-absorbing material adjacent to the antenna (step **7216**) may include applying the absorber precursor being a foam to a substrate adjacent to the antenna, such as the fifth ground plane **904***e*. In some implementations, the foam may be cured prior to affixing the foam to the fifth ground plane **904***e*. For example, the foam may be cured and

the one or more openings **6510** formed in the foam prior to disposing the foam around the radiators **6504** of the antenna (e.g., as shown and described in reference to FIG. **65**). In other implementations, affixing the EM-absorbing material adjacent to the antenna (step **7216**) may include spraying a mixture of the foam component parts onto the fifth ground plane **904***e* and allowing the foam component parts to polymerize to form a foam formed in place on the fifth ground plane **904***e*. In some implementations, excess foam formed in place on the fifth ground plane **904***e* may be removed, such as by cutting the foam.

(554) In some implementations, affixing the EM-absorbing material adjacent to the antenna (step **7216**) may include dipping one or more of the fifth ground plane **904***e* and the hollow waveguide **208** into absorber precursor such that the absorber precursor coats particular areas of the fifth ground plane **904***e* and the hollow waveguide (e.g., as shown and described in reference to FIG. **66**). In some implementations, affixing the EM-absorbing material to the fifth ground plane **904***e* (step **7216**) may include dipping the one or more of the fifth ground plane **904***e* and the hollow waveguide more than one time into absorber precursor until the EM-absorbing material disposed on the fifth ground plane **904***e* and/or the hollow waveguide reaches a desired thickness. In some implementations, the absorber precursor is allowed to cure between each dipping iteration.

(555) In one implementation, positioning an electromagnetic absorber around the antenna (step **7108**) does not include positioning the electromagnetic absorber between the radiator **6504** of the antenna and the hollow waveguide **208**.

ILLUSTRATIVE CLAUSES

(556) The following are illustrative clauses demonstrating non-limiting implementations of the present disclosure:

(557) Illustrative clause 1. A transmitter, comprising: a client-side input configured to receive one or more baseband signals having client data encoded therein; transmitter circuitry configured to receive the one or more baseband signals from the client-side input and generate one or more antenna feed signals based on the one or more baseband signals; and one or more antennas configured to receive the one or more antenna feed signals from the transmitter circuitry, generate one or more radiated signals based on the one or more antenna feed signals, and couple the one or more radiated signals into a hollow waveguide, each of the one or more radiated signals being radiated electromagnetic waves configured for coherent detection and having a frequency in a range between 300 Gigahertz (GHz) and 10 Terahertz (THz).

(558) Illustrative clause 2. The transmitter of illustrative clause 1, wherein the hollow waveguide has a hollow waveguide core having a refractive index in a range between 1.0 and 1.4.

(559) Illustrative clause 3. The transmitter of illustrative clause 1, wherein the hollow waveguide has a hollow waveguide core and a tubular sidewall surrounding the hollow waveguide core, the hollow waveguide core being filled with one of a gas, a vacuum, and a porous material having a porosity in a range between 25% and 99%.

(560) Illustrative clause 4. The transmitter of illustrative clause 3, wherein the tubular sidewall comprises a conductive layer.

(561) Illustrative clause 5. The transmitter of illustrative clause 4, wherein the tubular sidewall further comprises a support layer surrounding the conductive layer.

(562) Illustrative clause 6. The transmitter of illustrative clause 4, wherein the tubular sidewall further comprises a dielectric layer between the hollow waveguide core and the conductive layer.

(563) Illustrative clause 7. The transmitter of illustrative clause 3, wherein the tubular sidewall has one or more conductive layers and one or more dielectric layers, the one or more conductive layers interleaved with the one or more dielectric layers.

(564) Illustrative clause 8. The transmitter of illustrative clause 1, wherein each particular one of the one or more radiated signals has a bandwidth in a range between 10% and 40% of the frequency of the particular one of the one or more radiated signals.

(565) Illustrative clause 9. The transmitter of illustrative clause 1, wherein the hollow waveguide is

configured to support propagation of a single mode of the one or more radiated signals.
(566) Illustrative clause 10. The transmitter of illustrative clause 1, wherein the hollow waveguide is configured to support propagation of a plurality of modes of the one or more radiated signals.
(567) Illustrative clause 11. The transmitter of illustrative clause 1, the one or more antenna feed signals are provided to the one or more antennas on one or more transmission lines, each of the one or more transmission lines having two or more conductors.
(568) Illustrative clause 12. The transmitter of illustrative clause 11, wherein each of the one or more transmission lines have a first transmission loss and the hollow waveguide has a second transmission loss less than the first transmission loss, the second transmission loss being in a range between 0.001 and 20.00 decibels (dB) per meter (m) per Terabit (Tb) per second (s).
(569) Illustrative clause 13. The transmitter of illustrative clause 1, wherein two or more of the client-side input, the transmitter circuitry, and one or more antennas are disposed on a single substrate.
(570) Illustrative clause 14. The transmitter of illustrative clause 13, wherein at least two of the client-side input, the transmitter circuitry, and the one or more antennas are disposed on a multi-layer substrate having a plurality of layers, at least one of the client-side input, the transmitter circuitry, and the one or more antennas being disposed on a first layer of the plurality of layers, at least one of the client-side input, the transmitter circuitry, and the one or more antennas being disposed on a second layer of the plurality of layers.
(571) Illustrative clause 15. The transmitter of illustrative clause 13, wherein at least two of the client-side input, the transmitter circuitry, and the one or more antennas are integrated into a single monolithic semiconductor die.
(572) Illustrative clause 16. The transmitter of illustrative clause 1, wherein at least two of the client-side input, the transmitter circuitry, and the one or more antennas are disposed on a plurality of substrates, at least one of the client-side input, the transmitter circuitry, and the one or more antennas being disposed on a first substrate of the plurality of substrates, at least one of the client-side input, the transmitter circuitry, and the one or more antennas being disposed on a second substrate of the plurality of substrates.
(573) Illustrative clause 17. The transmitter of illustrative clause 16, wherein at least two of the plurality of substrates are in a stacked arrangement.
(574) Illustrative clause 18. The transmitter of illustrative clause 13, wherein at least one of the client-side input, the transmitter circuitry, and the one or more antennas are not disposed on the single substrate.
(575) Illustrative clause 19. The transmitter of illustrative clause 1, wherein each of the client-side input, the transmitter circuitry, and the one or more antennas are implemented using one or more of complementary metal-oxide semiconductor (CMOS) technology, silicon-germanium (SiGe) semiconductor technology, and Ill-V compound semiconductor technology.
(576) Illustrative clause 20. The transmitter of illustrative clause 1, wherein the client data is encoded in the one or more baseband signals using an encoding protocol conforming to requirements of one or more of return-to-zero (RZ) code, non-return-to-zero (NRZ) code, pulse-amplitude modulation (PAM), and quadrature-amplitude modulation (QAM).
(577) Illustrative clause 21. The transmitter of illustrative clause 1, wherein the client data is encoded in the one or more radiated signals using an encoding protocol conforming to requirements of one or more of return-to-zero (RZ) code, non-return-to-zero (NRZ) code, quadrature phase-shift keying (QPSK), quadrature-amplitude modulation (QAM), trellis coded modulation (TCM), and Bose-Chaudhuri-Hocquenghem (BCH) code.
(578) Illustrative clause 22. The transmitter of illustrative clause 1, wherein the one or more radiated signals are a plurality of radiated signals including a first complementary radiated signal having a first polarization and a second complementary radiated signal having a second polarization different from the first polarization, the one or more antennas being further configured

to generate the first complementary radiated signal and the second complementary radiated signal based on the one or more antenna feed signals.
(579) Illustrative clause 23. The transmitter of illustrative clause 22, wherein the first polarization is orthogonal to the second polarization.
(580) Illustrative clause 24. The transmitter of illustrative clause 23, wherein each of the first polarization and the second polarization is a linear polarization.
(581) Illustrative clause 25. The transmitter of illustrative clause 24, wherein each of the one or more antennas is one of a differential waveguide probe antenna, a differential tapered antenna, and a differential patch antenna.
(582) Illustrative clause 26. The transmitter of illustrative clause 23, wherein each of the first polarization and the second polarization is a circular polarization.
(583) Illustrative clause 27. The transmitter of illustrative clause 26, wherein each of the one or more antennas is one of a helix antenna and a spiral antenna.
(584) Illustrative clause 28. The transmitter of illustrative clause 1, wherein the one or more radiated signals are a plurality of radiated signals including a first complementary radiated signal having a first polarization, a second complementary radiated signal having a second polarization different from the first polarization, and a combined radiated signal, the one or more antennas being further configured to couple the first complementary radiated signal having the first polarization and the second complementary radiated signal having the second polarization in the hollow waveguide such that the first complementary radiated signal and the second complementary radiated signal interact in the hollow waveguide to form the combined radiated signal having a third polarization different from the first polarization and the second polarization.
(585) Illustrative clause 29. The transmitter of illustrative clause 28, wherein the one or more antennas are an antenna array comprising a plurality of antennas.
(586) Illustrative clause 30. The transmitter of illustrative clause 1, wherein the one or more baseband signals include a plurality of parallel baseband signals and a serial baseband signal, the transmitter further comprising a serializer configured to receive the plurality of parallel baseband signals and combine the plurality of parallel baseband signals into the serial baseband signal, the client-side input being configured to receive the serial baseband signal, the transmitter circuitry being configured to receive the serial baseband signal from the client-side input and generate the one or more antenna feed signals based on the serial baseband signal.
(587) Illustrative clause 31. The transmitter of illustrative clause 30, wherein combining the plurality of parallel baseband signals into the serial baseband signal utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM), and wavelength division multiplexing (WDM).
(588) Illustrative clause 32. The transmitter of illustrative clause 1, wherein the one or more baseband signals include a plurality of parallel baseband signals and a serial baseband signal, the transmitter further comprising a deserializer configured to receive the serial baseband signal and split the serial baseband signal into the plurality of parallel baseband signals, the client-side input being configured to receive the plurality of parallel baseband signals, the transmitter circuitry configured to receive the plurality of parallel baseband signals from the client-side input and generate the one or more antenna feed signals based on the plurality of parallel baseband signals.
(589) Illustrative clause 33. The transmitter of illustrative clause 32, wherein splitting the serial baseband signal into the plurality of parallel baseband signals utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM), and wavelength division multiplexing (WDM).
(590) Illustrative clause 34. The transmitter of illustrative clause 1, wherein the hollow waveguide core has a cross-section configured to support propagation of a plurality of polarizations.
(591) Illustrative clause 35. The transmitter of illustrative clause 34, wherein the cross-section of the hollow waveguide core has an elliptical or circular shape.

(592) Illustrative clause 36. The transmitter of illustrative clause 34, wherein the cross-section of the hollow waveguide core has a rectangular or square shape.
(593) Illustrative clause 37. The transmitter of illustrative clause 34, wherein the cross-section of the hollow waveguide core has a cross shape.
(594) Illustrative clause 38. The transmitter of illustrative clause 1, wherein the frequency of the one or more radiated signals is a transmission frequency, the transmitter circuitry comprising: one or more local oscillators configured to generate one or more carrier signals, each of the one or more carrier signals having a baseband frequency less than the transmission frequency; one or more modulation circuits configured to receive the one or more baseband signals from the client-side input and the one or more carrier signals from the one or more local oscillators and modulate the one or more baseband signals onto the one or more carrier signals to generate one or more modulated signals; and one or more up-conversion circuits configured to receive the one or more modulated signals from the one or more modulation circuits and up-convert the one or more modulated signals to generate the one or more antenna feed signals, each of the one or more antenna feed signals having the transmission frequency.
(595) Illustrative clause 39. The transmitter of illustrative clause 1, wherein the one or more baseband signals are a plurality of baseband signals, the one or more antenna feed signals being a plurality of antenna feed signals including a combined antenna feed signal, the one or more radiated signals including a combined radiated signal, the frequency of the one or more radiated signals being a transmission frequency, the transmitter circuitry comprising: a plurality of local oscillators configured to generate a plurality of carrier signals, each of the plurality of carrier signals having a baseband frequency less than the transmission frequency; a plurality of modulation circuits configured to receive the plurality of baseband signals from the client-side input and the plurality of carrier signals from the plurality of local oscillators and modulate the plurality of baseband signals onto the plurality of carrier signals to generate a plurality of modulated signals; a plurality of up-conversion circuits configured to receive the plurality of modulated signals from the plurality of modulation circuits and up-convert the plurality of modulated signals to generate a plurality of up-converted signals; and a combiner configured to receive the plurality of up-converted signals from the plurality of up-conversion circuits and combine the plurality of up-converted signals into the combined antenna feed signal; wherein the one or more antennas are configured to receive the combined antenna feed signal from the combiner, generate the combined radiated signal based on the combined antenna feed signal, and couple the combined radiated signal into the hollow waveguide.
(596) Illustrative clause 40. The transmitter of illustrative clause 39, wherein combining the plurality of up-converted signals into the combined antenna feed signal utilizes at least one of time division multiplexing (TDM) and wavelength division multiplexing (WDM).
(597) Illustrative clause 41. The transmitter of illustrative clause 1, wherein the one or more baseband signals are a plurality of baseband signals, the one or more antenna feed signals being a plurality of antenna feed signals, the one or more radiated signals being a plurality of radiated signals including a combined radiated signal, the frequency of the one or more radiated signals being a transmission frequency, the one or more antennas being an antenna array comprising a plurality of antennas, the transmitter circuitry comprising: a plurality of local oscillators configured to generate a plurality of carrier signals, each of the plurality of carrier signals having a baseband frequency less than the transmission frequency; a plurality of modulation circuits configured to receive the plurality of baseband signals from the client-side input and the plurality of carrier signals from the plurality of local oscillators and modulate the plurality of baseband signals onto the plurality of carrier signals to generate a plurality of modulated signals; and a plurality of up-conversion circuits configured to receive the plurality of modulated signals from the plurality of modulation circuits and up-convert the plurality of modulated signals to generate the plurality of antenna feed signals; wherein the plurality of antennas are configured to receive the plurality of

antenna feed signals from the plurality of up-conversion circuits, generate the plurality of radiated signals based on the plurality of antenna feed signals, and couple the plurality of radiated signals into the hollow waveguide such that the plurality of radiated signals interact in the hollow waveguide to form the combined radiated signal.
(598) Illustrative clause 42. The transmitter of illustrative clause 41, wherein coupling the plurality of radiated signals into the hollow waveguide such that the plurality of radiated signals interact in the hollow waveguide to form the combined radiated signal utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM) and wavelength division multiplexing (WDM).
(599) Illustrative clause 43. A receiver, comprising: one or more antennas configured to detect one or more radiated signals received from a hollow waveguide and generate one or more antenna output signals based on the one or more radiated signals, each of the one or more radiated signals being radiated electromagnetic waves configured for coherent detection, having a frequency in a range between 300 Gigahertz (GHz) and 10 Terahertz (THz), and having client data encoded therein; receiver circuitry configured to receive the one or more antenna output signals from the one or more antennas and generate one or more baseband signals based on the one or more antenna output signals; and a client-side output configured to receive the one or more baseband signals from the receiver circuitry and transmit the one or more baseband signals.
(600) Illustrative clause 44. The receiver of illustrative clause 43, wherein the hollow waveguide has a hollow waveguide core having a refractive index in a range between 1.0 and 1.4.
(601) Illustrative clause 45. The receiver of illustrative clause 43, wherein the hollow waveguide has a hollow waveguide core and a tubular sidewall surrounding the hollow waveguide core, the hollow waveguide core being filled with one of a gas, a vacuum, and a porous material having a porosity in a range between 25% and 99%.
(602) Illustrative clause 46. The receiver of illustrative clause 45, wherein the tubular sidewall comprises a conductive layer.
(603) Illustrative clause 47. The receiver of illustrative clause 46, wherein the tubular sidewall further comprises a support layer surrounding the conductive layer.
(604) Illustrative clause 48. The receiver of illustrative clause 46, wherein the tubular sidewall further comprises a dielectric layer between the hollow waveguide core and the conductive layer.
(605) Illustrative clause 49. The receiver of illustrative clause 45, wherein the tubular sidewall has one or more conductive layers and one or more dielectric layers, the one or more conductive layers interleaved with the one or more dielectric layers.
(606) Illustrative clause 50. The receiver of illustrative clause 43, wherein each particular one of the one or more radiated signals has a bandwidth in a range between 10% and 40% of the frequency of the particular one of the one or more radiated signals.
(607) Illustrative clause 51. The receiver of illustrative clause 43, wherein the hollow waveguide is configured to support propagation of a single mode of the one or more radiated signals.
(608) Illustrative clause 52. The receiver of illustrative clause 43, wherein the hollow waveguide is configured to support propagation of a plurality of modes of the one or more radiated signals.
(609) Illustrative clause 53. The receiver of illustrative clause 43, the one or more antenna output signals are received from the one or more antennas on one or more transmission lines, each of the one or more transmission lines having two or more conductors.
(610) Illustrative clause 54. The receiver of illustrative clause 53, wherein each of the one or more transmission lines have a first transmission loss and the hollow waveguide has a second transmission loss less than the first transmission loss, the second transmission loss being in a range between 0.001 and 20.00 decibels (dB) per meter (m) per Terabit (Tb) per second (s).
(611) Illustrative clause 55. The receiver of illustrative clause 43, wherein two or more of the client-side output, the receiver circuitry, and the one or more antennas are disposed on a single substrate.

(612) Illustrative clause 56. The receiver of illustrative clause 55, wherein at least two of the client-side output, the receiver circuitry, and the one or more antennas are disposed on a multi-layer substrate having a plurality of layers, at least one of the client-side output, the receiver circuitry, and the one or more antennas being disposed on a first layer of the plurality of layers, at least one of the client-side output, the receiver circuitry, and the one or more antennas being disposed on a second layer of the plurality of layers.
(613) Illustrative clause 57. The receiver of illustrative clause 55, wherein at least two of the client-side output, the receiver circuitry, and the one or more antennas are integrated into a single monolithic semiconductor die.
(614) Illustrative clause 58. The receiver of illustrative clause 43, wherein at least two of the client-side output, the receiver circuitry, and the one or more antennas are disposed on a plurality of substrates, at least one of the client-side output, the receiver circuitry, and the one or more antennas being disposed on a first substrate of the plurality of substrates, at least one of the client-side output, the receiver circuitry, and the one or more antennas being disposed on a second substrate of the plurality of substrates.
(615) Illustrative clause 59. The receiver of illustrative clause 58, wherein at least two of the plurality of substrates are in a stacked arrangement.
(616) Illustrative clause 60. The receiver of illustrative clause 55, wherein at least one of the client-side output, the receiver circuitry, and the one or more antennas are not disposed on the single substrate.
(617) Illustrative clause 61. The receiver of illustrative clause 43, wherein each of the client-side output, the receiver circuitry, and the one or more antennas are implemented using one or more of complementary metal-oxide semiconductor (CMOS) technology, silicon-germanium (SiGe) semiconductor technology, and Ill-V compound semiconductor technology.
(618) Illustrative clause 62. The receiver of illustrative clause 43, wherein the client data is encoded in the one or more baseband signals using an encoding conforming to one or more of return-to-zero (RZ) code, non-return-to-zero (NRZ) code, pulse-amplitude modulation (PAM), and quadrature-amplitude modulation (QAM).
(619) Illustrative clause 63. The receiver of illustrative clause 43, wherein the client data is encoded in the one or more radiated signals using an encoding conforming to one or more of return-to-zero (RZ) code, non-return-to-zero (NRZ) code, quadrature phase-shift keying (QPSK), quadrature-amplitude modulation (QAM), trellis coded modulation (TCM), and Bose-Chaudhuri-Hocquenghem (BCH) code.
(620) Illustrative clause 64. The receiver of illustrative clause 43, wherein the one or more radiated signals are a plurality of radiated signals including a first complementary radiated signal having a first polarization and a second complementary radiated signal having a second polarization different from the first polarization, the one or more antennas being further configured to generate the one or more antenna output signals based on the first complementary radiated signal and the second complementary radiated signal.
(621) Illustrative clause 65. The receiver of illustrative clause 64, wherein the first polarization is orthogonal to the second polarization.
(622) Illustrative clause 66. The receiver of illustrative clause 65, wherein each of the first polarization and the second polarization is a linear polarization.
(623) Illustrative clause 67. The receiver of illustrative clause 66, wherein each of the one or more antennas is one of a differential waveguide probe antenna, a differential tapered antenna, and a differential patch antenna.
(624) Illustrative clause 68. The receiver of illustrative clause 65, wherein each of the first polarization and the second polarization is a circular polarization.
(625) Illustrative clause 69. The receiver of illustrative clause 68, wherein each of the one or more antennas is one of a helix antenna and a spiral antenna.

(626) Illustrative clause 70. The receiver of illustrative clause 43, wherein the one or more radiated signals includes a first complementary radiated signal having a first polarization, a second complementary radiated signal having a second polarization different from the first polarization, and a combined radiated signal having a third polarization different from the first polarization and the second polarization, the combined radiated signal being formed by the first complementary radiated signal and the second complementary radiated signal interacting in the hollow waveguide, the one or more antennas being further configured to detect the combined radiated signal received from the hollow waveguide and generate the one or more antenna output signals based on the combined radiated signal.
(627) Illustrative clause 71. The receiver of illustrative clause 70, wherein the one or more antennas are an antenna array comprising a plurality of antennas.
(628) Illustrative clause 72. The receiver of illustrative clause 43, wherein the one or more baseband signals include a plurality of parallel baseband signals and a serial baseband signal, the receiver circuitry being configured to generate the serial baseband signal based on the one or more antenna output signals, the client-side output being configured to receive the serial baseband signal from the receiver circuitry and transmit the serial baseband signal, the receiver further comprising a deserializer configured to receive the serial baseband signal from the client-side output and split the serial baseband signal into the plurality of parallel baseband signals.
(629) Illustrative clause 73. The receiver of illustrative clause 72, wherein splitting the serial baseband signal into the plurality of parallel baseband signals utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM), and wavelength division multiplexing (WDM).
(630) Illustrative clause 74. The receiver of illustrative clause 43, wherein the one or more baseband signals include a plurality of parallel baseband signals and a serial baseband signal, the receiver circuitry being configured to generate the plurality of parallel baseband signals based on the one or more antenna output signals, the client-side output being configured to receive the plurality of parallel baseband signals from the receiver circuitry and transmit the plurality of parallel baseband signals, the receiver further comprising a serializer configured to receive the plurality of parallel baseband signals and combine the plurality of parallel baseband signals into the serial baseband signal.
(631) Illustrative clause 75. The receiver of illustrative clause 74, wherein combining the plurality of parallel baseband signals into the serial baseband signal utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM), and wavelength division multiplexing (WDM).
(632) Illustrative clause 76. The receiver of illustrative clause 43, wherein the hollow waveguide core has a cross-section configured to support propagation of a plurality of polarizations.
(633) Illustrative clause 77. The receiver of illustrative clause 76, wherein the cross-section of the hollow waveguide core has an elliptical or circular shape.
(634) Illustrative clause 78. The receiver of illustrative clause 76, wherein the cross-section of the hollow waveguide core has a rectangular or square shape.
(635) Illustrative clause 79. The receiver of illustrative clause 76, wherein the cross-section of the hollow waveguide core has a cross shape.
(636) Illustrative clause 80. The receiver of illustrative clause 43, wherein the frequency of the one or more radiated signals is a transmission frequency, the receiver circuitry comprising: one or more local oscillators configured to generate one or more reference signals, each of the one or more reference signals having a baseband frequency less than the transmission frequency; one or more down-conversion circuits configured to receive the one or more antenna output signals from the one or more antennas and the one or more reference signals from the one or more local oscillators and down-convert the one or more antenna output signals using the one or more reference signals to generate one or more modulated signals, each of the one or more modulated signals having the

baseband frequency; and one or more demodulation circuits configured to receive the one or more modulated signals from the one or more down-conversion circuits and demodulate the one or more modulated signals to generate the one or more baseband signals.

(637) Illustrative clause 81. The receiver of illustrative clause 43, wherein the one or more baseband signals are a plurality of baseband signals, the one or more antenna output signals being a plurality of antenna output signals including a combined antenna output signal, the one or more radiated signals including a combined radiated signal, the frequency of the one or more radiated signals being a transmission frequency, the one or more antennas being configured to detect the combined radiated signal received from the hollow waveguide and generate the combined antenna output signal based on the combined radiated signal, the receiver circuitry comprising: a splitter configured to receive the combined antenna output signal from the one or more antennas and split the combined antenna output signal into the plurality of antenna output signals; a plurality of local oscillators configured to generate a plurality of reference signals, each of the plurality of reference signals having a baseband frequency less than the transmission frequency; a plurality of down-conversion circuits configured to receive the plurality of antenna output signals from the splitter and the plurality of reference signals from the plurality of local oscillators and down-convert the plurality of antenna output signals using the plurality of reference signals to generate a plurality of modulated signals, each of the plurality of modulated signals having the baseband frequency; and a plurality of demodulation circuits configured to receive the plurality of modulated signals from the plurality of down-conversion circuits and demodulate the plurality of modulated signals to generate the plurality of baseband signals.

(638) Illustrative clause 82. The receiver of illustrative clause 81, wherein splitting the combined antenna output signal into the plurality of antenna output signals utilizes at least one of time division multiplexing (TDM) and wavelength division multiplexing (WDM).

(639) Illustrative clause 83. The receiver of illustrative clause 43, wherein the one or more baseband signals are a plurality of baseband signals, the one or more antenna output signals being a plurality of antenna output signals, the one or more radiated signals being a plurality of radiated signals including a first complementary radiated signal having a first polarization, a second complementary radiated signal having a second polarization different from the first polarization, and a combined radiated signal having a third polarization different from the first polarization and the second polarization, the combined radiated signal being formed by the first complementary radiated signal and the second complementary radiated signal interacting in the hollow waveguide, the frequency of the one or more radiated signals being a transmission frequency, the one or more antennas being an antenna array comprising a plurality of antennas, the plurality of antennas being configured to detect the first complementary radiated signal and the second complementary radiated signal based on the combined radiated signal received from the hollow waveguide and generate the plurality of antenna output signals based on the first complementary radiated signal and the second complementary radiated signal, the receiver circuitry comprising: a plurality of local oscillators configured to generate a plurality of reference signals, each of the plurality of reference signals having a baseband frequency less than the transmission frequency; a plurality of down-conversion circuits configured to receive the plurality of antenna output signals from the plurality of antennas and the plurality of reference signals from the plurality of local oscillators and down-convert the plurality of antenna output signals using the plurality of reference signals to generate a plurality of modulated signals, each of the plurality of modulated signals having the baseband frequency; and a plurality of demodulation circuits configured to receive the plurality of modulated signals from the plurality of down-conversion circuits and demodulate the plurality of modulated signals to generate the plurality of baseband signals.

(640) Illustrative clause 84. The receiver of illustrative clause 83, wherein detecting the first complementary radiated signal and the second complementary radiated signal based on the combined radiated signal received from the hollow waveguide utilizes at least one of polarization

division multiplexing (PDM), time division multiplexing (TDM) and wavelength division multiplexing (WDM).

(641) Illustrative clause 85. A transport network, comprising: one or more hollow waveguides; a transmitter, comprising: a client-side input configured to receive one or more first baseband signals having client data encoded therein; transmitter circuitry configured to receive the one or more first baseband signals from the client-side input and generate one or more antenna feed signals based on the one or more first baseband signals; and one or more first antennas configured to receive the one or more antenna feed signals from the transmitter circuitry, generate one or more radiated signals based on the one or more antenna feed signals, and couple the one or more radiated signals into at least one of the one or more hollow waveguides, each of the one or more radiated signals being radiated electromagnetic waves configured for coherent detection and having a frequency in a range between 300 Gigahertz (GHz) and 10 Terahertz (THz); and a receiver, comprising: one or more second antennas configured to detect the one or more radiated signals received from the at least one of the one or more hollow waveguides and generate one or more antenna output signals based on the one or more radiated signals; receiver circuitry configured to receive the one or more antenna output signals from the one or more second antennas and generate one or more second baseband signals based on the one or more antenna output signals, the one or more second baseband signals having the client data; and a client-side output configured to receive the one or more second baseband signals from the receiver circuitry and transmit the one or more second baseband signals.

(642) Illustrative clause 86. The transport network of illustrative clause 85, wherein the at least one of the one or more hollow waveguides has a hollow waveguide core having a refractive index in a range between 1.0 and 1.4.

(643) Illustrative clause 87. The transport network of illustrative clause 85, wherein the at least one of the one or more hollow waveguides has a hollow waveguide core and a tubular sidewall surrounding the hollow waveguide core, the hollow waveguide core being filled with one of a gas, a vacuum, and a porous material having a porosity in a range between 25% and 99%.

(644) Illustrative clause 88. The transport network of illustrative clause 87, wherein the tubular sidewall of the at least one of the one or more hollow waveguides comprises a conductive layer.

(645) Illustrative clause 89. The transport network of illustrative clause 88, wherein the tubular sidewall of the at least one of the one or more hollow waveguides further comprises a support layer surrounding the conductive layer.

(646) Illustrative clause 90. The transport network of illustrative clause 88, wherein the tubular sidewall of the at least one of the one or more hollow waveguides further comprises a dielectric layer between the hollow waveguide core and the conductive layer.

(647) Illustrative clause 91. The transport network of illustrative clause 87, wherein the tubular sidewall of the at least one of the one or more hollow waveguides has one or more conductive layers and one or more dielectric layers, the one or more conductive layers interleaved with the one or more dielectric layers.

(648) Illustrative clause 92. The transport network of illustrative clause 85, wherein each particular one of the one or more radiated signals has a bandwidth in a range between 10% and 40% of the frequency of the particular one of the one or more radiated signals.

(649) Illustrative clause 93. The transport network of illustrative clause 85, wherein the at least one of the one or more hollow waveguides is configured to support propagation of a single mode of the one or more radiated signals.

(650) Illustrative clause 94. The transport network of illustrative clause 85, wherein the at least one of the one or more hollow waveguides is configured to support propagation of a plurality of modes of the one or more radiated signals.

(651) Illustrative clause 95. The transport network of illustrative clause 85, the one or more antenna feed signals are provided to the one or more first antennas on one or more first transmission lines,

each of the one or more first transmission lines having two or more conductors.
(652) Illustrative clause 96. The transport network of illustrative clause 95, wherein each of the one or more first transmission lines have a first transmission loss and the at least one of the one or more hollow waveguides has a second transmission loss less than the first transmission loss, the second transmission loss being in a range between 0.001 and 20.00 decibels (dB) per meter (m) per Terabit (Tb) per second (s).
(653) Illustrative clause 97. The transport network of illustrative clause 85, the one or more antenna output signals are received from the one or more second antennas on one or more second transmission lines, each of the one or more second transmission lines having two or more conductors.
(654) Illustrative clause 98. The transport network of illustrative clause 97, wherein each of the one or more second transmission lines have a first transmission loss and the at least one of the one or more hollow waveguides has a second transmission loss less than the first transmission loss, the second transmission loss being in a range between 0.001 and 20.00 decibels (dB) per meter (m) per Terabit (Tb) per second (s).
(655) Illustrative clause 99. The transport network of illustrative clause 85, wherein two or more of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are disposed on a single substrate.
(656) Illustrative clause 100. The transport network of illustrative clause 99, wherein at least two of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are disposed on a multi-layer substrate having a plurality of layers, at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas being disposed on a first layer of the plurality of layers, at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas being disposed on a second layer of the plurality of layers.
(657) Illustrative clause 101. The transport network of illustrative clause 99, wherein at least two of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are integrated into a single monolithic semiconductor die.
(658) Illustrative clause 102. The transport network of illustrative clause 85, wherein at least two of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are disposed on a plurality of substrates, at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas being disposed on a first substrate of the plurality of substrates, at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas being disposed on a second substrate of the plurality of substrates.
(659) Illustrative clause 103. The transport network of illustrative clause 102, wherein at least two of the plurality of substrates are in a stacked arrangement.
(660) Illustrative clause 104. The transport network of illustrative clause 99, wherein at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are not disposed on the single substrate.
(661) Illustrative clause 105. The transport network of illustrative clause 85, wherein each of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are implemented using one or more of complementary metal-oxide semiconductor (CMOS) technology, silicon-germanium (SiGe) semiconductor technology, and Ill-V compound semiconductor technology.
(662) Illustrative clause 106. The transport network of illustrative clause 85, wherein the client data

is encoded in the one or more first baseband signals and the one or more second baseband signals using an encoding protocol conforming to requirements of one or more of return-to-zero (RZ) code, non-return-to-zero (NRZ) code, pulse-amplitude modulation (PAM), and quadrature-amplitude modulation (QAM).
(663) Illustrative clause 107. The transport network of illustrative clause 85, wherein the client data is encoded in the one or more radiated signals using an encoding protocol conforming to requirements of one or more of return-to-zero (RZ) code, non-return-to-zero (NRZ) code, quadrature phase-shift keying (QPSK), quadrature-amplitude modulation (QAM), trellis coded modulation (TCM), and Bose-Chaudhuri-Hocquenghem (BCH) code.
(664) Illustrative clause 108. The transport network of illustrative clause 85, wherein the one or more radiated signals are a plurality of radiated signals including a first complementary radiated signal having a first polarization and a second complementary radiated signal having a second polarization different from the first polarization, the one or more first antennas being further configured to generate the first complementary radiated signal and the second complementary radiated signal based on the one or more antenna feed signals, the one or more second antennas being further configured to generate the one or more antenna output signals based on the first complementary radiated signal and the second complementary radiated signal.
(665) Illustrative clause 109. The transport network of illustrative clause 108, wherein the first polarization is orthogonal to the second polarization.
(666) Illustrative clause 110. The transport network of illustrative clause 109, wherein each of the first polarization and the second polarization is a linear polarization.
(667) Illustrative clause 111. The transport network of illustrative clause 110, wherein each of the one or more first antennas and the one or more second antennas is one of a differential waveguide probe antenna, a differential tapered antennas, and a differential patch antenna.
(668) Illustrative clause 112. The transport network of illustrative clause 109, wherein each of the first polarization and the second polarization is a circular polarization.
(669) Illustrative clause 113. The transport network of illustrative clause 112, wherein each the one or more first antennas and the one or more second antennas is one of a helix antenna and a spiral antenna.
(670) Illustrative clause 114. The transport network of illustrative clause 85, wherein the one or more radiated signals are a plurality of radiated signals including a first complementary radiated signal having a first polarization, a second complementary radiated signal having a second polarization different from the first polarization, and a combined radiated signal, the one or more first antennas being further configured to couple the first complementary radiated signal having the first polarization and the second complementary radiated signal having the second polarization in the at least one of the one or more hollow waveguides such that the first complementary radiated signal and the second complementary radiated signal interact in the at least one of the one or more hollow waveguides to form the combined radiated signal having a third polarization different from the first polarization and the second polarization, the one or more second antennas being further configured to detect the combined radiated signal received from the at least one of the one or more hollow waveguides and generate the one or more antenna output signals based on the combined radiated signal.
(671) Illustrative clause 115. The transport network of illustrative clause 114, wherein the one or more first antennas are a first antenna array comprising a first plurality of antennas and the one or more second antennas are a second antenna array comprising a second plurality of antennas.
(672) Illustrative clause 116. The transport network of illustrative clause 85, wherein the one or more first baseband signals include a plurality of first parallel baseband signals and a first serial baseband signal and the one or more second baseband signals include a plurality of second parallel baseband signals and a second serial baseband signal, the transmitter further comprising a serializer configured to receive the plurality of first parallel baseband signals and combine the plurality of

first parallel baseband signals into the first serial baseband signal, the client-side input being configured to receive the first serial baseband signal, the transmitter circuitry being configured to receive the first serial baseband signal from the client-side input and generate the one or more antenna feed signals based on the first serial baseband signal, the receiver circuitry being configured to generate the second serial baseband signal based on the one or more antenna output signals, the client-side output being configured to receive the second serial baseband signal from the receiver circuitry and transmit the second serial baseband signal, the receiver further comprising a deserializer configured to receive the second serial baseband signal from the client-side output and split the second serial baseband signal into the plurality of second parallel baseband signals.
(673) Illustrative clause 117. The transport network of illustrative clause 116, wherein combining the plurality of first parallel baseband signals into the first serial baseband signal and splitting the second serial baseband signal into the plurality of second parallel baseband signals utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM), and wavelength division multiplexing (WDM).
(674) Illustrative clause 118. The transport network of illustrative clause 85, wherein the one or more first baseband signals include a plurality of first parallel baseband signals and a first serial baseband signal and the one or more second baseband signals include a plurality of second parallel baseband signals and a second serial baseband signal, the transmitter further comprising a deserializer configured to receive the first serial baseband signal and split the first serial baseband signal into the plurality of first parallel baseband signals, the client-side input being configured to receive the plurality of first parallel baseband signals, the transmitter circuitry configured to receive the plurality of first parallel baseband signals from the client-side input and generate the one or more antenna feed signals based on the plurality of first parallel baseband signals, the receiver circuitry being configured to generate the plurality of second parallel baseband signals based on the one or more antenna output signals, the client-side output being configured to receive the plurality of second parallel baseband signals from the receiver circuitry and transmit the plurality of second parallel baseband signals, the receiver further comprising a serializer configured to receive the plurality of second parallel baseband signals and combine the plurality of second parallel baseband signals into the second serial baseband signal.
(675) Illustrative clause 119. The transport network of illustrative clause 118, wherein splitting the first serial baseband signal into the plurality of first parallel baseband signals and combining the plurality of second parallel baseband signals into the second serial baseband signal utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM), and wavelength division multiplexing (WDM).
(676) Illustrative clause 120. The transport network of illustrative clause 85, wherein the hollow waveguide core of the at least one of the one or more hollow waveguides has a cross-section configured to support propagation of a plurality of polarizations.
(677) Illustrative clause 121. The transport network of illustrative clause 120, wherein the cross-section of the hollow waveguide core of the at least one of the one or more hollow waveguides has an elliptical or circular shape.
(678) Illustrative clause 122. The transport network of illustrative clause 120, wherein the cross-section of the hollow waveguide core of the at least one of the one or more hollow waveguides has a rectangular or square shape.
(679) Illustrative clause 123. The transport network of illustrative clause 120, wherein the cross-section of the hollow waveguide core of the at least one of the one or more hollow waveguides has a cross shape.
(680) Illustrative clause 124. The transport network of illustrative clause 85, wherein the frequency of the one or more radiated signals is a transmission frequency, the transmitter circuitry comprising: one or more local oscillators configured to generate one or more carrier signals, each

of the one or more carrier signals having a first baseband frequency less than the transmission frequency; one or more modulation circuits configured to receive the one or more first baseband signals from the client-side input and the one or more carrier signals from the one or more local oscillators and modulate the one or more first baseband signals onto the one or more carrier signals to generate one or more modulated signals; and one or more up-conversion circuits configured to receive the one or more modulated signals from the one or more modulation circuits and up-convert the one or more modulated signals to generate the one or more antenna feed signals, each of the one or more antenna feed signals having the transmission frequency.
(681) Illustrative clause 125. The transport network of illustrative clause 85, wherein the frequency of the one or more radiated signals is a transmission frequency, the receiver circuitry comprising: one or more local oscillators configured to generate one or more reference signals, each of the one or more reference signals having a baseband frequency less than the transmission frequency; one or more down-conversion circuits configured to receive the one or more antenna output signals from the one or more second antennas and the one or more reference signals from the one or more local oscillators and down-convert the one or more antenna output signals using the one or more reference signals to generate one or more modulated signals, each of the one or more modulated signals having the baseband frequency; and one or more demodulation circuits configured to receive the one or more modulated signals from the one or more down-conversion circuits and demodulate the one or more modulated signals to generate the one or more second baseband signals.
(682) Illustrative clause 126. The transport network of illustrative clause 85, wherein the one or more first baseband signals are a plurality of first baseband signals, the one or more antenna feed signals being a plurality of antenna feed signals including a combined antenna feed signal, the one or more radiated signals including a combined radiated signal, the frequency of the one or more radiated signals being a transmission frequency, the transmitter circuitry comprising: a plurality of local oscillators configured to generate a plurality of carrier signals, each of the plurality of carrier signals having a baseband frequency less than the transmission frequency; a plurality of modulation circuits configured to receive the plurality of first baseband signals from the client-side input and the plurality of carrier signals from the plurality of local oscillators and modulate the plurality of first baseband signals onto the plurality of carrier signals to generate a plurality of modulated signals; a plurality of up-conversion circuits configured to receive the plurality of modulated signals from the plurality of modulation circuits and up-convert the plurality of modulated signals to generate a plurality of up-converted signals; and a combiner configured to receive the plurality of up-converted signals from the plurality of up-conversion circuits and combine the plurality of up-converted signals into the combined antenna feed signal; wherein the one or more first antennas are configured to receive the combined antenna feed signal from the combiner, generate the combined radiated signal based on the combined antenna feed signal, and couple the combined radiated signal into the at least one of the one or more hollow waveguides.
(683) Illustrative clause 127. The transport network of illustrative clause 126, wherein combining the plurality of up-converted signals into the combined antenna feed signal utilizes at least one of time division multiplexing (TDM) and wavelength division multiplexing (WDM).
(684) Illustrative clause 128. The transport network of illustrative clause 85, wherein the one or more second baseband signals are a plurality of second baseband signals, the one or more antenna output signals being a plurality of antenna output signals including a combined antenna output signal, the one or more radiated signals including a combined radiated signal, the frequency of the one or more radiated signals being a transmission frequency, the one or more second antennas being configured to detect the combined radiated signal received from the one or more hollow waveguides and generate the combined antenna output signal based on the combined radiated signal, the receiver circuitry comprising: a splitter configured to receive the combined antenna output signal from the one or more second antennas and split the combined antenna output signal

into the plurality of antenna output signals; a plurality of local oscillators configured to generate a plurality of reference signals, each of the plurality of reference signals having a baseband frequency less than the transmission frequency; a plurality of down-conversion circuits configured to receive the plurality of antenna output signals from the splitter and the plurality of reference signals from the plurality of local oscillators and down-convert the plurality of antenna output signals using the plurality of reference signals to generate a plurality of modulated signals, each of the plurality of modulated signals having the baseband frequency; and a plurality of demodulation circuits configured to receive the plurality of modulated signals from the plurality of down-conversion circuits and demodulate the plurality of modulated signals to generate the plurality of second baseband signals.

(685) Illustrative clause 129. The transport network of illustrative clause 128, wherein splitting the combined antenna output signal into the plurality of antenna output signals utilizes at least one of time division multiplexing (TDM) and wavelength division multiplexing (WDM).

(686) Illustrative clause 130. The transport network of illustrative clause 85, wherein the one or more first baseband signals are a plurality of first baseband signals, the one or more antenna feed signals being a plurality of antenna feed signals, the one or more radiated signals being a plurality of radiated signals including a combined radiated signal, the frequency of the one or more radiated signals being a transmission frequency, the one or more first antennas being a first antenna array comprising a plurality of first antennas, the transmitter circuitry comprising: a plurality of local oscillators configured to generate a plurality of carrier signals, each of the plurality of carrier signals having a baseband frequency less than the transmission frequency; a plurality of modulation circuits configured to receive the plurality of first baseband signals from the client-side input and the plurality of carrier signals from the plurality of local oscillators and modulate the plurality of first baseband signals onto the plurality of carrier signals to generate a plurality of modulated signals; and a plurality of up-conversion circuits configured to receive the plurality of modulated signals from the plurality of modulation circuits and up-convert the plurality of modulated signals to generate the plurality of antenna feed signals; wherein the plurality of first antennas are configured to receive the plurality of antenna feed signals from the plurality of up-conversion circuits, generate the plurality of radiated signals based on the plurality of antenna feed signals, and couple the plurality of radiated signals into the at least one of the one or more hollow waveguides such that the plurality of radiated signals interact in the at least one of the one or more hollow waveguides to form the combined radiated signal.

(687) Illustrative clause 131. The transport network of illustrative clause 130, wherein coupling the plurality of radiated signals into the at least one of the one or more hollow waveguides such that the plurality of radiated signals interact in the at least one of the one or more hollow waveguides to form the combined radiated signal utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM) and wavelength division multiplexing (WDM).

(688) Illustrative clause 132. The transport network of illustrative clause 85, wherein the one or more second baseband signals are a plurality of second baseband signals, the one or more antenna output signals being a plurality of antenna output signals, the one or more radiated signals being a plurality of radiated signals including a first complementary radiated signal, a second complementary radiated signal, and a combined radiated signal formed by the first complementary radiated signal and the second complementary radiated signal interacting in the at least one of the one or more hollow waveguides, the frequency of the one or more radiated signals being a transmission frequency, the one or more second antennas being an antenna array comprising a plurality of antennas, the plurality of antennas being configured to detect the first complementary radiated signal and the second complementary radiated signal based on the combined radiated signal received from the at least one of the one or more hollow waveguides and generate the plurality of antenna output signals based on the first complementary radiated signal and the second complementary radiated signal, the receiver circuitry comprising: a plurality of local oscillators

configured to generate a plurality of reference signals, each of the plurality of reference signals having a baseband frequency less than the transmission frequency; a plurality of down-conversion circuits configured to receive the plurality of antenna output signals from the plurality of antennas and the plurality of reference signals from the plurality of local oscillators and down-convert the plurality of antenna output signals using the plurality of reference signals to generate a plurality of modulated signals, each of the plurality of modulated signals having the baseband frequency; and a plurality of demodulation circuits configured to receive the plurality of modulated signals from the plurality of down-conversion circuits and demodulate the plurality of modulated signals to generate the plurality of second baseband signals.
(689) Illustrative clause 133. The transport network of illustrative clause 132, wherein detecting the first complementary radiated signal and the second complementary radiated signal based on the combined radiated signal received from the at least one of the one or more hollow waveguides utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM) and wavelength division multiplexing (WDM).
(690) Illustrative clause 134. A transceiver, comprising: a transmitter, comprising: a client-side input configured to receive one or more first baseband signals having first client data; transmitter circuitry configured to receive the one or more first baseband signals from the client-side input and generate one or more antenna feed signals based on the one or more first baseband signals; and one or more first antennas configured to receive the one or more antenna feed signals from the transmitter circuitry, generate one or more first radiated signals based on the one or more antenna feed signals, and couple the one or more first radiated signals into a first hollow waveguide, each of the one or more first radiated signals being radiated electromagnetic waves configured for coherent detection and having a first frequency in a range between 300 Gigahertz (GHz) and 10 Terahertz (THz); and a receiver, comprising: one or more second antennas configured to detect one or more second radiated signals received from one of the first hollow waveguide and a second hollow waveguide and generate one or more antenna output signals based on the one or more second radiated signals, each of the one or more second radiated signals being radiated electromagnetic waves configured for coherent detection, having a second frequency in a range between 300 GHz and 10 THz, and having second client data; receiver circuitry configured to receive the one or more antenna output signals from the one or more second antennas and generate one or more second baseband signals based on the one or more antenna output signals; and a client-side output configured to receive the one or more second baseband signals from the receiver circuitry and transmit the one or more second baseband signals.
(691) Illustrative clause 135. The transceiver of illustrative clause 134, wherein each of the first hollow waveguide and the second hollow waveguide has a hollow waveguide core having a refractive index in a range between 1.0 and 1.4.
(692) Illustrative clause 136. The transceiver of illustrative clause 134, wherein each of the first hollow waveguide and the second hollow waveguide has a hollow waveguide core and a tubular sidewall surrounding the hollow waveguide core, the hollow waveguide core being filled with one of a gas, a vacuum, and a porous material having a porosity in a range between 25% and 99%.
(693) Illustrative clause 137. The transceiver of illustrative clause 136, wherein the tubular sidewall of each of the first hollow waveguide and the second hollow waveguide comprises a conductive layer.
(694) Illustrative clause 138. The transceiver of illustrative clause 137, wherein the tubular sidewall of each of the first hollow waveguide and the second hollow waveguide further comprises a support layer surrounding the conductive layer.
(695) Illustrative clause 139. The transceiver of illustrative clause 137, wherein the tubular sidewall of each of the first hollow waveguide and the second hollow waveguide further comprises a dielectric layer between the hollow waveguide core and the conductive layer.
(696) Illustrative clause 140. The transceiver of illustrative clause 136, wherein the tubular sidewall

of each of the first hollow waveguide and the second hollow waveguide has one or more conductive layers and one or more dielectric layers, the one or more conductive layers interleaved with the one or more dielectric layers.

(697) Illustrative clause 141. The transceiver of illustrative clause 134, wherein each particular one of the one or more first radiated signals and has a first bandwidth in a range between 10% and 40% of the first frequency of the particular one of the one or more first radiated signals and each particular one of the one or more second radiated signals and has a second bandwidth in a range between 10% and 40% of the second frequency of the particular one of the one or more second radiated signals.

(698) Illustrative clause 142. The transceiver of illustrative clause 134, wherein each of the first hollow waveguide is configured to support propagation of a single mode of the one or more first radiated signals and the second hollow waveguide is configured to support propagation of a single more of the one or more second radiated signals.

(699) Illustrative clause 143. The transceiver of illustrative clause 134, wherein the first hollow waveguide is configured to support propagation of a plurality of first modes of the one or more first radiated signals and the second hollow waveguide is configured to support propagation of a plurality of second modes of the one or more second radiated signals.

(700) Illustrative clause 144. The transceiver of illustrative clause 134, the one or more antenna feed signals are provided to the one or more first antennas on one or more first transmission lines and the one or more antenna output signals are received from the one or more second antennas on one or more second transmission lines, each of the one or more first transmission lines and the one or more second transmission lines having two or more conductors.

(701) Illustrative clause 145. The transceiver of illustrative clause 144, wherein each of the one or more first transmission lines and the one or more second transmission lines have a first transmission loss and each of the first hollow waveguide and the second hollow waveguide has a second transmission loss less than the first transmission loss, the second transmission loss being in a range between 0.001 and 20.00 decibels (dB) per meter (m) per Terabit (Tb) per second (s).

(702) Illustrative clause 146. The transceiver of illustrative clause 134, wherein two or more of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are disposed on a single substrate.

(703) Illustrative clause 147. The transceiver of illustrative clause 146, wherein at least two of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are disposed on a multi-layer substrate having a plurality of layers, at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas being disposed on a first layer of the plurality of layers, at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas being disposed on a second layer of the plurality of layers.

(704) Illustrative clause 148. The transceiver of illustrative clause 146, wherein at least two of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are integrated into a single monolithic semiconductor die.

(705) Illustrative clause 149. The transceiver of illustrative clause 134, wherein at least two of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are disposed on a plurality of substrates, at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas being disposed on a first substrate of the plurality of substrates, at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more

second antennas being disposed on a second substrate of the plurality of substrates.
(706) Illustrative clause 150. The transceiver of illustrative clause 149, wherein at least two of the plurality of substrates are in a stacked arrangement.
(707) Illustrative clause 151. The transceiver of illustrative clause 146, wherein at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are not disposed on the single substrate.
(708) Illustrative clause 152. The transceiver of illustrative clause 134, wherein each of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are implemented using one or more of complementary metal-oxide semiconductor (CMOS) technology, silicon-germanium (SiGe) semiconductor technology, and Ill-V compound semiconductor technology.
(709) Illustrative clause 153. The transceiver of illustrative clause 134, wherein the first client data is encoded in the one or more first baseband signals and the second client data is encoded in the one or more second baseband signals using an encoding protocol conforming to requirements of one or more of return-to-zero (RZ) code, non-return-to-zero (NRZ) code, pulse-amplitude modulation (PAM), and quadrature-amplitude modulation (QAM).
(710) Illustrative clause 154. The transceiver of illustrative clause 134, wherein the first client data is encoded in the one or more first radiated signals and the second client data is encoded in the one or more second radiated signals using an encoding protocol conforming to requirements of one or more of return-to-zero (RZ) code, non-return-to-zero (NRZ) code, quadrature phase-shift keying (QPSK), quadrature-amplitude modulation (QAM), trellis coded modulation (TCM), and Bose-Chaudhuri-Hocquenghem (BCH) code.
(711) Illustrative clause 155. The transceiver of illustrative clause 134, wherein the one or more first radiated signals are a plurality of first radiated signals including a first complementary radiated signal having a first polarization and a second complementary radiated signal having a second polarization different from the first polarization and the one or more second radiated signals are a plurality of second radiated signals including a third complementary radiated signal having a third polarization and a fourth complementary radiated signal having a fourth polarization different from the third polarization, the one or more first antennas being further configured to generate the first complementary radiated signal and the second complementary radiated signal based on the one or more antenna feed signals, the one or more second antennas being further configured to generate the one or more antenna output signals based on the third complementary radiated signal and the fourth complementary radiated signal.
(712) Illustrative clause 156. The transceiver of illustrative clause 155, wherein the first polarization is orthogonal to the second polarization and the third polarization is orthogonal to the fourth polarization.
(713) Illustrative clause 157. The transceiver of illustrative clause 156, wherein each of the first polarization, the second polarization, the third polarization, and the fourth polarization is a linear polarization.
(714) Illustrative clause 158. The transceiver of illustrative clause 157, wherein each of the one or more first antennas and the one or more second antennas is one of a differential waveguide probe antenna, a differential tapered antenna, and a differential patch antenna.
(715) Illustrative clause 159. The transceiver of illustrative clause 156, wherein each of the first polarization, the second polarization, the third polarization, and the fourth polarization is a circular polarization.
(716) Illustrative clause 160. The transceiver of illustrative clause 159, wherein each of the one or more first antennas and the one or more second antennas is one of a helix antenna and a spiral antenna.
(717) Illustrative clause 161. The transceiver of illustrative clause 134, wherein the one or more first radiated signals are a plurality of first radiated signals including a first complementary radiated

signal having a first polarization, a second complementary radiated signal having a second polarization different from the first polarization, and a first combined radiated signal and the one or more second radiated signals are a plurality of second radiated signals including a third complementary radiated signal having a third polarization, a fourth complementary radiated signal having a fourth polarization different from the third polarization, and a second combined radiated signal having a fifth polarization different from the third polarization and the fourth polarization, the second combined radiated signal being formed by the third complementary radiated signal and the fourth complementary radiated signal interacting in the second hollow waveguide, the one or more first antennas being further configured to couple the first complementary radiated signal having the first polarization and the second complementary radiated signal having the second polarization in the first hollow waveguide such that the first complementary radiated signal and the second complementary radiated signal interact in the first hollow waveguide to form the first combined radiated signal having a sixth polarization different from the first polarization and the second polarization, the one or more second antennas being further configured to detect the second combined radiated signal received from the one of the first hollow waveguide and the second hollow waveguide and generate the one or more antenna output signals based on the second combined radiated signal.

(718) Illustrative clause 162. The transceiver of illustrative clause 161, wherein the one or more first antennas are a first antenna array comprising a plurality of first antennas, and the one or more second antennas are a second antenna array comprising a plurality of second antennas.

(719) Illustrative clause 163. The transceiver of illustrative clause 134, wherein the one or more first baseband signals include a plurality of first parallel baseband signals and a first serial baseband signal and the one or more second baseband signals include a plurality of second parallel baseband signals and a second serial baseband signal, the transmitter further comprising a serializer configured to receive the plurality of first parallel baseband signals and combine the plurality of first parallel baseband signals into the first serial baseband signal, the client-side input being configured to receive the first serial baseband signal, the transmitter circuitry being configured to receive the first serial baseband signal from the client-side input and generate the one or more antenna feed signals based on the first serial baseband signal, the receiver circuitry being configured to generate the second serial baseband signal based on the one or more antenna output signals, the client-side output being configured to receive the second serial baseband signal from the receiver circuitry and transmit the second serial baseband signal, the receiver further comprising a deserializer configured to receive the second serial baseband signal from the client-side output and split the second serial baseband signal into the plurality of second parallel baseband signals.

(720) Illustrative clause 164. The transceiver of illustrative clause 163, wherein combining the plurality of first parallel baseband signals into the first serial baseband signal and splitting the second serial baseband signal into the plurality of second parallel baseband signals utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM), and wavelength division multiplexing (WDM).

(721) Illustrative clause 165. The transceiver of illustrative clause 134, wherein the one or more first baseband signals include a plurality of first parallel baseband signals and a first serial baseband signal and the one or more second baseband signals include a plurality of second parallel baseband signals and a second serial baseband signal, the transmitter further comprising a deserializer configured to receive the first serial baseband signal and split the first serial baseband signal into the plurality of first parallel baseband signals, the client-side input being configured to receive the plurality of first parallel baseband signals, the transmitter circuitry configured to receive the plurality of first parallel baseband signals from the client-side input and generate the one or more antenna feed signals based on the plurality of first parallel baseband signals, the receiver circuitry being configured to generate the plurality of second parallel baseband signals based on the

one or more antenna output signals, the client-side output being configured to receive the plurality of second parallel baseband signals from the receiver circuitry and transmit the plurality of second parallel baseband signals, the receiver further comprising a serializer configured to receive the plurality of second parallel baseband signals and combine the plurality of second parallel baseband signals into the second serial baseband signal.

(722) Illustrative clause 166. The transceiver of illustrative clause 165, wherein splitting the first serial baseband signal into the plurality of first parallel baseband signals and combining the plurality of second parallel baseband signals into the second serial baseband signal utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM), and wavelength division multiplexing (WDM).

(723) Illustrative clause 167. The transceiver of illustrative clause 134, wherein the hollow waveguide core of each of the first hollow waveguide and the second hollow waveguide has a cross-section configured to support propagation of a plurality of polarizations.

(724) Illustrative clause 168. The transceiver of illustrative clause 167, wherein the cross-section of the hollow waveguide core of each of the first hollow waveguide and the second hollow waveguide has an elliptical or circular shape.

(725) Illustrative clause 169. The transceiver of illustrative clause 167, wherein the cross-section of the hollow waveguide core of each of the first hollow waveguide and the second hollow waveguide has a rectangular or square shape.

(726) Illustrative clause 170. The transceiver of illustrative clause 167, wherein the cross-section of the hollow waveguide core of each of the first hollow waveguide and the second hollow waveguide has a cross shape.

(727) Illustrative clause 171. The transceiver of illustrative clause 134, wherein the first frequency of the one or more first radiated signals is a transmission frequency, the transmitter circuitry comprising: one or more local oscillators configured to generate one or more carrier signals, each of the one or more carrier signals having a baseband frequency less than the transmission frequency; one or more modulation circuits configured to receive the one or more first baseband signals from the client-side input and the one or more carrier signals from the one or more local oscillators and modulate the one or more first baseband signals onto the one or more carrier signals to generate one or more modulated signals; and one or more up-conversion circuits configured to receive the one or more modulated signals from the one or more modulation circuits and up-convert the one or more modulated signals to generate the one or more antenna feed signals, each of the one or more antenna feed signals having the transmission frequency.

(728) Illustrative clause 172. The transceiver of illustrative clause 134, wherein the second frequency of the one or more second radiated signals is a transmission frequency, the receiver circuitry comprising: one or more local oscillators configured to generate one or more reference signals, each of the one or more reference signals having a baseband frequency less than the transmission frequency; one or more down-conversion circuits configured to receive the one or more antenna output signals from the one or more second antennas and the one or more reference signals from the one or more local oscillators and down-convert the one or more antenna output signals using the one or more reference signals to generate one or more modulated signals, each of the one or more modulated signals having the baseband frequency; and one or more demodulation circuits configured to receive the one or more modulated signals from the one or more down-conversion circuits and demodulate the one or more modulated signals to generate the one or more second baseband signals.

(729) Illustrative clause 173. The transceiver of illustrative clause 134, wherein the one or more first baseband signals are a plurality of first baseband signals, the one or more antenna feed signals being a plurality of antenna feed signals including a combined antenna feed signal, the one or more first radiated signals including a first combined radiated signal, the first frequency of the one or more first radiated signals being a transmission frequency, the transmitter circuitry comprising: a

plurality of local oscillators configured to generate a plurality of carrier signals, each of the plurality of carrier signals having a baseband frequency less than the transmission frequency; a plurality of modulation circuits configured to receive the plurality of first baseband signals from the client-side input and the plurality of carrier signals from the plurality of local oscillators and modulate the plurality of first baseband signals onto the plurality of carrier signals to generate a plurality of modulated signals; a plurality of up-conversion circuits configured to receive the plurality of modulated signals from the plurality of modulation circuits and up-convert the plurality of modulated signals to generate a plurality of up-converted signals; and a combiner configured to receive the plurality of up-converted signals from the plurality of up-conversion circuits and combine the plurality of up-converted signals into the combined antenna feed signal; wherein the one or more first antennas are configured to receive the combined antenna feed signal from the combiner, generate the first combined radiated signal based on the combined antenna feed signal, and couple the first combined radiated signal into the first hollow waveguide.

(730) Illustrative clause 174. The transceiver of illustrative clause 173, wherein combining the plurality of up-converted signals into the combined antenna feed signal utilizes at least one of time division multiplexing (TDM) and wavelength division multiplexing (WDM).

(731) Illustrative clause 175. The transceiver of illustrative clause 134, wherein the one or more second baseband signals are a plurality of second baseband signals, the one or more antenna output signals being a plurality of antenna output signals including a combined antenna output signal, the one or more second radiated signals including a second combined radiated signal, the second frequency of the one or more second radiated signals being a transmission frequency, the one or more second antennas being configured to detect the second combined radiated signal received from the one of the first hollow waveguide and the second hollow waveguide and generate the combined antenna output signal based on the second combined radiated signal, the receiver circuitry comprising: a splitter configured to receive the combined antenna output signal from the one or more second antennas and split the combined antenna output signal into the plurality of antenna output signals; a plurality of local oscillators configured to generate a plurality of reference signals, each of the plurality of reference signals having a baseband frequency less than the transmission frequency; a plurality of down-conversion circuits configured to receive the plurality of antenna output signals from the splitter and the plurality of reference signals from the plurality of local oscillators and down-convert the plurality of antenna output signals using the plurality of reference signals to generate a plurality of modulated signals, each of the plurality of modulated signals having the baseband frequency; and a plurality of demodulation circuits configured to receive the plurality of modulated signals from the plurality of down-conversion circuits and demodulate the plurality of modulated signals to generate the plurality of second baseband signals.

(732) Illustrative clause 176. The transceiver of illustrative clause 175, wherein splitting the combined antenna output signal into the plurality of antenna output signals utilizes at least one of time division multiplexing (TDM) and wavelength division multiplexing (WDM).

(733) Illustrative clause 177. The transceiver of illustrative clause 134, wherein the one or more first baseband signals are a plurality of first baseband signals, the one or more antenna feed signals being a plurality of antenna feed signals, the one or more first radiated signals being a plurality of first radiated signals including a first combined radiated signal, the first frequency of the first radiated signals being a transmission frequency, the one or more first antennas being a first antenna array comprising a plurality of first antennas, the transmitter circuitry comprising: a plurality of local oscillators configured to generate a plurality of carrier signals, each of the plurality of carrier signals having a baseband frequency less than the transmission frequency; a plurality of modulation circuits configured to receive the plurality of first baseband signals from the client-side input and the plurality of carrier signals from the plurality of local oscillators and modulate the plurality of first baseband signals onto the plurality of carrier signals to generate a plurality of modulated signals; and a plurality of up-conversion circuits configured to receive the plurality of modulated

signals from the plurality of modulation circuits and up-convert the plurality of modulated signals to generate the plurality of antenna feed signals; wherein the plurality of first antennas are configured to receive the plurality of antenna feed signals from the plurality of up-conversion circuits, generate the plurality of first radiated signals based on the plurality of antenna feed signals, and couple the plurality of first radiated signals into the first hollow waveguide such that the plurality of first radiated signals interact in the first hollow waveguide to form the first combined radiated signal.

(734) Illustrative clause 178. The transceiver of illustrative clause 177, wherein coupling the plurality of first radiated signals into the first hollow waveguide such that the plurality of first radiated signals interact in the first hollow waveguide to form the first combined radiated signal utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM) and wavelength division multiplexing (WDM).

(735) Illustrative clause 179. The transceiver of illustrative clause 134, wherein the one or more second baseband signals are a plurality of second baseband signals, the one or more antenna output signals being a plurality of antenna output signals, the one or more second radiated signals being a plurality of second radiated signals including a first complementary radiated signal, a second complementary radiated signal, and a second combined radiated signal formed by the first complementary radiated signal and the second complementary radiated signal interacting in the second hollow waveguide, the second frequency of the one or more second radiated signals being a transmission frequency, the one or more second antennas being a second antenna array comprising a plurality of second antennas, the plurality of second antennas being configured to detect the first complementary radiated signal and the second complementary radiated signal based on the second combined radiated signal received from the one of the first hollow waveguide and the second hollow waveguide and generate the plurality of antenna output signals based on the first complementary radiated signal and the second complementary radiated signal, the receiver circuitry comprising: a plurality of local oscillators configured to generate a plurality of reference signals, each of the plurality of reference signals having a baseband frequency less than the transmission frequency; a plurality of down-conversion circuits configured to receive the plurality of antenna output signals from the plurality of second antennas and the plurality of reference signals from the plurality of local oscillators and down-convert the plurality of antenna output signals using the plurality of reference signals to generate a plurality of modulated signals, each of the plurality of modulated signals having the baseband frequency; and a plurality of demodulation circuits configured to receive the plurality of modulated signals from the plurality of down-conversion circuits and demodulate the plurality of modulated signals to generate the plurality of second baseband signals.

(736) Illustrative clause 180. The transceiver of illustrative clause 179, wherein detecting the first complementary radiated signal and the second complementary radiated signal based on the second combined radiated signal received from the one of the first hollow waveguide and the second hollow waveguide utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM) and wavelength division multiplexing (WDM).

CONCLUSION

(737) The foregoing description provides illustration and description, but is not intended to be exhaustive or to limit the inventive concepts to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of the methodologies set forth in the present disclosure.

(738) Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one other claim, the disclosure includes each dependent claim in combination with every other claim in the

claim set.

(739) No element, act, or instruction used in the present application should be construed as critical or essential to the invention unless explicitly described as such outside of the preferred implementation. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

Claims

1. A transport network, comprising: one or more hollow waveguides; a transmitter, comprising: a client-side input configured to receive one or more first baseband signals having client data encoded therein; transmitter circuitry configured to receive the one or more first baseband signals from the client-side input and generate one or more antenna feed signals based on the one or more first baseband signals; and one or more first antennas configured to receive the one or more antenna feed signals from the transmitter circuitry, generate one or more radiated signals based on the one or more antenna feed signals, and couple the one or more radiated signals into at least one of the one or more hollow waveguides, each of the one or more radiated signals being radiated electromagnetic waves configured for coherent detection and having a frequency in a range between 300 Gigahertz (GHz) and 10 Terahertz (THz); and a receiver, comprising: one or more second antennas configured to detect the one or more radiated signals received from the at least one of the one or more hollow waveguides and generate one or more antenna output signals based on the one or more radiated signals; receiver circuitry configured to receive the one or more antenna output signals from the one or more second antennas and generate one or more second baseband signals based on the one or more antenna output signals, the one or more second baseband signals having the client data; and a client-side output configured to receive the one or more second baseband signals from the receiver circuitry and transmit the one or more second baseband signals.
2. The transport network of claim 1, wherein the at least one of the one or more hollow waveguides has a hollow waveguide core having a refractive index in a range between 1.0 and 1.4.
3. The transport network of claim 1, wherein the at least one of the one or more hollow waveguides has a hollow waveguide core and a tubular sidewall surrounding the hollow waveguide core, the hollow waveguide core being filled with one of a gas, a vacuum, and a porous material having a porosity in a range between 25% and 99%.
4. The transport network of claim 3, wherein the tubular sidewall of the at least one of the one or more hollow waveguides further comprises a conductive layer.
5. The transport network of claim 4, wherein the tubular sidewall of the at least one of the one or more hollow waveguides further comprises a support layer surrounding the conductive layer.
6. The transport network of claim 4, wherein the tubular sidewall of the at least one of the one or more hollow waveguides further comprises a dielectric layer between the hollow waveguide core and the conductive layer.
7. The transport network of claim 3, wherein the tubular sidewall of the at least one of the one or more hollow waveguides has one or more conductive layers and one or more dielectric layers, the one or more conductive layers interleaved with the one or more dielectric layers.
8. The transport network of claim 1, wherein the at least one of the one or more hollow waveguides is configured to support propagation of a single mode of the one or more radiated signals.
9. The transport network of claim 1, wherein the at least one of the one or more hollow waveguides is configured to support propagation of a plurality of modes of the one or more radiated signals.
10. The transport network of claim 1, wherein the one or more antenna feed signals are provided to the one or more first antennas on one or more first transmission lines, each of the one or more first transmission lines having two or more conductors.
11. The transport network of claim 10, wherein each of the one or more first transmission lines have a first transmission loss and the at least one of the one or more hollow waveguides has a

second transmission loss less than the first transmission loss, the second transmission loss being in a range between **0.001** and **20.00** decibels (dB) per meter (m) per Terabit (Tb) per second(s).

12. The transport network of claim 1, wherein the one or more antenna output signals are received from the one or more second antennas on one or more second transmission lines, each of the one or more second transmission lines having two or more conductors.

13. The transport network of claim 12, wherein each of the one or more second transmission lines have a first transmission loss and the at least one of the one or more hollow waveguides has a second transmission loss less than the first transmission loss, the second transmission loss being in a range between **0.001** and **20.00** decibels (dB) per meter (m) per Terabit (Tb) per second(s).

14. The transport network of claim 1, wherein two or more of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are disposed on a single substrate.

15. The transport network of claim 12, wherein at least two of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are disposed on a multi-layer substrate having a plurality of layers, at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas being disposed on a first layer of the plurality of layers, at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas being disposed on a second layer of the plurality of layers.

16. The transport network of claim 14, wherein at least two of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are integrated into a single monolithic semiconductor die.

17. The transport network of claim 14, wherein at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are not disposed on the single substrate.

18. The transport network of claim 1, wherein at least two of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are disposed on a plurality of substrates, at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas being disposed on a first substrate of the plurality of substrates, at least one of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas being disposed on a second substrate of the plurality of substrates.

19. The transport network of claim 18, wherein at least two of the plurality of substrates are in a stacked arrangement.

20. The transport network of claim 1, wherein each of the client-side input, the transmitter circuitry, the one or more first antennas, the client-side output, the receiver circuitry, and the one or more second antennas are implemented using one or more of complementary metal-oxide semiconductor (CMOS) technology, silicon-germanium (SiGe) semiconductor technology, and Ill-V compound semiconductor technology.

21. The transport network of claim 1, wherein the client data is encoded in the one or more radiated signals using an encoding protocol conforming to requirements of one or more of return-to-zero (RZ) code, non-return-to-zero (NRZ) code, quadrature phase-shift keying (QPSK), quadrature-amplitude modulation (QAM), trellis coded modulation (TCM), and Bose-Chaudhuri-Hocquenghem (BCH) code.

22. The transport network of claim 1, wherein the one or more radiated signals are a plurality of radiated signals including a first complementary radiated signal having a first polarization and a second complementary radiated signal having a second polarization different from the first polarization, the one or more first antennas being further configured to generate the first

complementary radiated signal and the second complementary radiated signal based on the one or more antenna feed signals, the one or more second antennas being further configured to generate the one or more antenna output signals based on the first complementary radiated signal and the second complementary radiated signal.

23. The transport network of claim 22, wherein the first polarization is orthogonal to the second polarization.

24. The transport network of claim 23, wherein each of the first polarization and the second polarization is a linear polarization.

25. The transport network of claim 24, wherein each of the one or more first antennas and the one or more second antennas is one of a differential waveguide probe antenna, a differential tapered antenna, and a differential patch antenna.

26. The transport network of claim 23, wherein each of the first polarization and the second polarization is a circular polarization.

27. The transport network of claim 26, wherein each the one or more first antennas and the one or more second antennas is one of a helix antenna and a spiral antenna.

28. The transport network of claim 1, wherein the one or more first baseband signals include a plurality of first parallel baseband signals and a first serial baseband signal and the one or more second baseband signals include a plurality of second parallel baseband signals and a second serial baseband signal, the transmitter further comprising a serializer configured to receive the plurality of first parallel baseband signals and combine the plurality of first parallel baseband signals into the first serial baseband signal, the client-side input being configured to receive the first serial baseband signal, the transmitter circuitry being configured to receive the first serial baseband signal from the client-side input and generate the one or more antenna feed signals based on the first serial baseband signal, the receiver circuitry being configured to generate the second serial baseband signal based on the one or more antenna output signals, the client-side output being configured to receive the second serial baseband signal from the receiver circuitry and transmit the second serial baseband signal, the receiver further comprising a deserializer configured to receive the second serial baseband signal from the client-side output and split the second serial baseband signal into the plurality of second parallel baseband signals.

29. The transport network of claim 28, wherein combining the plurality of first parallel baseband signals into the first serial baseband signal and splitting the second serial baseband signal into the plurality of second parallel baseband signals utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM), and wavelength division multiplexing (WDM).

30. The transport network of claim 1, wherein the one or more first baseband signals include a plurality of first parallel baseband signals and a first serial baseband signal and the one or more second baseband signals include a plurality of second parallel baseband signals and a second serial baseband signal, the transmitter further comprising a deserializer configured to receive the first serial baseband signal and split the first serial baseband signal into the plurality of first parallel baseband signals, the client-side input being configured to receive the plurality of first parallel baseband signals, the transmitter circuitry configured to receive the plurality of first parallel baseband signals from the client-side input and generate the one or more antenna feed signals based on the plurality of first parallel baseband signals, the receiver circuitry being configured to generate the plurality of second parallel baseband signals based on the one or more antenna output signals, the client-side output being configured to receive the plurality of second parallel baseband signals from the receiver circuitry and transmit the plurality of second parallel baseband signals, the receiver further comprising a serializer configured to receive the plurality of second parallel baseband signals and combine the plurality of second parallel baseband signals into the second serial baseband signal.

31. The transport network of claim 30, wherein splitting the first serial baseband signal into the

plurality of first parallel baseband signals and combining the plurality of second parallel baseband signals into the second serial baseband signal utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM), and wavelength division multiplexing (WDM).

32. The transport network of claim 1, wherein at least one of the one or more hollow waveguides has a hollow waveguide core having a cross-section configured to support propagation of a plurality of polarizations.

33. The transport network of claim 32, wherein the cross-section of the hollow waveguide core of the at least one of the one or more hollow waveguides has a rectangular or square shape.

34. The transport network of claim 32, wherein the cross-section of the hollow waveguide core of the at least one of the one or more hollow waveguides has a cross shape.

35. The transport network of claim 32, wherein the cross-section of the hollow waveguide core of the at least one of the one or more hollow waveguides has an elliptical or circular shape.

36. The transport network of claim 35, wherein at least one of the one or more hollow waveguides is a multi-mode waveguide.

37. The transport network of claim 36, wherein the at least one of the one or more hollow waveguides has a hollow waveguide core having a refractive index in a range between 1.0 and 1.4.

38. The transport network of claim 36, wherein the at least one of the one or more hollow waveguides has a hollow waveguide core and a tubular sidewall surrounding the hollow waveguide core, the hollow waveguide core being filled with one of a gas, a vacuum, and a porous material having a porosity in a range between 25% and 99%.

39. The transport network of claim 1, wherein the frequency of the one or more radiated signals is a transmission frequency, the transmitter circuitry comprising: one or more local oscillators configured to generate one or more carrier signals, each of the one or more carrier signals having a first baseband frequency less than the transmission frequency; one or more modulation circuits configured to receive the one or more first baseband signals from the client-side input and the one or more carrier signals from the one or more local oscillators and modulate the one or more first baseband signals onto the one or more carrier signals to generate one or more modulated signals; and one or more up-conversion circuits configured to receive the one or more modulated signals from the one or more modulation circuits and up-convert the one or more modulated signals to generate the one or more antenna feed signals, each of the one or more antenna feed signals having the transmission frequency.

40. The transport network of claim 1, wherein the frequency of the one or more radiated signals is a transmission frequency, the receiver circuitry comprising: one or more local oscillators configured to generate one or more reference signals, each of the one or more reference signals having a baseband frequency less than the transmission frequency; one or more down-conversion circuits configured to receive the one or more antenna output signals from the one or more second antennas and the one or more reference signals from the one or more local oscillators and down-convert the one or more antenna output signals using the one or more reference signals to generate one or more modulated signals, each of the one or more modulated signals having the baseband frequency; and one or more demodulation circuits configured to receive the one or more modulated signals from the one or more down-conversion circuits and demodulate the one or more modulated signals to generate the one or more second baseband signals.

41. The transport network of claim 1, wherein the one or more first baseband signals are a plurality of first baseband signals, the one or more antenna feed signals being a plurality of antenna feed signals including a combined antenna feed signal, the one or more radiated signals including a combined radiated signal, the frequency of the one or more radiated signals being a transmission frequency, the transmitter circuitry comprising: a plurality of local oscillators configured to generate a plurality of carrier signals, each of the plurality of carrier signals having a baseband frequency less than the transmission frequency; a plurality of modulation circuits configured to

receive the plurality of first baseband signals from the client-side input and the plurality of carrier signals from the plurality of local oscillators and modulate the plurality of first baseband signals onto the plurality of carrier signals to generate a plurality of modulated signals; a plurality of up-conversion circuits configured to receive the plurality of modulated signals from the plurality of modulation circuits and up-convert the plurality of modulated signals to generate a plurality of up-converted signals; and a combiner configured to receive the plurality of up-converted signals from the plurality of up-conversion circuits and combine the plurality of up-converted signals into the combined antenna feed signal; and wherein the one or more first antennas are configured to receive the combined antenna feed signal from the combiner, generate the combined radiated signal based on the combined antenna feed signal, and couple the combined radiated signal into the at least one of the one or more hollow waveguides.
42. The transport network of claim 41, wherein combining the plurality of up-converted signals into the combined antenna feed signal utilizes at least one of time division multiplexing (TDM) and wavelength division multiplexing (WDM).
43. The transport network of claim 1, wherein the one or more second baseband signals are a plurality of second baseband signals, the one or more antenna output signals being a plurality of antenna output signals including a combined antenna output signal, the one or more radiated signals including a combined radiated signal, the frequency of the one or more radiated signals being a transmission frequency, the one or more second antennas being configured to detect the combined radiated signal received from the one or more hollow waveguides and generate the combined antenna output signal based on the combined radiated signal, the receiver circuitry comprising: a splitter configured to receive the combined antenna output signal from the one or more second antennas and split the combined antenna output signal into the plurality of antenna output signals; a plurality of local oscillators configured to generate a plurality of reference signals, each of the plurality of reference signals having a baseband frequency less than the transmission frequency; a plurality of down-conversion circuits configured to receive the plurality of antenna output signals from the splitter and the plurality of reference signals from the plurality of local oscillators and down-convert the plurality of antenna output signals using the plurality of reference signals to generate a plurality of modulated signals, each of the plurality of modulated signals having the baseband frequency; and a plurality of demodulation circuits configured to receive the plurality of modulated signals from the plurality of down-conversion circuits and demodulate the plurality of modulated signals to generate the plurality of second baseband signals.
44. The transport network of claim 43, wherein splitting the combined antenna output signal into the plurality of antenna output signals utilizes at least one of time division multiplexing (TDM) and wavelength division multiplexing (WDM).
45. The transport network of claim 1, wherein the one or more first baseband signals are a plurality of first baseband signals, the one or more antenna feed signals being a plurality of antenna feed signals, the one or more radiated signals being a plurality of radiated signals including a combined radiated signal, the frequency of the one or more radiated signals being a transmission frequency, the one or more first antennas being a first antenna array comprising a plurality of first antennas, the transmitter circuitry comprising: a plurality of local oscillators configured to generate a plurality of carrier signals, each of the plurality of carrier signals having a baseband frequency less than the transmission frequency; a plurality of modulation circuits configured to receive the plurality of first baseband signals from the client-side input and the plurality of carrier signals from the plurality of local oscillators and modulate the plurality of first baseband signals onto the plurality of carrier signals to generate a plurality of modulated signals; a plurality of up-conversion circuits configured to receive the plurality of modulated signals from the plurality of modulation circuits and up-convert the plurality of modulated signals to generate the plurality of antenna feed signals; and wherein the plurality of first antennas are configured to receive the plurality of antenna feed signals from the plurality of up-conversion circuits, generate the plurality of radiated signals

based on the plurality of antenna feed signals, and couple the plurality of radiated signals into the at least one of the one or more hollow waveguides such that the plurality of radiated signals interact in the at least one of the one or more hollow waveguides to form the combined radiated signal.
46. The transport network of claim 45, wherein coupling the plurality of radiated signals into the at least one of the one or more hollow waveguides such that the plurality of radiated signals interact in the at least one of the one or more hollow waveguides to form the combined radiated signal utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM) and wavelength division multiplexing (WDM).
47. The transport network of claim 1, wherein the one or more second baseband signals are a plurality of second baseband signals, the one or more antenna output signals being a plurality of antenna output signals, the one or more radiated signals being a plurality of radiated signals including a first complementary radiated signal, a second complementary radiated signal, and a combined radiated signal formed by the first complementary radiated signal and the second complementary radiated signal interacting in the at least one of the one or more hollow waveguides, the frequency of the one or more radiated signals being a transmission frequency, the one or more second antennas being an antenna array comprising a plurality of antennas, the plurality of antennas being configured to detect the first complementary radiated signal and the second complementary radiated signal based on the combined radiated signal received from the at least one of the one or more hollow waveguides and generate the plurality of antenna output signals based on the first complementary radiated signal and the second complementary radiated signal, the receiver circuitry comprising: a plurality of local oscillators configured to generate a plurality of reference signals, each of the plurality of reference signals having a baseband frequency less than the transmission frequency; a plurality of down-conversion circuits configured to receive the plurality of antenna output signals from the plurality of antennas and the plurality of reference signals from the plurality of local oscillators and down-convert the plurality of antenna output signals using the plurality of reference signals to generate a plurality of modulated signals, each of the plurality of modulated signals having the baseband frequency; and a plurality of demodulation circuits configured to receive the plurality of modulated signals from the plurality of down-conversion circuits and demodulate the plurality of modulated signals to generate the plurality of second baseband signals.
48. The transport network of claim 47, wherein detecting the first complementary radiated signal and the second complementary radiated signal based on the combined radiated signal received from the at least one of the one or more hollow waveguides utilizes at least one of polarization division multiplexing (PDM), time division multiplexing (TDM) and wavelength division multiplexing (WDM).
49. The transport network claim 1, wherein the client data is provided by a client in connection with a data communication service between two or more servers.
50. The transport network of claim 1, wherein each particular one of the one or more radiated signals has a bandwidth in a range between **10**% and **40**% of the frequency of the particular one of the one or more radiated signals.
51. The transport network of claim 1, wherein the client data is encoded in the one or more first baseband signals and the one or more second baseband signals using an encoding protocol conforming to requirements of one or more of return-to-zero (RZ) code, non-return-to- zero (NRZ) code, pulse-amplitude modulation (PAM), and quadrature-amplitude modulation (QAM).
52. The transport network of claim 1, wherein the one or more radiated signals are a plurality of radiated signals including a first complementary radiated signal having a first polarization and a second complementary radiated signal having a second polarization different from the first polarization, the one or more first antennas being further configured to couple the first complementary radiated signal having the first polarization and the second complementary radiated signal having the second polarization in the at least one of the one or more hollow waveguides such

that the first complementary radiated signal and the second complementary radiated signal interact in the at least one of the one or more hollow waveguides to form a combined radiated signal having a third polarization different from the first polarization and the second polarization, the one or more second antennas being further configured to detect the combined radiated signal received from the at least one of the one or more hollow waveguides and generate the one or more antenna output signals based on the combined radiated signal.

53. The transport network of claim 52, wherein the one or more first antennas are a first antenna array comprising a first plurality of antennas and the one or more second antennas are a second antenna array comprising a second plurality of antennas.

54. The transport network of claim 1, wherein the client-side input is configured to carry the first baseband signals in a differential format.

55. The transport network of claim 1, wherein the frequency of the one or more radiated signals is a transmission frequency, the transmitter circuitry comprising: one or more local oscillators configured to generate one or more carrier signals, each of the one or more carrier signals having a baseband frequency less than the transmission frequency; one or more modulation circuits configured to receive the one or more first baseband signals from the client-side input and the one or more carrier signals from the one or more local oscillators and modulate the one or more first baseband signals onto the one or more carrier signals to generate one or more modulated signals; and an RF interface configured to receive the one or more modulated signals and generate the one or more antenna feed signals.

56. The transport network of claim 1, wherein the transmitter circuitry is configured to generate one or more in-phase (I) antenna feed signals and one or more quadrature (Q) antenna feed signals; and wherein the transmitter circuitry includes a power combiner configured to receive the one or more I antenna feed signals and the one or more Q antenna feed signals and combine the one or more I antenna feed signals and the one or more Q antenna feed signals to produce the one or more antenna feed signals having an I component and a Q component.

57. The transport network of claim 1, wherein the one or more first baseband signals are a plurality of first baseband signals, and wherein the transmitter circuitry includes a plurality of modulators encoding the plurality of first baseband signals to form a plurality of different channel signals, and a combiner to combine the plurality of different channel signals to produce the one or more antenna feed signals.

58. The transport network of claim 57, wherein the plurality of modulators includes a first modulator configured to modulate one or more first baseband signals into a first carrier frequency with a first modulation format to generate a first channel signal and a second modulator configured to modulate one or more first baseband signals into a second carrier frequency with a second modulation format to generate a second channel signal.

59. The transport network of claim 58, wherein the first carrier frequency and the second carrier frequency are different and the one or more antenna feed signals are in a wavelength division multiplexed (WDM) format.

60. The transport network of claim 58, wherein the first modulation format is different that the second modulation format.

61. The transport network of claim 58, wherein at least one of the one or more hollow waveguides is a multi-mode waveguide.

62. The transport network of claim 58, wherein the first carrier frequency and the second carrier frequency are the same and the one or more antenna feed signals are in a polarization division multiplexed (PDM) format.

63. The transport network of claim 1, wherein the one or more first antennas include a slot antenna.

64. The transport network of claim 63, wherein the slot antenna includes a ground plane having a slot and a back reflector adjacent to the ground plane.

65. The transport network of claim 64, wherein the ground plane is between the back reflector and

at least one of the one or more hollow waveguides.
66. The transport network of claim 64, wherein the slot antenna is a double slot antenna having a pair of slots.
67. The transport network of claim 1, further comprising an intermediary waveguide positioned between the one or more first antennas and at least one of the one or more hollow waveguides, the intermediary waveguide configured to propagate the one or more radiated signals to the at least one of the one or more hollow waveguides.
68. The transport network of claim 67, wherein the intermediary waveguide has one or more first intended waveguide modes and the at least one of the one or more hollow waveguides has one or more second intended waveguide modes, and wherein the one or more first intended waveguide modes of the intermediary waveguide sufficiently matches the one or more second intended waveguide modes of the at least one of the one or more hollow waveguides such that a coupling loss between the intermediary waveguide and the at least one of the one or more hollow waveguides is in a range between 0.1 dB and 5.0 dB.
69. The transport network of claim 67, wherein the intermediary waveguide has a flared end coupled to the at least one of the one or more hollow waveguides and configured to facilitate a mode transition between the intermediary waveguide and the at least one of the one or more hollow waveguides.
70. The transport network of claim 69, wherein the flared end comprises a horn separate from the intermediary waveguide.
71. The transport network of claim 70, wherein the horn has a first end abutting the intermediary waveguide, a second end opposite the first end, and a curved surface extending between the first end and the second end.
72. The transport network of claim 71, wherein at least one of the one or more hollow waveguides has a hollow waveguide core and a tubular sidewall surrounding the hollow waveguide core, the hollow waveguide core being filled with one of a gas, a vacuum, and a porous material having a porosity in a range between 25% and 99%.
73. The transport network of claim 1, wherein the radiated electromagnetic waves configured for coherent detection have a frequency in a range between 700 Gigahertz (GHz) and 5 Terahertz (THz).
74. The transport network of claim 1, wherein the client data is provided by a client in connection with a telecommunication service.
75. The transport network of claim 1, wherein the client data is provided by a client in connection with a storage service.
76. The transport network of claim 1, wherein the transmitter and receiver operate entirely in the electrical domain.
77. The transport network of claim 1, wherein at least one of the one or more hollow waveguides is a solid rod fiber.
78. The transport network of claim 1, wherein at least one of the one or more hollow waveguides has a hollow waveguide core comprising silicon.
79. The transport network of claim 78, wherein the hollow waveguide core comprises silica glass (SiO_2).
80. The transport network of claim 1, wherein at least one of the one or more hollow waveguides has a hollow waveguide core composed of a polymer.
